Pediatrics

General

Which of the following measures would decrease the incidence of Sudden Infant Death Syndrome?
A. Pacifier at bedtime
B. Sleep in prone position
C. Soft sleeping mattress
D. Sleep positon support device

Answer: A

Which of the following is the most common cause of dehydration in children?
A. Heat stroke
B. Food poisoning
C. Gastroenteritis
D. Drugs side effects

Answer: C
A 5-year-old child presents to Emergency Room with vomiting and diarrhea for 4 days. On examination, he looks dehydrated, hypoactive with sunken eyes, doughy skin and dry mucous membrane. The capillary refill is 4 seconds.
Blood pressure 90/70 mmH
Heart rate 130 /min
Respiratory rate 22 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following is the expected serum sodium level based on clinical finding?
A. Low
B. High
C. Normal
D. No correlation

Answer: B
A 5-year-old child presents to Emergency Room with vomiting and diarrhea for 4 days. On examination, he looks dehydrated, hypoactive with sunken eyes and delayed skin turger with dry mucous membrane and capillary refill is 4 seconds. The child connected to cardiopulmonary monitor and IV access attained, blood sample sent for electrolytes.
Blood pressure 60/40 mmH
Heart rate 134 /min
Respiratory rate 23 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following is the most appropriate next action?
A. Start inotropes
B. Give 5% albumin bolus
C. 20 ml per kg normal saline bolus
D. Start maintenance fluid waiting for serum electrolytes

Answer: C

A 2-month-old infant found dead at his bed. According to the mother, he was feeding normally with usual activity. No history of fever or other abnormality. On examination, no sign of trauma or bruises on the skin. Baby gram was negative for fracture.
Which of the following history's part is important to make the diagnosis?
A. Social history
B. Allergic history
C. Blood transfusion history
D. Materal Medication history during pregnancy

Answer: A
A 2-month-old infant found dead at his bed. According to the mother, he was feeding normally with usual activity. No history of fever or other abnormality. On examination, no sign of trauma or bruises on the skin. Baby gram was negative for fracture. Mother is divorced and known smoker.
Which of the following is the most likely diagnosis?
A. Child abuse syndrome
B. Maternal smoking syndrome
C. Sudden infant death syndrome
D. Apparent life threatening events

Answer: C

Which of the following measures would decrease the incidence of Sudden Infant Death Syndrome?

A. Pacifier at bedtime
B. Sleep in prone potion
C. Soft sleeping mattress
D. Sleep position support device

Answer: A

A 3-month-old girl presented dead to the hospital by her parents. History revealed that the girl was preterm with low birth weight and chronic lung disease. She was sleeping with her parents on their bed and both parents are smoker.
Which of the following describes her death?
A. Acute lung failure syndrome
B. Sudden infant death syndrome
C. Apparent life-threatening syndrome
D. Unexpected oxygen insufficiency syndrome

Answer: B

Sudden infant death syndrome (SIDS) defined as unexpected death of an infant for which no adequate cause is found after postmortem examination. The incidence of SIDS has fallen during the last 20 years. Which of the following contribute the most for fallen in the incidence?
A. Infant to be put to sleep on their back
B. Parents avoid smoking near their infant
C. Infant to be placed in the feet to foot position
D. Avoid overheating by heavy rapping or high room temperature

Answer: A

A 6-month-old infant presents to the Emergency Room with sudden apnea and color change. The mother gave history of recurrent vomiting, and muitiple chest infection. On examination, he looks failing to thrive with dystonic neck posturing while crying.
Which of the following considers as high-risk case for the above disease?
A. Obese baby
B. Preterm infant
C. Post pertussis infection
D. Following inguinal hernia repair

Answer: B

ER/ICU/fluid replacement

A 4-year-old girl is brought to the Pediatrics Emergency Department with vomiting and lethargy. The parents report a weight loss of 1 kg over the last 2 weeks. There has been little appetite, but a lot of thirst and frequent urination. The child appeared sleepy and irritable. Physical examination confirmed dry mucous membranes and a generally reduced skin tone. After running several exams, the pediatrician started IV treatment with ringer lactate plus added electrolytes and a low-dose insulin infusion (see lab results).
Test Result Normal Values
pH 7.2 7.36-7.45
Glucose 20 3.5-5.5 mmol/L
Urinalysis Result Normal Value
Ketones positive Negative
Which of the following complications should be observed?
A. Hypoglycemia
B. Kidney failure
C. Cerebral edema
D. Metabolic alkalosis

Answer: C
A 10-year-old boy presents to the Paediatric Clinic with weight loss of 4 kg over the last 3 weeks. The parents report that he had had little appetite and had wet his bed several times in the last week. The child has previously been healthy and there is no history of any major illness. The child appears lethargic and irritable. Physical examination confirms reduced skin tone and dry mucous membranes (see lab results).
Blood pressure 100/60 mmH
Heart rate 80 /min
Respiratory rate 50 /min
Temperature 3t.0° C
Weight 28 kg (10th percentile)
Height 104 cm (50th percentile)
Test Result Normal Values
pH 7.2 7.36-7.45
Glucose 20 3.5-5.5 mol/L
Urinalysis Result Normal Values
Ketones positive Negative
Which of the following is the initial line of management?
A. Electrolyte replacement
B. Bicarbonate treatment
C. Fluid replacement
D. Insulin infusion

Answer: C

A 6-year-old boy is brought the Emergency Department after falling down 1 flight of stairs and hitting the floor with his head. On examination, he is alert, oriented and has no focal neurological findings. There is blood coming out of his left ear (see report).

Otoscopy:
Confirms a ruptured eardrum on the left side with bleeding. The external auditory canal is intact.
Which of the following is the most likely diagnosis?
A. Basilar skull fracture
B. Subdural hematoma
C. Subarachnoid haemorrhage
D. Intraventricular hemorrhage

Answer: A

A 3-year-old child presents to the Paediatric Emergency Department a half hour after ingesting a caustic material. The child is crying, drooling, and does not look well.
Which of the following is the most appropriate initial management?
A. Establish the airway
B. Activated charcoal
C. Identify the agent
D. Immediate endoscopy

Answer: A

A 7-month-old infant is brought to clinic with a 2-day history of vomiting and diarrhea. The stool is watery without mucous or blood. examination reveals that the infant is only mildly dehydrated.
Which of the following is the most appropriate management?
A. ORS1, and cefixime
B. ORS and reassurance
C. Ant-emetcs and ORS
D. Ant-emetcs, and cotrimoxazole

Answer: B

A 5-month-old infant presents with acute onset of severe vomiting and diarrhea for the last 2 days. On examination, the baby is lethargic; had an overall poor and ill appearance, dry mucous membranes, and crying with tears, while capillary refill is 3 seconds (see lab results).
Heart rate 130 /min
Respiratory rate 25 /min
Temperature 38.6° C
Test Result Normal Values
Sodium 149 134-146 mmol/L
Glucose 2.9 3.3-5.5 mmol/L
Which of the following fluids should be administered?
A. IV isotonic fluid 20 ml/Kg
B. Pedialyte 10 ml over five min
C. Dextrose 10 % in 1/2 normal saline
D. ORS which has 1:1 rato of Na: glucose

Answer: C
A 6-month-old child presents to the Pediatric Outpatient Clinic with a 2-day history of gastroenteritis. The examination reveals decreased skin turgor, a depressed anterior fontanel, and sunken eyes.
What is the most likely percentage of dehydration?
A. 5
B. 10
C. 15

D. 20

Answer: B

A 2-month-old girl has diarrhea without vomiting. The mother is worried that she has dehydration and asks for advice on management.
Which of the following would be recommended?
A. Changing milk
B. Ant-diarrheal agent
C. Oral rehydration fluid
D. Intravenous saline infusion

Answer: C

A 3-year-old child is brought to the Emergency Room after swallowing a battery that is now lodged in the esophagus.
Which of the following is the most appropriate intervention?
A. Discharge home with reassurance
B. Administering intravenous glucagon
C. Observing the patent for 12 hours
D. Immediate endoscopic removal

Answer: D

A 4-year-old healthy child is found in the kitchen, comatose and cyanotic with peanuts in his hands
Which of the following is the most likely cause?
A. Choking secondary to aspiraton
B. Trauma with CNS bleed
C. Overwhelming sepsis
D. Postictal state

Answer: A

A 1-year-old child with high fever, poor feeding and inactivity has not urinated for 4 hours. On examination, there is a generalized petechial rash and the extremities are cold.
Blood pressure 74/40 mmHg
Heart rate 180 /min
Respiratory rate 34 /min
Temperature 37.4° C rectal
Which of the following is the most likely diagnosis?

A. Septic shock
B. Renal failure
C. Acute myocarditis
D. Idiopathic thrombocytopenic purpura

Answer: A

A 2-month-old baby is brought to the clinic. The mother reports her baby had been crying so much for the last 2 weeks and all her efforts to calm him had failed. The child was delivered normally at 36 weeks gestation. He has been breast fed since birth, but the mother recently introduced an evening bottle to help calm the baby. On examination, he is crying and irritable. rcentle on the growth curve and apart from abdominal distension there are

What is the daily fluid maintenance requirement in ml for a nine month-old child whose weight is 10 Kg?
A. 500
B. 1,000
C. 1,500
D. 2,000

Answer: B

A 6-month-old infant is brought to the clinic with a 3-day history of watery diarrhoea. On examination, the infant is awake and alert but restless and eager to drink, had a depressed fontanel, tachycardia, normal pulses and sunken eyes.
What would be the expected percentage of dehydration?
A.<1%
B. 1-3%
C. 4-9%
D. >9%

Answer: D

A 10-month-old child has a 3-day history of green watery stools. She has also been vomiting for 1 day. Physical examination confirms an afebrile, irritable baby with dry mucous membranes and sunken eyes.
Which of the following is the most important management strategy?
A. Start antibiotics
B. Antdiarrheal drugs
C. Order a stool culture
D.Fluid and electrolyte replacement

Answer: D

A 14-month-old girt is brought to the clinic with vomiting and diarrhoea. The mother reports her child is less active. On examination, she has mild dehydration with an estimated fluid deficit of 5% (see lab results).
Weigh 10 Kg
Test Result Normal Values
Sodium 128 134-146 mmol/L
Potassium 3.3 3.5-5.1 mmol/L
Chloride 94 97-108 mmol/L
Bicarbonate 12 21-28 mmol/L

How much intravenous fluid (mi/day) should be given?
A. 1000
B. 1250
C. 1500
D. 1750

Answer: C

A 5-year-old child presents to the Emergency Department with a 4-day history of malaise, poor oral intake, diarrhea, and decreased urine output. Physical examination confirms a tachycardia,

xerostomia and sunken eyes (see lab results).
Test Results Normal Values
Blood urea nitrogen 14.9 1.8 to 6.4 mmol/L
Creatinine 180 27-62 mol/L
Potassium 5.5 3.5-5.1 mmol/L
Bicarbonate 14 21-28 mmol/L
Which of the following is the most appropriate management?
A. Dialysis
B. Diuretcs
C. Normal saline
D. NaHCO3 soluton

Answer: C

A 2-month-old infant presents to Emergency Room with history of apnea. He is bottle fed with good weight gain. The mother gave history of breath cessation and cyanosis 10 minutes after each bottle feed that aborted spontaneously. He has normal examination and growth parameter. Blood pressure 80/50 mmg
Heart rate 110 /min Respiratory rate 22 /min Temperature 36.6 °C Oxygen saturation 95 %
Which of the following is most likely admission diagnosis?
A. Acute life threatening event
B. Apnea to roll out sepsis
C. Infant with seizure
D. Syncopal attack

Answer: A

Which of the following is the most common cause of shock in children?
A. Septic
B. Hypovolemic
C. Cardiogenic
D. Obstructive

Answer: B
An 18-month-old child with cerebral palsy admitted to Pediatrics Intensive Care Unit for respiratory support following event of vomiting and cyanosis. She is connected to mechanical ventilator with FIO2 of 100% (see image and lab results).
Blood pressure 100/70 mmH
Heart rate 120 /min Respiratory rate 18 /min Temperature 36.6 °C Oxygen saturation 95 %
etCO2 35 mmHg
Test Result Normal Values
ABG HCO3- 21 13-22 mmol/L
ABG PCO2 35 27-40 mmHg
pH 7.37 7.36-7.45
ABG PO2 43 8-24 mmHg (Newbor)
55-80 mmlg (Child)
Base access 2 -2 to 2 mmol/L
Which of the following is the most likely diagnosis?
A. Hypoxic respiratory failure
B. Hypocapnic respiratory failure
C. Neurogenic respiratory failure
D. Hypercapnic respiratory failure

Answer: A
A 12-year-old child brought by an ambulance to Emergency Room after fall down from height. The primary survey reveals awake child, breathing with open wound in left thigh covered with blood-sucked gauze. The capillary refill is 4 seconds.
Blood pressure 70/50 mmHg
Heart rate 118 /min
Respiratory rate 22 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following is the most appropriate next action?
A. Send for imaging
B. Give morphine bolus
C. Call surgeon on duty
D. Give normal saline bolus

Answer: D

A 3-year-old child presents to Emergency Department with fever, poor feeding and vomiting for 3 days after a flu-like illness. On examination, she looks sick, lethargic and capillary refill is 4 seconds. Cardiopulmonary monitor attached and intravenous access attained.
Blood pressure 62/46 mmH
Heart rate 156 /min
Respiratory rate 48 /min
Temperature 38.2 °C
Oxygen saturation 92%
Which of the following is the most likely diagnosis?
A. Sepsis
B. Bronchiolitis
C. Encephilitis
D. Septic shock

Answer: D

A 2-month-old infant presents to Emergency Room with bronchiolitis required non-invasive mechanical ventilation as a respiratory support. Which of the following is the most appropriate destination?
A. Admission to paediatric word
B. Admission to neonatal care unit
C. Observation in emergency department
D. Admission to pediatrics critical care unit

Answer: D
An 8-month-old infant admitted to Pediatrics Critical Care for close observation. Which of the following standers continuous monitoring in pediatrics critical care measured as trans-cutaneous?
A. Blood pressure
B. Fluid balance
C. Glasgow coma scale
D. Arterial oxygen saturation

Answer: D

A 2-year-old girl presents to the Emergency Room with fever, vomiting and diarrhea for the last 4 days. On examination, she looks severely dehydrated, lethargic with altered conscious stat, with mild

abdominal pain. The family mentioned that they tried to use oral rehydration solution at home but she is not drinking well. A rapid correction by intravenous normal saline fluid established to manage her case. After 30 minutes, she starts to have abnormal movement and went in coma, which required resuscitation intubation and admission to Pediatric Critical Care Unit (see lab results).
Blood pressure 75/35 mmg Heart rate 110 /min Respiratory rate 28 /min Temperature 39 °C
Oxygen saturation 95 %
Test Result Normal Values
Sodium 162 134-146 mol/L
Potassium 3.8 3.5-5.1 mmol/L
Chloride 115 97-108 mmol/L
Bicarbonate 14 21-28 mmol/L
which of the following is the most likely cause her deterioration?
A. The case complicated with meningitis
B. Rapid correction lead to cerebral edema
C. Delayed in starting fluid and antibiotics
D. Sudden intraventricular hemorrhage because the high sodium

Answer: B
A 2-year-old child presents to the Emergency Room with fever, vomiting and diarrhea for the last 4 days. On examination, she looks severely dehydrated, lethargic with altered conscious stat, with mild abdominal pain. The family mentioned that they tried to use oral rehydration solution at home but she is not drinking well. A rapid correction by intravenous normal saline fluid established to manage her case. After 30 minutes, she starts to have abnormal movement and went in coma, which required resuscitation intubation and admission to Pediatric Critical Care Unit (see lab results).
Blood pressure 75/45 mmH
Heart rate 120 /min
Respiratory rate 28 /min
Temperature 39 °C
Oxygen saturation 95 %
Test Result Normal Values
Sodium 162 134-146 mmol/L
Potassium 3.6 3.5-5.1 mmol/L
Chloride 115 97-108 mmol/L
Bicarbonate 14 21-28 mmol/L
which of the following is the most appropriate management?
A. Fluid deficit should be replaced over 48 hours
B. Fluid deficit should be replaced over 24 hours
C. Fluid deficit should be replaced over 12 hours
D. Fluid deficit should be replaced over 6 hours

Answer: A

A 5-year-old girl presented to Emergency Room with high-grade fever, vomiting and severe diarrhea for the last 2 days (see lab results).
Blood pressure 110/70 mmHg
Heart rate 130 /min
Respiratory rate 30 /min Temperature 39.2 °C
Oxygen saturation 96 %
Test Result Normal Values ABG HCO3- 18 13-22 mmol/L ABG PCO2 25 27-40 mmHg pH 7.32 7.36-7.45
Base access -4 -2 to 2 mmol/L
Which of the following describes her shock status?
A. Early compensated shock

B. Late compensated shock
C. Late uncompensated shock
D. Early uncompensated shock

Answer: A
A 4-year-old boy presented to Emergency Room with fever, sever vomiting and diarrhea. On examination, she is tachypnic, tachycardic and has mottled skin with decrease skin turgor (see lab results).
Blood pressure 110/70 mmHg
Heart rate 125 /min Respiratory rate 28 /min Temperature 38.5 °C
Oxygen saturation 97 %
Test Result Normal Values ABG HCO3- 19 13-22 mmol/L ABG PCO2 25 27-40 mmHg pH 7.30 7.36-7.45
Base access -3-2 to 2 mmol/L
Which of the following clinical sign can be observed?
A. Blue peripheries
B. Kussmaul breathing
C. Decrease urine output
D. Depressed cerebral state

Answer: C
A 6-year-old boy presented to Emergency Department with severe vomiting and diarrhea. Blood pressure 110/70 mmHg
Heart rate 120 /min
Respiratory rate 22 /min
Temperature 38 °C
Oxygen saturation 98 %
Weight 18 kg
After stabilization and fluid resuscitation, how much the volume of maintenance fluid intake per hour she required?
A. 47
B. 56
C. 65
D. 70

Answer: B
Children with septic shock need to be rapidly stabilized and resuscitated for their life saving. Which of the following has lowest evidence of benefit in their management?
A. Antibiotics
B. Steroids
C. Inotropic support
D. Fluid resuscitation

Answer: B
A 7-year-old boy presented to the Emergency Room with acute stridor, tachypnea, pallor and hypotension. In quick history, the mother mentioned he was attending a party in the school and that he has food allergy. Which of the following is the first step in management?
A. Steroid
B. Antibiotics
C. Adrenaline
D. Fluid bolus

Answer: C

A child is brought to the Emergency Department after ingesting a potentially toxic dose relative's medication.
Which of the following is the next step in management?
A. Activated charcoal
B. Gastric lavage
C. Ipecac syrup
D. Cathartics

Answer: A
A 12-year-old patent presents to the clinic with diarrhoea, vomiting, miosis, bronchoconstricton, sweating, and excessive salivation. On examination, the breath has a garlic odour.
Which of the following is the most likely diagnosis?
A. Organophosphate poisoning
B. Diabetic ketoacidosis
C. Cyanide ingestion
D. Cocaine overdose

Answer: A
A child is brought to the Emergency Department after accidentally consuming a large dose of nutritional supplements 2 hours before. Now has developed nausea, vomiting, abdominal pain, and diarrhoea. The vomit is black.
Blood pressure 90/60 mmHg
Heart rate 100 /min
Respiratory rate 18 /min
Temperature 36.6° C
Which of the following is the most likely diagnosis?
A. Obstruction
B. Iron poisoning
C. Hypervitaminosis
D. Hyperproteinemia

Answer: B

A 3-year-old boy presented with his family to the Emergency Room after ingestion unknown amount of paracetamol tablets. He is quite, with accepted vital sign and normal examination. According to his mother, the bottle was containing small amount of the syrup, and she found some of it spelled over his clothe. He is observed for 4 hours and clinically there is no change on his condition.
Which of the following is the proper disposition for him?
A. Performed gastric lavage
B. Monitor for another 4 hours
C. Discharge home with instruction
D. Start intravenous acetylcysteine

Answer: C
An 8-year-old girl presents to the Emergency Room by her mother after ingestion of large amount of acetaminophen tablets. According to the mother, she ingested full 2 packs, containing 20 tablets. 1 day ago after she had fight with her. On examination, she has right upper quadrant abdominal pain, with nausea and vomiting
Blood pressure 95/50 mmg, Heart rate 135 /min, Respiratory rate 25 /min Temperature 36.6 °C
Oxygen saturation 95 %
Which phase of acetaminophen toxicity she is at?
A. Phase 1
B. Phase 2

C. Phase 3
D. Phase 4

Answer: B
An 8-year-old girl presents to the Emergency Room by her mother after ingestion of large amount of acetaminophen tablets. According to the mother, she ingested full 2 packs, containing 20 tablets, 1 day ago after she had fight with her. On examination, she has right upper quadrant abdominal pain, with nausea and vomiting.
Blood pressure 95/55 mmHg Heart rate 130 /min Respiratory rate 25 /min Temperature 36.6 °C Oxygen saturation 96 %
Which of the following is the most important antidote according to her stage of toxicity?
A. Desferrioxamine
B. Activated charcoal
C. N Acetylcysteine
D. Antidote is not indicated at this stage

Answer: C
A 6-year-old girl presents to the Emergency Room with her father after ingestion of large amount of iron tablets. She ingested full bottle containing >20 tablets, 3 hours ago. On examination, she has acute abdominal pain, with nausea, vomiting and diarrhea, and she passed 2 times melaena.
Blood pressure 100/60 mmHg Heart rate 120 /min Respiratory rate 26 /min Temperature 36.6 °C
Which of the following is the most appropriate treatment option at this stage?
A. Gastric lavage
B. N acetylcysteine
C. Intravenous desferrioxamine
D. No antidote required at this stage

Answer: C
A 5-year-old child presents with drowsiness, abdominal pain, nausea and bloody diarrhea after ingesting an unknown number of iron tablets several hours before (see lab result).
Test Result Normal Value
Iron 94 9-21.5 umol/L
Which of the following is the most appropriate treatment?
A.Haemodialysis
B.Gastric lavage
C. Oral activated charcoal
D.Intravenous deferoxamine

Answer: D

A young child is brought to the Emergency Department after ingesting a large amount of baby aspirin.
Which of the following is the most likely acid-base imbalance?
A. Respiratory acidosis followed by metabolic acidosis
B. Metabolic alkalosis followed by respiratory alkalosis
C. Metabolic acidosis followed by respiratory alkalosis
D. Respiratory alkalosis followed by metabolic acidosis

Answer: D

A 6-month-old girl presents to the Emergency Room by an ambulance with unexplained loss of conscious. Her mother appeared to have learning difficulties, and her father explained that suddenly

the child went all floppy and asleep. CT brain showed subdural hemorrhage, chest x- ray showed old fractures, and eye exam showed retinal hemorrhage.
Which of the following is the most likely diagnosis?
A. Child abuse
B. Bone disease
C. Blood disease
D. Metabolic disease

Answer: A
A 6-month-old girl presents to Emergency Room by ambulance with unexplained loss of conscious. Her mother appeared to have learning difficulties, and her father explained that suddenly the child went all floppy and asleep. CT brain showed subdural hemorrhage, chest x- ray showed old fractures, and eye exam showed retinal hemorrhage. After short time, the child start to cry and the parents want go home against the advice for admission.
Which of the following is the most appropriate action?
A. Discharge on parent responsibilities
B. Discharge with follow up after 2 weeks with neurosurgery
C. Convince the family to continue observing her 4 hours more
D. Contact child protection agency and force the family to stay

Answer: D

Development/ psychiatry

An apparently normal child is being evaluated. He runs around, tells short stories and plays a "father role". He draws his brother with a head, hands and feet. He undresses himself for the examination.
What is the most likely age of this child?
A. 2
B. 3
C. 4
D. 5

Answer: C

Which of the following is the most typical milestone for a 4-year-old child?
A. Write the first name
B. Copy a square and triangle
C. Speak clearly in sentences
D. Prefer solitary or parallel play

Answer:C

A 5-month-old baby referred to the doctor from the Well-Baby Clinic. His mother is concerned, because he could not pass the developmental screening tests done by the nurse.
What is the best developmental test would help the doctor to reassure the mother?
A. Waves goodbye
B. Has pincer grasp
C. Sits independently
D. Reaches for objects

Answer: D
Which of the following can the average 3-year-old child perform?
A. With alternating feet catch a ball
B. Know days of the week
C. Trace a triangle
D. Climb stairs

Answer: D
While in the clinic, a child runs towards the doctor, walks upstairs with one hand held and pours out pellets from a bottle. The child does not handle a spoon well and cannot put 3 words together.
What is the approximate age in months?
A. 12
B. 15
C. 18
D. 24

Answer: C
During a health supervision examination, a mother reports that her son can say his name, feeds himself, separates easily from parents and rides a tricycle.
What is the developmental age in years?
A. 2
B. 3
C. 4

D. 5

Answer: B
While at the clinic, an infant craw toward the doctor, rolls over, sits briefly, transfers objects from hand to hand and grasps objects using his radial palm. The infant is unable to use a pincer grasp.
What is the approximate age in months?
A. 3
B. 5
C. 7
D. 9

Answer: C
A child walks with one hand held, says a few words besides "mama" and "dada", and picks up pellets with an unassisted pincer grasp, but does not insert them into a bottle.
What is the approximate age in months?
A. 7
B. 9
C. 12
D. 15

Answer:C
While in the dinic, a child demonstrates an ability to run, is able to open a door and climb onto furniture. The child uses 3-word sentences, but does not know his full name and does not climb up stairs using alternate feet.
Which of the following is the approximate age in years?
A. 1
B. 2
C. 3
D. 4

Answer: B
An infant lies with his head to one side and raises his head slightly, but the head lags when pulling up to a sitting position. The infant smiles when involved in social interactions and follows moving objects 180 degrees, but does not attempt to reach for them.
What is the approximate developmental age in weeks?
A. 4
B. 8
C. 12
D. 16

Answer: B
A child's developmental assessment conforms the following (see table).
Motor Walks alone, crawls up stairs
Adaptive Makes a tower of three cubes, makes a line with crayons
Language Jargon, follows simple commands
Social Indicates desire by pointing
Which of the following is the most likely age in months?
A. 6
B. 9
C. 15
D. 24

Answer: C

Which of the following is a developmental screening tool that relies only on information from the parents?
A. Brigance screens – II
B. Ages and stages questionnaire
C. Bayley scales of infant development
D. Safety word inventory and literacy screener

Answer: B
When a mother placed her infant on the examining table, the infant laughs out loudly. As the doctor approaches him, he squeals and reaches for his mother.
What is the most likely developmental month for this child?
A. 2
B. 4
C. 6
D. 8

Answer: B
A child accompanied by his parents walked into the examination room. He is holding a ball. The doctor asks to let him see the ball; he threw the ball towards the doctor. Then he reaches out to catch it when the doctor threw the ball back to him. He draws a straight line on the paper and then picks up a few blocks from the floor and stacks one on the top of the other.
What is the most likely developmental month for this child?
A. 9
B. 12
C. 15
D. 18

Answer: D
A child enters the clinic and says "hi" to the doctor then turns to the mother and says "milk". The mother gives her a bottle of milk, and the child tries to give her doll a drink. The mother shakes her head, and the child says "no". After which the child smiles and points to her own mouth and says "mouth". When her mother wipes the doll's face, the child says "me" and begins to imitate the mother's action and gives the doll a hug.
What developmental age in months is this most consistent with?
A. 12
B. 15
C. 18
D. 24

Answer: C

During a routine check-up, a child shakes hands, knows his name, gender and age. He can dress with help and ride a tricycle. He points to red colours when asked to do so but had difficulty copying a square.
What is the age in years this child most likely at?
A. 2
B. 3
C. 4
D. 5

Answer: B
An infant sits in the tripod position, can roll from prone to supine and vice versa and attempts to reach a toy beyond reach but cannot pick up cereals between 2 fingers.

What developmental age in months is this most consistent with?
A. 2
B. 4
C. 6
D. 9

Answer: C
A mother of a 3-year-old child asks about his language development. He easily communicates with other children and 75% of his speech is understood by strangers. He asks questions frequently and follows 2-part commands.
What is the best description of this language development?
A. Receptive language delay
B. Developmental disfluency
C. Expressive language delay
D. Normal development for age

Answer: D
In a prone position, an infant can lift his head 45" off the examination table and moves his head from side to side while following an object. He smiles and coos when the doctor plays with him, but cannot maintain a seated position.
What is the age in months this child most likely at?
A. 1
B. 3
C. 6
D. 9

Answer: B
A 7-day-old baby is brought to the clinic for a check-up. The mother raised the concern that his current weight is 3.1 Kg while his birth weight was 3.5 Kg. She is breastfeeding him every 2 to 3 hours for 10-15 min with good sucking. Physical examination is normal.
Which of the following is the most appropriate management?
A. Maternal nutrition counselling
B. Reassure the mother all is normal
C. Admit baby for further investigation
D. routine tests and reassess in one month

Answer: B

Which of the following language milestones is appropriate for a 4-year-old child?
A. Speaks 50-100 words
B. Can use the future tense
C. Speaks 2-word sentences
D. Can count to 4 and tell a story

Answer: D

By what age in months should a child smile?
A. 1
B. 2
C. 4
D. 6

Answer: C

A child is brought for a well-child visit. He is healthy and developing normally. The child can name 4 colours, define 5 words, understand prepositions, hops on 1 foot, and gets dressed with little assistance.
What developmental age in months is this most consistent with?
A.48
B. 36
C. 24
D. 18

Answer: A
A 6-month-old child is brought to the paediatrician for a well-child visit. The infant was bom at full term after a normal pregnancy and spontaneous vaginal delivery. The linear growth and the amount of weight gained over the past 6 months are both appropriate. The mother began feeding the infant pureed baby foods approximately 1 month before while also continuing breast feeding. The infant has been well since birth. The physical examination is normal for age. developmental screening exam is performed.
Which of the following gross motor skills is most appropriate at his age?
A. Sits without support
B. Stands while holding onto support
C. Rolls from prone to supine position
D. Rolls from supine to prone position

Answer: D
By what age in months should a child learn to speak a few words?
A. 3
B. 6
C. 12
D. 24

Answer: C
When assessing an adolescent girl with a body mass index at the 95th percentile for age and sex. Which of the following actions should be taken?
A. Avoid discussion of weight
B. Initiate a multidisciplinary plan
C. Provide a list of low-calorie foods
D. Confirm size larger than those of same age

Answer: B
A child is brought for a health maintenance check. The parent states that the child sits up alone without support, pulls himself into a standing position, explores around furniture, crawls without difficulty. grasps objects with a thumb and forefinger, uses repetitive consonants such as "dada", and responds to the sound of his own name.
What age in months is the child most probably at?
A. 8 months
B. 10 months
C. 12 months
D. 14 months

Answer: B

A 4-month-old girl is brought to the Paediatric Clinic for a routine monthly exam. The baby smiles back at people and imitates facial expressions. She uses eyes and hands together and reaches out with one hand for a toy. There is babbling along with a facial expression. The gift pushes up to her elbows when lying on the tummy.
Weight 7300 grams (=50th percentile)
Length 67 cm (=75th percentile)
Which additional developmental milestone is most likely to have?
A. Hold head steady without support
B. Roll over in both directions
C. Make crawling attempts
D. Sit without support

Answer: A

Which of the following is a known side effect of beclomethasone use in children?
A. Intraocular hypertension
B. Excitable behaviour
C. Sleep disturbance
D. Growth retardation

Answer: D
An 18-month-old infant presents to the Pediatric Outpatient Clinic for evaluation of delay in walking. It was observed that there is widening of the lower ends of long bones (see lab results).
Height 78 cm
Weight 10 Kg
Test Result Normal Values
Calcium total 8.1 2.2-2.7 mmol/L
Phosphate 2.1 1.3-2.3 mmol/L
Alkaline phosphatase 1020 100-320 U/L
Which of the following is the most likely cause?
A. Hypophosphatasia
B. Renal osteodystrophy
C. Vitamin D deficient rickets
D. Familial hypophosphatemic rickets

Answer: C
A 2-year-old child presents to the clinic with chalky discoloration on both sides of the teeth. The child sleeps with the milk bottle in his mouth. On examination, there are opaque discolorations on the backsides of the central maxillary incisors and the first molars.
Which of the following is the most likely diagnosis?
A. Herpetic gingivostomatitis
B. Nursing bottle caries
C. Gingival hyperplasia
D. Congenital syphilis

Answer: B

A 36-month-old girl seen in the clinic. Her growth parameters as plotted (see image). Which of the following is the most likely diagnosis?
A. Indeterminate
B. Failure to thrive
C. Constitutional delay
D. Normal growth pattern

Answer: B
An 18-month-old boy is seen in screening clinic. The mother concerns about his growth. Which of the following findings need referral for further management?
A. Persistence weight curve on 15% percentile
B. Persistence weight curve on 10 % percentile
C. Weight curve moves from 5% to 15% percentile
D. Weight curve drops one percentile but still within normal ranges

Answer: D

Which of the following statements is normal weight gain in infants?
A. Double their birth weight at 10 months
B. Mean weight gain by 10 g/day in the first 3 month
C. Mean weight gain by 15 g/day in the first 3 month
D. Mean weight gain by 30 g/day in the first 3 month

Answer: D

A 2-month-old infant is brought to Well Baby Clinic by her mother, who is concems about her being not sleeping overnight and keeps crying intensely recently. These crying episodes are 2 to 3 times per week, last for 1 to 2 hours since the age of 1 month. Which of the following is the typical age of this condition?
A. First 6 weeks of life
B. First 2 months of life
C. First 3 months of life
D. First 6 months of life

Answer: C
A 2-month-old infant is brought to Well Baby Clinic by her mother, who is concems about her being not sleeping overnight and keeps crying intensely recently. These crying episodes are 2 to 3 times per week, last for 1 to 2 hours since the age of 1 month Which of the following is the most likely diagnosis?
A. Child abuse
B. Temper tantrum
C. Seizure attack
D. Infantile colic

Answer:D
A 2-month-old infant is brought to Well Baby Clinic by her mother because she is crying intensely recently and is not sleeping well overnight. The mother gave history of normal feeding. no fever or vomiting. These crying episodes are 2 to 3 times per week, last for 1 to 2 hours since the age of 1 month. Which of the following is the recommended revealing measure?
A. Changing formula
B. Oral colic drops
C. Getting a baby sitter
D. Behavioural adaptation

Answer: D
A 3-year-old girl is seen in the clinic. The mother gives history of yelling, hitting and saying "no" with other oppositional behaviours. Which of the following is the most likely diagnosis?
A. Depression
B. Anxiety disorder

C. Temper tantrums
D. Mental retardation

Answer: C
A 2.5-year-old girl seen in the screening clinic. The mother gave history of yelling, hitting and saying "no" with other oppositional behaviours. Which of the following is the most appropriate counselling should be given?
A. Ignorance behaviour
B. Antidepressant agent
C. Strict and firmness behaviour
D. Reinforcing positive behaviours

Answer: D

What is the recommended age (in months) to screen for haemoglobin level in infant with no risk factors?
A. 6
B. 12
C. 18
D. 24

Answer: B

A 2-week-old infant seen in the clinic. Rubbing side of the mouth with tip of the little finger is one of the primitive reflexes (see image). Which of the following is the name of this reflex?
A. Moro
B. Rooting
C. Sucking
D. Parachute

Answer: B
A 2-week-old infant seen in the clinic. Rubbing side of his mouth with tip of the little finger is one of the primitive reflexes (see image). What is expected age (in months) for this reflex to disappear?
A. 2
B. 4
C. 6
D. 8

Answer: B
A 1-week-old infant seen in screening clinic (see image). Which of the following is the name of this primitive reflex?
A. Rooting
B. Stepping
C. Parachute
D. Lateral neck

Answer: B
A 2-week-old infant seen in screening clinic (see image). What is expected age (in months) for this reflex to disappear?
A. 2
B. 4
C. 6
D. 8

Answer: A

An infant presents to Well Baby Clinic. She can keep her head up unsupported and need back support to set. She can smile and can reach mouth. What is the developmental age in months?
A. 3
B. 5
C. 7
D. 9

Answer: B
A 2-month-old infant with no risk factor screened in Well Baby Clinic. Which of the following measures need to be screened?
A. Autism
B. Lead level
C. Body mass index
D. Head circumference

Answer: D

A child who can jump on 2 feet, build a tower of 6 blocks, use 2 word phrases and compares 2 items. What is the developmental age?
A. 1 year
B. 2 years
C. 3 years
D. 4 years

Answer: B
A 4-year-old boy present to the clinic with a worried mother, for general assessment and evaluation of his development. He can hop on 1 foot but not able to balance for 10 seconds, draw square, tells a story but not able to use the future tens and has a preferred friend. Which of the following is the most appropriate response to his mother?
A. He is growing normally according to his age
B. He has delayed development in gross and fine motor aspect
C. He is growing normally but delayed in the speech and language
D. He is delayed in all aspect and required further investigation

Answer: A

A 15-month-old child brought to the clinic by his parents. He was born at 36 weeks by normal vaginal delivery. At 8 month' test for hearing was done, which was normal. The parents say that he babbles, but does not speak 2-3 word phrases yet. On physical examination, the child appears normal. Which of the following is the appropriate action?
A. Refer to ENT
B. Reassess at 24 months
C. Arrange for audiometry testing
D. Reassure the parents that this is a normal variant

Answer: D

A 10-year-old boy comes with his uncle for follow-up after major car accident where he lost his mother and his 2 brothers. The uncle is worried after receiving teacher's note regarding child behavior in the class. Teacher wrote that the child becomes unresponsive sometimes, has a lapse in

attention. He stares at nothing, and act as if he just woke up in response to being called to attention.Which of the following term describe child's response mechanism?
A. Blame
B. Anger
C. Denial
D. Disassociation

Answer: D
A 15-year-old girl presents to the Emergency Department with chest pain, nausea, dizziness, and sweating. She states that she is afraid he is going to die. History includes 3 previous episodes during the past 2 months, each episode lasting about 15 minutes. She has been afraid to visit the doctor but she has become more afraid during the past month of having yet another episode. Which of the following is the most likely diagnosis?
A. Malingering
B. Agoraphobia
C. Panic attack
D. Factitious disorder

Answer: C

A 10-years-old boy diagnosed attention deficit hyperactivity disorders, presents with his mother the clinic. She complains about his behavior. She is advised to identify and reward positive behaviors and ignore disruptive. Which of the following helps to achieve this advice?
A. Systemic family therapy
B. Parent training programs
C. Behavior management techniques
D. Time-out from positive reinforcement

Answer: C
A 2-year-old boy presents with his mother complains that her child rebelling constantly with temper tantrums. What is the most appropriate counseling should be given?
A. Use time out technique
B. Set limits on the child's behavior
C. Punish the child every time the child says "no*
D. Allow the behavior because this is normal at this age period

Answer: B
Advising a mother regarding introducing solid foods to her child diet. Which of the following is the most appropriate advice?
A. Mix different foods together
B. Begins with one tablespoon of food
C. Eliminate a refused food from the diet
D. Introduces each new food 4 to 7 days apart

Answer: D
When assessing development in a 9-month-old infant, which of the following is expected to observe at this age?
A. Creeping along the floor
B. Speaking in 2-word sentences
C. Grasping objects with palmar grasp
D. Beginning to use a spoon rather sloppily

Answer: C

A 7-month-old infant visits Well Baby Clinic. Which of the following this infant should be able to do?
A. Release cubes into a cup
B. Hold crayon and draw a circle
C. Transfer objects from one hand to the other
D. Use thumb and index finger in crude pincer grasp

Answer: C

Genetics

A newborn presents with a low weight and hypotonia. Tongue protrusion was observed on multiple occasions and has a single palmar crease. Cardiac auscultation confirms a systolic murmur and echocardiography confirms the presence of a ventricular septal defect.
Which of the following syndromes is the most likely diagnosis?
A. Down
B. Alport
C. Klinefelter
D. Prader-Willi

Answer: A
An infant girl, born to a 24-year-old woman, diagnosed clinically as Down's syndrome. The mother asked about the risk in any future pregnancy.
Which of the following is the appropriate investigation to determine this risk?
A. Karyotyping for this infant
B. Amniocentesis in next pregnancy
C. Ultrasound during next pregnancy
D. Karyotyping for this infant and mother

Answer: D

A 3-year-old child presents to the screening clinic (see image), Which of the following is the most likely diagnosis?
A. Trisomy 21 syndrome
B. Trisomy 18 syndrome
C. Trisomy 13 syndrome
D. Normal ethnic features

Answer: A
Which of the following is the most common congenital heart defect in Down syndrome?
A. Cushion defect
B. Atrial septal defect
C. Coarctation of aorta
D. Ventricular septal defect

Answer: A
A 4-month-old boy with Down syndrome seen in the screening clinic. Which growth chart should be used?
A. No special charts
B. Individualized chart
C. Down syndrome charts
D. Standard growth chart

Answer: C
A 4-day-old boy with Down syndrome seen in the screening clinic. Which of the following is the most striking feature?
A. Hypotonia
B. Doughy skin
C. Simian creases
D. Curved fifth little finger

Answer: A
What are the most common endocrine diseases associated with Down syndrome?
A. Hypothyroidism
B. Hyperthyroidism
C. Hypoparathyroidism
D. Panhypopituitarism

Answer: A
A 1-year-old diagnosed with Down syndrome, seen in the screening clinic. The mother who is 27-year-old is asking about risk of recurrence in future pregnancy. What is the most appropriate counselling should be given?
A. Carry no further risk
B. Depends on mother age
C. Depends on parents and child genetics profile
D. Father genetics has no role in recurrence rate

Answer: C

A 4-year-old boy brought to the clinic. During the physical exam, a 7 cafe-au-lait macule over his trunk. The mother stated she and many of her family members have the same "birth-marks". Which of the following is the best next step?
A. Order MRI brain
B. Family Counselling about neurofibromatosis type 1
C. Referral to medical geneticist for further evaluation and genetic testing
D. Referral to genetic counsellor for further education about neurofibromatosis type 1

Answer: C
A 3-year-old boy is diagnosed of tuberous sclerosis. There is no previous history of same disease in the family. Which of the following is the best option of genetic testing?
A. Exome sequencing
B. Single-gene testing
C. Chromosomal microarray
D. Genetic testing panel of multiple genes

Answer: B
A 3-year-old boy is diagnosed of tuberous sclerosis. His mother has the same disease and was tested positive for a pathogenic mutation in TSC1 gene. which of the following is the most appropriate genetic testing to be performed?
A. Exome sequencing
B. Single-gene testing
C. Genetic testing panel of multiple gene
D. Targeted testing for the familial variant

Answer: D

A 3-year-old boy presents to the clinic with marked obesity. The mother gave history of persistent hyperphagia, and that he does not control his hunger. On examination, he is found to have dysmorphic facial features, hypotonia and undescended testes Which of the following is most likely diagnosis?
A. Noonan syndrome
B. Williams syndrome
C. DiGeorge syndrome
D. Prader-Willi Syndrome

Answer: D

Which of the following has the lower incidence to result in Down syndrome?
A. Mosaicism
B. Translocation
C. Meiotic disjunction
D. Meiotic non disjunction

Answer: A
A newborn in nursery with multiple congenital anomalies in form of prominent occipit, overlapping fingers, rocker-bottom feet and cardiac malformation. Chromosomal analysis sent awaiting the result.
Which of the following is the most likely diagnosis?
A. Down syndrome
B. Edward syndrome
C. Patau syndrome
D. Turner syndrome

Answer: B
A 10-year-old child presents to the clinic with cutaneous manifestations. She had 7 cafe-au lait spots, 2 neurofibroma and axillary freckles. The mother mentioned a presence of nephew who had similar lesions. What is mode of inheritance of this disease?
A. Autosomal dominant inheritance
B. Autosomal recessive inheritance
C. X-linked dominant inheritance
D. X-linked recessive inheritance

Answer: A
A 10-years-old child presents to the clinic with cutaneous manifestations. She had 7 cafe-aulait spots. 2 neurofibroma and axiliary freckles. The mother mentioned a presence of nephew who had similar lesions. Which of the following is the most likely diagnosis?
A. Tuberous sclerosis
B. Angiofibromata type 2
C. Sturge-Weber syndrome
D. Neurofibromatosis type 1

Answer: D
While reviewing expanded family pedigree for a child presented to the clinic for investigation and diagnosis. It found that there is a 25% risk for 2 carrier parents to have affected child and the risk increase with consanguinity. Which of the following is the inheritance?
A. Autosomal recessive
B. Autosomal dominant
C. X-linked recessive
D. X-linked dominant

Answer: A
An 11-year-old presents to the clinic with the following features, macrocephaly, macro-orchidism long face, large everted ears and ECHO revealed the presence of mitral valve prolapse. He has significant joint laxity and moderate learning difficulties. Which of the following is the most likely diagnosis?
A. Hunter syndrome
B. Hurler syndrome

C. Friedreich ataxia
D. Fragile X syndrome

Answer: D

Immunology

A 3-year-old child presented to Emergency Room with difficulty breathing, lip swelling and spreading skin rash. On examination, marked airway swelling observed with rash cover all the body. The mother said he was attending a party at his friend's house and she was not with him Which o of the followings will likely reveals the diagnosis?
A. Family history
B. Social history
C. Allergy history
D. Nutritional history

Answer: C
A 4-year-old child presented to Emergency Room with difficulty breathing. lip swelling and spreading skin rash. On examination, marked airway swelling observed with rash cover all the body. The mother said he was attending a party at his friend house and she was not with him. Which of the following is the most likely diagnosis?
A. Trauma
B. Insect bites
C. Food allergy
D. Drug reaction

Answer: C

A 4-year-old child presented with itchy and scratching skin lesions that started on head and trunk then progress to peripheries. On examination, he was febrile with skin lesions appear as crops of papules, vesicles with surrounding erythema. The mother gave history of similar incident happen last week in his school. The parents are worried since there is an immunodeficient child at home.
What would be the best action plan to protect the immunodeficient kid?
A. Avoid skin contact
B. It is a self-limiting disease with no risk
C. Oral antiviral therapy for the immunodeficient child
D. Specific immunoglobulin for the immunodeficient child

Answer: D

A 7-year-old boy has had recurrent upper and lower respiratory tract bacterial infections since infancy. On examination, he has eczema and areas of ecchymosis on his skin. His brother and a paternal uncle are similarly affected (see lab results).
Test Result Normal Values
Hb 108 112-165 g/L
WBC t.6 4.5-10.5 x 109/L
Platelets count 40 150-400 × 109/L
Which of the following is the most likely diagnosis?
A. Severe combined immunodeficiency
B. Bruton agammaglobulinemia
C. Wiskott-Aldrich syndrome
DiGeorge syndrome

Answer: C
A 14-month-old boy is brought to the clinic with severe gram-positive bacterial pneumonia, was referred to the hospital by the family physician. This is his third infection in 4 months. He has 2 healthy sisters aged 3 and 5 years. The family lost a boy at 10 months of age to bacterial pneumonia 6 years ago. The family doctor has sent along some blood test results that shows low all classes of

serum immunoglobulin levels, few B cells, but normal numbers and functioning of T cells.
Which of the following is the most likely diagnosis?
A. Transient hypogammaglobulinemia of infancy
B. Common variable hypogammaglobulinemia
C. CD40 Genetic/Metabolic/Metabolic defect
D. X-linked hypogammaglobulinemia

Answer: D

Vaccines

Which of the following vaccines is used for immunization against Hepatitis B virus?
A. Toxoid
B. Inactivated
C. Recombinant
D. Live-attenuated

Answer: B
Which of the following viral gastroenteritis can be prevented by vaccination?
A. Adenovirus infection
B. Astrovirus infection
C. Norovirus infection
D. Rotavirus infection

Answer: D

A 2-year-old boy with sickle cell anemia presents to the Paediatric Outpatient Clinic for a second dose of hepatitis A vaccine. According to the mother, he was admitted 3 weeks before with a vaso-occlusive crisis and received a blood transfusion for anaemia.
Which of the following is the most appropriate approach for vaccination?
A. Vaccinate as per schedule
B. Delay vaccine 3 months
C. Delay vaccine until 6 months
D. First, evaluate hepatts A serology

Answer: A

What is the best way to provide immunologic protection against tetanus neonatorum in new- borns?
A. Inject the newborn with tetanus toxoid
B. Inject the infant with human tetanus antitoxin
C. Inject the mother with toxoid within 72 hours of the birth of her child.
D. Immunize the mother with tetanus toxoid before or early in pregnancy

Answer: D
During a well-baby vaccination supervision visit, the mother reports that her 9-month-old girl had received all her immunizations since birth and she is due for another vaccination.
Which of the following vaccines would be recommended?
A. Oral polio and hepatitis B
B. Measles, mumps and rubella
C. Diphtheria, tetanus and pertussis
D. Measles and meningococcal conjugate

Answer: D
A 2-month-old baby is brought for his first visit in the Well-Baby Clinic. He was born at 35 weeks by a normal vaginal delivery, and was 1.t Kg at birth and was managed in an incubator for 2 weeks. He has done well since and is gaining almost 50 grams each day.
Which of the following is the best immunisation plan for this visit?
A. Delay for 2 months
B. TdaP instead of DaP
C. Usual schedule but half dose
D. Usual schedule and usual dose

Answer: D
A 1-year-old boy is brought to the clinic for failure to thrive, recurrent oral thrush and hepatosplenomegaly. He had a brother who died at the age of 3 years with severe infection and septic shock.
Which of the following vaccines must be avoided until a diagnosis is made?
A. Varicella
B. Influenza
C. Injectable polio
D. Pneumococcal conjugate

Answer: A
A 4-month-old girl is brought to the clinic for vaccination. She was born preterm at 32 weeks' gestation due to maternal pre-eclampsia.
Which of the following is the best approach for the patent's immunisation schedule?
A. Reduce the vaccine dose for preterm infants
B. Use the adjusted age to schedule vaccinations
C. Use the calculated weight to schedule vaccinations
D. Follow chronological age for administering vaccines

Answer: D
A 4-year-old girl was involved in a motor-vehicle accident and required splenectomy after severe abdominal trauma with splenic tears.
What additional specific vaccination should she receive?
A. Human papilloma virus vaccine
B. Meningococcal vaccine
C. Rota virus vaccine
D. Tetanus vaccine

Answer: B
What is the most common minor side effect with routine DTP vaccination?
A. Injection site erythema
B. Generalised pain
C. Fever of 38.3°C
D. vomiting

Answer: A
A 6-month-old boy presents to the Paediatric Outpatient Clinic to receive scheduled 6-month vaccinations. His mother reports that he required immediate admission after the 4 month vaccinations with anaphylaxis. She is not aware which vaccine is responsible for this reaction. Which of the following is the most appropriate way of management?
A. Screen for allergens first
B. Vaccinate, except with DTP
C. Vaccinate and give reassurance
D. Vaccinate and observe for 1 hour

Answer: A
A 5-month-old boy presents to the clinic after missing his 4-month vaccinations. He appears to be well and thriving. The parents missed his 4-month visit due to transportation difficulties. What would be the best way to modify his immunisation schedule?
A. Arrange to give all missed vaccinations
B. Disregard missed vaccinations, Continue to next
C. Give missed vaccinations, and next appointment

D. Give missed vaccinations at 6 months with others

Answer: C
A 6-month-old girl is brought to the clinic for routine vaccination. The mother reports that she had developed fever associated with a local area of redness and swelling, after the previous 4 month vaccinations and is concerned.
What modification, if any, shall be taken to the vaccination schedule?
A. None, but give reassurance
B. Do not give the DPT vaccines
C. First, refer for allergen screening
D. Antipyretic and antihistamine prophylaxis

Answer: A
A 1-year-old boy has come for vaccination. The mother explains that his 6-year-old sister has received a renal transplant 2 months ago and is currently on immunosuppressive medications. Which of the following vaccines should be avoided?
A. Varicella
B. Oral polio
C. Injectable Influenza
D. Measles, mumps and rubella

Answer: B
A 4-month-old boy is brought to the clinic for routine vaccinations. The mother reports that he has been having 2 days of diarrhoea; described as watery, 3-4 times per day. He is still breastfeeding with no episodes of vomiting.
Blood pressure 88/68 mmHg
Heart rate 120 /min
Respiratory rate 35 /min
Temperature 37.3° C
Which of the following is the best decision to take regarding vaccination?
A. Delay vaccines
B. Give all vaccines
C. Give Hepatitis B vaccine
D. Give all vaccines except OPV

Answer: B

A 7-year-old boy is brought to the clinic to receive the injectable influenza vaccine. The father said that last year he had taken the vaccine, but still got the flu. He thought he got flu because he had received the vaccine, and is sceptical about its value.
Which of the following is the best advice?
A. Killed vaccine have no such risk
B. Attenuated vaccine has some risk
C. Live vaccines have a risk of infection
D. The nature of this vaccine may cause flu

Answer: A

A 1-year-old boy is admitted for investigation of failure to thrive, recurrent oral thrush and hepatosplenomegaly. He had a brother who died at the age of 3 years with severe infection and septic shock.
What advice should be given to the parents regarding their child's vaccinations?
A. Wait until child is 3 years-old

B. No live vaccines to be given to this child
C. Important that all vaccinations be given
D. No killed vaccines until brother diagnosed

Answer: B

A 1-year-old boy is seen on a routine examination. He has had vertically transmitted HIV infection and now has a low T-cell count.
Which of the following vaccines would be contraindicated?
A. Varicella
B. Hepatitis B
C. Pneumococcal
D. Diphtheria, Tetanus, acellular Pertussis

Answer: A

Which of the following vaccine is contraindicated in a child with egg allergy and should be assessed by an allergy/immunology specialist before vaccination?
A. MMR
B. Influenza
C. Yellow fever
D. Pneumococcus

Answer: C

A 6-month-old boy presents to the Well Baby Clinic for the 6-month vaccination. The mother states that he has developed febrile seizure with his last DTaP vaccine. Which of the following is the best answer to the mother concerns?
A. Delay the vaccination until he is 18 month old
B. Febrile seizure is true contraindication to DaP vaccination
C. He has to have the vaccine regardless of the adverse reactions
D. Febrile seizure does not have long-term sequelae and is not contraindication for subsequent vaccines

Answer: D
A 6-month-old boy presents to the Well Baby Clinic for the 6-month vaccination. The mother states that he has developed fever of 41 "C with his last DTaP vaccine. Which of the following is the best answer for the mother concern?
A. Delay the vaccinations until he become 18 month old
B. This is a very high fever and he should not have subsequent DaP vaccines
C. Fever is not contraindication for subsequent vaccines as long as it is not associated with seizure
D. High fever does not have any long term sequelae and is not contraindication for subsequent vaccines

Answer: D
Which of the following is an absolute contraindication to DTaP vaccination?
A. Seizure
B. Hypotonic-hyporesponsive episodes
C. Prolonged sever crying for 3 hours or more
D. Encephalopathy occurring within 7 days of previous dose not attributed to another cause

Answer: D

A 12-year-old girl diagnosed with systemic lupus erythematosus 10 months ago. She is currently on high doses of immunosuppressive treatment. Her immunization status is not known. Which of the following vaccines should be recommended?
A. MMR
B. BCG
C. Varicella
D. Annual Influenza vaccine

Answer: D

Newborn with normal physical examination. Upon reviewing the history with the mother, she mentioned that she had a daughter died with suspicion of immunodeficiency disease. The newborn is to be sent home and the parents ask about birth vaccines. Which of the following is the most appropriate management plan?
A. Receive Hepatitis B vaccine only
B. Receive routine birth vaccines
C. Order the birth vaccines and follow-up with Immunology team
D. Defer vaccines until full assessment by immunology team

Answer: A

A 3-hour-old newborn seen in the Nursery for assessment. The mother gave a family history of early infancy death secondary to primary immunodeficiency. Which of the following is the recommended first vaccination (Saudi program)?
A. Do not give BCG vaccine
B. Give BCG and follow up in clinic
C. Give vaccines as per national program
D. Start intravenous immunoglobulin

Answer: A

A full-term baby delivered normal vaginal delivery, received initial care and ready to be discharged home. Which of the following vaccines should be given before discharge?
A. BCG and HBV
B. HBV and MMR
C. BCG and DaP
D. MMR and DTaP

Answer: A
A 6-month-old infant presents to Vaccination Clinic. The mother gave history of flu-like symptoms and she is hesitant to give the vaccines as her child has a fever. On examination, the child looks healthy with normal exam.
Heart rate 107 /min
Respiratory rate 18 /min
Temperature 38 °C
Oxygen saturation 95 %
Which of the following is the best next action?
A. Start antibiotic course
B. Re-schedule after 2 weeks
C. Proceed with vaccination
D. Give antipyretics and discharge

Answer: C
A 2-month-old infant presents to Vaccination Clinic. What are the recommended vaccines according to national program?
A. IPV.HBV.DTP HIB.MMR
B. IPV,HBV,DTP,HAV,PCV
C. IPV.HBV.DTP.HIB,PCV
D. IPV.HBV.DTP HiB

Answer: C
A 1-year-old infant presents to Vaccination Clinic with normal exam and growth What are the recommended vaccines according to national program?
A. OPV. MMR, HAV, PCV
B. Varecilla, PCV,HIB
C. OPV,MMR, varecilla, PCV
D. OPV,MMR, varecilla, DTP

Answer: C
According to national Saudi program of vaccination for children up to primary school admission, which of the following disease is effectively prevented?
A. Q-fever
B. Japanese encephalitis
C. Herpes simplex encephalitis
D. Haemophilus influenza meningitis

Answer: D
A 6-year-old child seen in pre-school screening clinic with normal growth and general health status. What are the recommended vaccines at this stage according to national program in Saudi Arabia?
A. MMR.DTP.OPV
B. MMR.HBV.DTP.HAV
C. MMR,varicella, DTP.OPV
D. MMR,varicella.OPV.PCV

Answer: C
A full term girl was born to a mother who is hepatitis B surface Ag positive (HbAg +ve). The baby weighs 3.2 kg and is asymptomatic.
What is the appropriate immunoprophylaxis that should be considered in the first 12 hours?
A. Administer hepatitis B vaccine
B. Give hepatitis B immunoglobulins
C. Give hepatitis B vaccine and immunoglobulins
D.Measure hepatitis B surface antigen in the newborn

Answer: C
An 18-month-old girl presents to the clinic to receive her scheduled vaccines. Her mother reports that she is currently completing a course of antibiotics for tonsillitis, diagnosed 4 days ago. initially she had fever, irritability, and poor oral intake. Today she is better and started to play and feed normally. examination reveals a hyperaemic pharynx.
Blood pressure 85/65 mmHg Heart rate 100 /min Respiratory rate 25 /min Temperature 37.3° C
Which of the following is the best decision about the vaccination?
A. Vaccinate as per schedule
B. Wait until antibiotics finished
C. Postpone the Oral Polio Vaccine
D. Do not administer diphteria-pertussis-tetanus

Answer: A
A 14-year-old has received all immunizations except chickenpox; but he has never had chickenpox. What varicella vaccination plan should be used?
A. No vaccination required
B. 1 dose of varicella vaccine
C. 2 doses given 4 weeks apart
D. 3 doses at 0, 1 and 6 months

Answer: C
A 5-year-old child is about to receive a flu vaccination, and the doctor asked the mother about any food allergies the child might have.
Which of the following is the most important to note?
A. Chickens
B. Shellfish
C. Eggs
D. Nuts

Answer: C
An adolescent has missed the second scheduled dose of a varicella vaccination, with the first injection have been administered 1 year ago.
How should full immunity be achieved?
A. Test for antibodies prior to revaccination
B. Administer a double dose of vaccine
C. Administer the second dose
D. Repeat the entire series

Answer: C

Infectious/ dermatology

A 9-month-old child is brought to the Emergency Room due to low-grade fever, cough, and nasal congestion for several days. The previous night, her cough sounded "barky" like a seal and her parents are now worried about her ability to breathe. She has more difficulty breathing when she becomes upset and cries. On examination, she appears well hydrated without acute respiratory distress. However, audible breath sounds are noted when she begins to cry with high-pitched barking cough.
Which of the following is expected to be heard upon pulmonary auscultation?
A. Overall decreased breath sounds
B. Prominent expiratory wheezing and prolonged expiration
C. Prominent inspiratory sounds due to subglottic airway narrowing
D. Prominent inspiratory sounds due to mucus deposition in the large airways

Answer: C

A 6-year-old boy presents with fever. On examination, he shows small white vesicles around his mouth and on his gum. Small discrete cervical lymph nodes are also palpable.
Which of the following viruses is more likely to be associated with these symptoms?
A. Epstein bar virus
B. Herpes simplex virus
C. Varicella zoster virus
D. Human papilloma virus

Answer: B

A 4-year-old girl is brought to the clinic for prolonged diarrhoea. On examination, she looks pale. She is drinking 3 pints of goat's milk/day, and a very fussy eater (see lab results).
Test Result Normal Values
Hb 59 112-165 g/L
MCV 102 80-95 f1
MCHC 390 260-340 g/L
Which of the following is the most significant in the past medical history?
A. Hypothyroidism
B. Aplastic anaemia
C. Giardia parasitic infection
D. Psychological deprivation

Answer: C

A 10-month-old presents with fever, vomiting and copious watery diarrhea for the past 3 days. examination reveals a lethargic child with cool, dry skin, sunken eyes, tachycardia and mild tachypnea.
Which of the following tests is the most likely confirm the diagnosis?
A. Serum viral culture
B. Serum antibody
C. Stool leukocyte
D. Stool antigen

Answer: D

A 6-year-old child is brought to the Paediatric Clinic. The parent reports intermittent fever, vomiting, diarrhea, and diffuse abdominal pain for the past week. The parent indicated that the child's eyes

also appears to have a yellowish skin stain for the past 2 days. The liver is palpable 5 cm below costal margin and there is jaundice, scleral icterus, and mild abdominal tenderness on examination.
Blood pressure 110/70 mmlg
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 38.0° C
Which of the following is the most likely type of viral hepatitis?
A. A
B. B
C. C
D. D

Answer: A

A 15-month-old child is brought to the clinic with one day of fever and vomiting. A lumber puncture is performed (see lab results and report).
CSF exam
Test Result Normal Values
Total protein 300 150-450 mg/L
Glucose 3 2.50-4.8 mmol/L
Cells 1200 0-3 /pL
CSF Culture:
Gram positive cocci in chains, pus cells and WBCs with a predominance of Neutrophils.
Which of the following is the most appropriate antibiotic or antibiotic combinations?
A. Ampicillin
B. Vancomycin
C. Ampicillin and gentamicine
D. Ceftriaxone and vancomycin
Answer: D

Answer:
A 9-year-old boy who presents to the Emergency Department with fever, headache, sore throat, and vomiting. He recently arrived from Africa with his parents. On examination, he shows abnormal reflexes, stiff neck and difficulty in lifting his head and legs when he was lying flat on his back. His laboratory investigations show a clear CSF with lymphocytosis, normal glucose
level, and normal protein level. No bacteria were seen.
Which of the following is the most likely organism responsible for his illness?
A. Poliovirus
B. Coronavirus
C. Cytomegalovirus
D. Epstein-Barr virus

Answer: A
A 3-day-old baby is brought to the clinic with fever, vomiting and hypothermia. Prior to labour his mother was febrile. He was examined and lumbar puncture was performed (see report).
Cerebrospinal Fluid Gram stain:
Showed Gram positive bacilli and on culture it was Beta hemolytic and catalase positive.
Which of the following antibiotics should be started empirically?
A. Meropenem
B. Cefotaxime
C. Gentamicin
D. Ampicillin

Answer: D
Group B streptococci are a common cause of neonatal meningitis. Vaginal swab for group B streptococci is recommended in pregnant mother.
At which gestational age should the vaginal swab be collected?
A. 15 weeks
B. 25 weeks
C. 35 weeks
D. 40 weeks

Answer: C
A 7-year-old child is brought to the clinic with fever, headache and vomiting of 2 days. On examination, there is meningeal irritation. A lumbar puncture is done (see lab results and report).
CSF Analysis
Test Result Normal Values
Total Protein 560 150-450 mg/L
Glucose 2.9 2.50-4.8 mmol/L
Bacteriology Report:
Gram positive cocci in pairs.
Which of the following antibiotic combination would be most appropriate?
A. Penicillin and vancomycin
B. Vancomycin and rifampin
C. Ampicillin and gentamicin
D. Ceftriaxone and vancomycin

Answer: D
A full-term infant developed bilateral conjunctivitis at 2 weeks of age. The conjunctivitis is followed by severe coughing but the infant remained febrile. When the infant was 4 weeks old, a chest X-ray showed bilateral symmetrical interstitial infiltrates (see lab results).
Test Result Normal Values
WBC 14 4.5-10.5 x 109 /L
Neutrophils 32 40-60%
Lymphocytes 15 20-40 %
Monocytes 2 2-8 %
Eosinophils 4 1-4 %
IgG 30 7.5-19 g/L
IgA 8 1-4 g/L
IgM 3.5 0.52-1.52 g/L
Which of the following is the most probable infectious agent?
A. Streptococcus agalactiae
B. Haemophilus influenzae
C. Chlamydia trachomatis
D. Adenovirus

Answer: C

A 3-year-old girl presents with a high-grade fever and chills. Physical examination, confirms that she is anemic with splenomegaly. Clinical Image with giemsa stain is shown (see image).
Which of the following is the most likely etiology?
A. P. falciparum
B. P. malariae
C. P. ovale

D. P. vivax

Answer: A

A full term boy was born to a mother diagnosed antenatally with hepatitis B surface Ag positive (HBAg positive). Her delivery was unremarkable and the newborn weighed 3.4 kg and was normal on assessment. He received the hepatitis B vaccine and hepatitis B Immunoglobulins within the first 4 hours. The mother is very enthusiastic to breastfeed her child as she did with her previous children.
Which of the following is the best advice?
A. Wait 12 hours before
B. Encourage breastfeeding
C. Advise not to breastfeed
D. Measure HbsAg, then decide

Answer: B

A 4-year-old child suddenly developed a high fever, chills, and vomiting. examination shows nuchal rigidity and a positive Kemig's sign. Petechial lesions are noted on both extremities and over the abdomen.
Which of the following is the most likely diagnosis?
A. Meningococcal meningitis
B. Pneumococcal meningitis
C. Aseptic meningitis
D. Encephalitis

Answer: A
A 2-year-old, severely ill child has been irritable and pyrexial for 2 days. the child started vomiting. appeared drowsy, and has deteriorated rapidly during the day. In addition, a petechial rash started today which has spread extremely quickly around the body.
Which of the following is the most likely diagnosis?
A. Rocky mountain fever
B. Kawasaki syndrome
C. Meningococcemia
D. Measles

Answer: C
Which of the following organs is most commonly affected by mumps in a 4-year-old boy?
A. Skin
B. Lungs
C. Testes
D. Parotid

Answer: D
A child is brought to Paediatric Clinic with cough, fever, rhinorrhoea, malaise, with conjunctival erythema. There are small, grayish, irregular lesions surrounded by an erythematous base, on the buccal mucus membrane near the second molar teeth.
Which of the following is the most likely diagnosis?
A. Measles
B. Rubella
C. Parainfluenza
D. Respiratory syncytial infection

Answer: A
A 6-year-old boy presents to the Medical Care Centre with high fever and cough. examination shows maculo-papular rash on his face and small white spots appeared on the inner part of his cheeks.
Which of the following agents is most likely responsible?
A. Mumps virus
B. Measles virus
C. Varicella zoster virus
D. Herpes simplex virus type-1

Answer: B
A 12-year-old girl presents to the Emergency Department with mild fever. examination shows generalized vesicular rash on her chest, trunk and on her arms and legs.
Which of the following laboratory tests have a higher diagnostic value?
A. Mumps virus IgM
B. HSV-1 IgM
C. HSV-2 IgM
D. VZV IgM

Answer: D
A 1-month-old infant referred to the dental clinic with presentation of white plaques visible on the surfaces of the dorsal tongue and hard palate, which can be wiped off. A clinical diagnosis was made and the infant was started on nystatin 4 times a day for a period of 10 days.
Which of the following is the most likely diagnosis?
A. Erythematous candidiasis
B. Gingivostomatitis
C. Leukoplakia
D. Oral thrush

Answer: D

A 15-year-old child presents to the Outpatient Clinic suffering from abdominal pain, diarrhoea, and tenesmus for the last 2 weeks. There is mucous and blood present in the stool.
Which of the following is the most likely diagnosis?
A. Amoebiasis
B. Tania saginata
C. Bile salt deficiency
D. Intestinal ascariasis

Answer: A
A 4-year-old has a 3-day history of fever, imitability, and anorexia. examination confirms painful vesicles over the mucosa of the soft palate and posterior pharynx, sparing the gingiva and anterior oral cavity.
Blood pressure 110/70 mmHg
Heart rate 120 /min
Respiratory rate 30 /min
Temperature 39.0° C
Which of the following is the most likely diagnosis?
A. Congenital epulis
B. Aphthous ulcer
C. Gingival cyst
D. Herpangina

Answer: D

A 12-year-old child was prescribed antibiotics for treatment of a chest infection. 6 weeks after cessation of antibiotics, the patient presents with fever, abdominal pain, and profuse watery stools for the past month. investigation confirms fecal leukocytes.
Which of the following is the most likely pathogen?
A. Clostridium perfringens
B. Clostridium difficile
C. Giardia lamblia
D. Escherichia coli

Answer: B
A 7-year-old child is brought to the Emergency Department with a history of fever and vomiting for 2 days. On examination, he is dehydrated, comatose and acidotic (see lab results).
CSF Test Result Normal Values
Cells 20 0-3 / uL
Total Protein 0.45 0.22-0.33 g/L
Glucose 3.5 2.50-4.8 mmol/L
Which of the following is the most likely the diagnosis?
A. Tubercular meningitis
B. Salicylate poisoning
C. Diabetic coma
D. Viral meningitis

Answer: D

An 18-month-old child presents to the Emergency Department after a brief, generalized tonic clonic seizure. He has had a large, watery stool that had both blood and mucus in it. He is now postictal.
Blood pressure 100/60 mmHg
Heart rate 110 /min
Respiratory rate 28 /min
Temperature 40.0° C
Which of the following is the most likely cause?
A. Campylobacter
B. Salmonella
C. Rotavirus
D. Shigella

Answer: D

A 1-year-old girl presents to the Paediatrics Emergency Department at night with episodes of loud cough and refusal to feed. There is a history of upper airway infection with nasal congestion and sneezing over the last week. She has not received any routine immunizations. During the physical examination, there are repetitive violent fits of a hacking cough that are followed by forceful, high-pitched inhalations and accompanied by facial cyanosis. Between attacks the girl appeared exhausted and apathetic.
Heart rate 150 /min
Respiratory rate 60 /min
Temperature 36.6° C
Oxygen saturation 92% on room air
Which of the following is the most likely cause?
A. Bordetella pertussis
B. Respiratory syncytial virus
C. Streptococcus pneumonia
D. Haemophilus influenza type B

Answer: A
A 1-year-old girl presents to the Pediatrics Emergency Department at night with episodes of loud coughing and refusal to feed. There is a history of upper airway infection with nasal congestion and sneezing over the last week. She has not received any routine immunizations. During the physical examination, there are repetitive violent fits of a hacking cough that are followed by forceful, high-pitched inhalations accompanied by facial cyanosis. Between attacks the girl appears exhausted and apathetic.
Heart rate 150 /min
Respiratory rate 60 /min Temperature 36.6° C
Oxygen saturation 92% on room air
Which of the following diagnostic test would be most appropriate?
A. Nasopharyngeal swabbing
B. Sweat chloride test
C. Chest radiography
D. Serologic testing

Answer: A
A 5-year-old childis brought to the clinic with a febrile illness with a cough and separate irregularly shaped (blotchy) rashes. He has not had any routine vaccinations. On physical examination, there is enlarged tender cervical lymph nodes. There are also erythematous maculopapular rashes behind the ear, face and neck and to a lesser extent on the trunk. Which of the following is the most likely diagnosis?
A. Chickenpox
B. Measles
C. Mumps
D. Rubella

Answer: B
A 6-year-old child presents with a low-grade fever, malaise, conjunctivitis, coryza, and a cough. Recently, a rash of discrete pink macules began on the face and extends to the hands and feet. Which of the following is the most likely diagnosis?
A. Gonococcal bacteremia
B. Toxic shock syndrome
C. Reiter's syndrome
D. Rubella

Answer: D
A 6-month-old infant develops a cough, coryza, fever, expiratory wheezing, grunting, tachypnea, rib retractions, and air trapping.
Which of the following is optimal immediate treatment?
A. Erythromycin
B. Penicillin
C. Steroids
D. Oxygen

Answer: D
A 6-year-old child presents with malaise, fever, cough, nasal discharge, and conjunctivitis. Several days later the patient develops a maculopapular rash starting on the face and rapidly spreading to the rest of the body.
Which of the following viruses is most likely the cause?
A. Epstein-barr
B. Coxsackie

C. Vaccinia
D. Rubella

Answer: D
A 3-year-old child presents with a 1-week history of malaise, anorexia, nasal discharge, and conjunctivitis. The parent states that the child has also had 2 days of paroxysmal coughing spells with a stridor "whoop."
Which of the following is the most likely diagnosis?
A. Pertussis
B. Epiglottis
C. Psittacosis
D. Mycoplasma

Answer: A

A 3-year-old boy presents to the Paediatric Emergency Department with fever, headache and generalized muscular pain over the last 24 hours. The child appears sleepy and irritable. There are several spots on the buttocks and legs. Physical examination confirms stiffness of the neck. After performing a lumbar puncture, an intravenous infusion of penicillin G is started (see image, lab result and report).
Blood pressure 90/55 mmHg Heart rate 120 /min Respiratory rate 38 /min Temperature 39° C
Oxygen saturation 95% on room air
Test Result Normal Value
WBC 25 4.5-10.5 x 109/L
CSF Culture:
Gram Negative diplococci.
Which of the following is the most likely late complication?
A. Ataxia
B Seizures
C. Deafness
D Cranial nerve palsy

Answer: C
A 3-year-old boy presents to the Pediatric Emergency Department with fever, headache and generalized muscular pain over the last 24 hours. The child appears sleepy and irritable. There are several spots on the buttocks and legs. Physical examination confirms stiffness of the neck. After performing a lumbar puncture, an intravenous infusion of penicillin G is started. The parents also report that the child's 6-year-old brother had been in close contact with this boy and ask how best to look after him (see image and report).
Blood pressure 90/55 mmH
Heart rate 120 /min
Respiratory rate 38 /min
Temperature 39° C
Oxygen saturaton 95% on room air
CSF analysis:
Elevated cell count and Gram Negative diplococci.
What would be the best management for the older brother?
A. Hospital admission and monitoring
B. Intravenous penicillin G
C. Active immunisation
D. Oral rifampicin

Answer: D

A 5-year-old boy presents with a sore throat and refusal to eat. The mother reported that the child has had an increased temperature for the last 3 days. Several children that attended the same play group have developed similar symptoms. Physical examination shows spots and blisters on the palm of both hands as well as on the buttocks. There are small ulcers on the palate (see image).
Heart rate 110 /min
Respiratory rate 30 /min
Temperature 38.6° C
Which of the following additional physical findings is most likely to have?
A. Spots on the soles
B. Swollen tonsils
C. Scratch marks
D. Diarrhoea

Answer: A
A 10-year-old boy presents to the Paediatric Clinic with a skin rash. The parents report that the symptoms had started with a single red spot over a small scratch and have spread over the 4 days. There are several lesions on the cheeks, the trunk and the upper limbs. Physical examination confirms swollen lymph nodes in the arm pits and groins (see image).
Blood pressure 100/65 mg
Heart rate 80 /min
Respiratory rate 24 /min
Temperature 37.4° C
Which of the following is the most appropriate treatment?
A. Steroids
B. Acyclovir
C. antibiotics
D. Antiseptics

Answer: C
A 10-year-old boy presents to the Paediatric Clinic with a skin rash. The parents report that the symptoms had started with a single red spot over a small scratch and have spread over the 4 days. There are several lesions on the cheeks, the trunk and the upper limbs. Physical examination confirms swollen lymph nodes in the arm pits and groins (see image).
Blood pressure 100/65 mmHg
Heart rate 80 /min
Respiratory rate 24 /min
Temperature 37.4° C
Which of the following is the most likely diagnosis?
A. Impetigo
B. Candidiasis
C. Herpes simplex
D. Dermatitis herpetiformis

Answer: A
A 6-year-old girl presents with ear pain and swelling of the pinna. She had returned the previous day from swimming camp and developed these complains 1 day before. On examination, she has significant pain when moving the pinna and a foul smelling discharge from the ear canal. Which of the following is the most appropriate management?
A. Oral amoxicillin
B. Oral cefuroxime
C. Topical neomycin
D. Topical dexametasone

Answer: C

An 18-month-old child is brought to the Emergency Department with fever, irritability and refusal to move the left lower extremity. Physical examination confirms a swollen, erythematous knee with resistance to passive mobilizaton.
Which of the following would be the single most important test?
A. Blood culture
B. X-ray of the knee
C. Complete blood count
D. examination of joint fluid

Answer: D

A 3-year-old child presents to the Emergency Department with a history of bloody diarrhea and oliguria. The parents report that the child had started a fever, vomiting, diarrhea and abdominal pain 7 days before, shortly after they ate at a fast-food restaurant. On physical examination, the child appears ill and in distress. She is also pale and lethargic (see lab results).
Test Results Normal Values
Hb 83 112-165 g/L
WBC 22 4.5-10.5 x 109/L
Platelets count 50 150-400 x 109 /L
Urinalysis Results Normal Values
Proteins + Absent
Erythrocytes 10 0-2 per high power field
Which of the following is the most appropriate management strategy?
A. Steroids
B. antibiotics
C. Supportive care
D. Watchful waiting at home

Answer: C

A 6-year-old child presents with a high-grade fever, malaise, conjunctivitis, coryza, and a cough. A rash of discrete pink macules recently began on the face and extends to the hands and feet. The child also had small, whitish spots inside the mouth before the rash appeared.
Which of the following is the most likely diagnosis?
A. Toxic shock syndrome
B. Gonococcal bacteremia
C. Measles
D. Rubella

Answer: C
A 4-month-old boy presents to the clinic with rash over the neck for 3 days. The rash is not itchy (see image).
Which of the following is the most likely diagnosis?
A. Cellulitis
B. Candidal intertrigo
C. Erythema Multiforme
D. Seborrheic dermatitis

Answer: B

Which of the following is the most frequent agent causing acute bronchiolitis?
A. Adenovirus.

B. Parainfluenza
C. Mycoplasma pneumonia
D. Respiratory syncytial virus

Answer: D

An 8-year-old sickle cell child arrives at the Emergency Department in respiratory distress. Over the previous 3 days, the child has become progressively tired and pale (see lab result).
Test Result Normal Value
Hb 31 112-165 g/L
Which of the following is the most likely cause?
A. Cytomegalovirus
B. Parvovirus B19
C. Coxsackie A16
D. Echovirus 11

Answer: B
A 3-year-old alert and oriented giri presents to the Paediatric Emergency Department at 2 a.m. with breathing difficulties and a brassy, barking cough. There is inspiratory stridor at rest and a hoarse voice. The giri has been suffering from a viral upper airway infection for the past week. A physical examination shows moderate chest wall retractions. On auscultation, there is decreased air entry. The extremities and buccal membranes appear pink and moist.
Heart rate 180 /min
Respiratory rate 60 /min
Temperature 39.9 C rectal
Oxygen saturation 93% on room air
Which of the following is the most appropriate management?
A. Broad spectrum antibiotic treatment
B. Nebulized epinephrine and oral steroids
C. Inhalation with salbutamol and budesonide
D. Cough suppression with codeine hydrochloride

Answer: B
A 7-month-old presents with low-grade fever and a dry cough for the last 3 days. On examination, there is wheezing over both lung fields (see report).
Heart Rate 127 /min
Respiratory Rate 47 /min
Temperature 38.9° C
Chest X-ray:
Hyperinflation and some infiltrate.
Which of the following is the most likely diagnosis?
A. Asthma
B. Sinusitis
C. Influenza
D. Bronchiolitis

Answer: D
A 2-year-old is brought to the clinic with sore throat and high-grade fever for 5 days. On examination, he is febrile and mildly dehydrated. The pharynx and tonsillar area is congested along with inflammation of the mouth and lips associated with gingivitis. Lesions in the mouth and lips appear as whitish-yellow plaques with erythematous base.
Which of the following is the most likely cause?
A. Adenovirus

B. Coxsackie virus
C. Herpes simplex virus
D. Epstein-barr virus

Answer: C
\An 8-year-old girl with presents with sore a throat and a temperature of 38°C. History reveals, she has not received any childhood immunisation. A clinical examination confirms a severely swollen and edematous neck region with enlarged cervical lymph nodes. Her pharynx is erythematous with bleeding spots when a swab is taken for culture. An inspiratory stridor is noted upon auscultation of the lungs.
Which of the following is the most likely diagnosis?
A. Streptococcal pharyngitis
B. Diphtheria
C. Pertussis
D. Thrush

Answer: A

A 1-year-old girl is brought to the Paediatric Emergency Department in the early morning with breathing difficulties. She has a dry cough that sounds like the barking of a sea lion and gets worse when she is crying. Every time the infant breathes in, there is a high-pitched sound There is also a 4-day history of a mild upper airway infection.
Blood pressure 80/65 mmg
Heart rate 120 /min
Respiratory rate 40 /min
Temperature 38.5° C
Which of the following is the most appropriate way to establish the diagnosis?
A. Chest X-ray
B. Spirometry
C. Pharyngeal swab
D. Clinical examination

Answer: D
A 6-month-old boy presents to the paediatrician with a loud barking cough and a rectal temperature of 38° C. The parents report a cold for the last 3 days that worsens at night. The coughing is aggravated with crying and there is a high-pitched, rasping sound on inhalation. Physical examination confirms a reddened throat and a nasal obstruction (see report).
Chest X-ray:
Normal.
Which of the following is the most likely cause?
A. Parainfluenza virus
B. Mycoplasma pneumonia
C. Respiratory syncytial virus
D. Haemophilus influenzae type B

Answer: A
A 3-year-old girl presents to the Paediatric Emergency Department at night with a barking, dry cough and laboured breathing. There is a 3-day history of nasal obstruction and cough that had worsened over the last 2 hours. The child appears anxious and cries with a hoarse voice. There is a sharp, high-pitched sound on inspiration. Physical examination confirms indrawing of the chest wail and a reddened throat. A single treatment with nebulized epinephrine results.in a quick improvement of the airway symptoms. 30 minutes after the inhalation the coughing and the breathing difficulties

reappear.
Which of the following is the next step in management?
A. Observation in clinic
B. intravenous steroids
C. Oxygen administration
D. Repeat epinephrine inhalation

Answer: D

An 8-month-old child presents to the Pediatric Emergency Department in the early morning hours. He has laboured breathing and a barking cough that becomes worse when crying. There is a 2-day history of nasal congestion and dry cough. Physical examination confirms nasal flaring and a reddened throat. There is also a harsh, high-pitched sound when the child breathes in. auscultation over the lung fields is normal. The boy is anxious and cries with a hoarse voice.
Heart rate 110 /min
Respiratory rate 30 /min
Temperature 39° C
Oxygen saturation 94% on room air
Which of the following is the most likely cause?
A. Viral
B. Allergy
C. Bacterial
D. Foreign body

Answer: A
A 12-month-old boy presents to the Pediatric Emergency Department in the early morning hours with nasal congestion and a barking cough and 3 days history of a mild airway infection. Physical examination shows suprasternal retractions and nasal flaring. The back of the throat is reddened. Lung fields are clear on auscultation. The inhalation of nebulized epinephrine temporarily improves the symptoms.
Heart rate 110 /min
Respiratory rate 30 /min
Temperature 38.5° C
Oxygen saturation 94% on room air
Which of the following is the most appropriate next step in management?
A. Steroids
B. antibiotics
C. Anthistamine
D. Bronchodilators

Answer: A
A 4-year-old boy presents to the Paediatric Emergency Department at 3 a.m. with a barking cough and mild respiratory distress. The parents report that he had an airway infection over the last 2 days that had suddenly worsened. The boy appears irritable. Physical examination confirms an inspiratory stridor. The child speaks with a hoarse voice.
Blood pressure 90/65 mmH
Heart rate 110 /min
Respiratory rate 30 /min
Temperature 38° C
Oxygen saturation 92% on room air
Which additional finding would be cause for concern?
A. Tachypnea
B. Nasal flaring

C. Expiratory stridor
D Bluish coloured lips

Answer: D
A 3-year-old girl presents to the Paediatric Emergency Department in the early morning hours with a barking cough, that worsens with crying. There is a history of mild cough and temperatures around 38° C (rectal) over the last 3 days. Physical examination confirms an inflamed throat and mild suprasternal retractions.
Blood pressure 95/65 mmH
Heart rate 90 /min
Respiratory rate 34 /min
Temperature 39.3° C
Which of the following is the most likely at auscultatory finding?
A. Crepitation
B. Wheezing
C. Stridor
D. Gasp

Answer: C

A 13-year-old girl presented to the clinic with fever and painful cervical lymph nodes. On examination, tonsils were covered by white exudate. Temperature 39 °C
Which of the following is the most likely complication to develop?
A. pneumonia
B. pharyngitis
C. scarlet fever
D. glomerulonephritis

Answer: D

A 4-year-old child presented with anorexia, low-grade fever, watery bloodless diarrhea, vomiting, abdominal cramps and dehydration 2 days after exposure to a case of diarrhea in day care center.
Which of the following is the most appropriate treatment?
A. Doxycycline
B. Ceftriaxone
C. Metronidazole
D. Supportive management

Answer: D

A 2-month-old infant present to Emergency Room with cough, sneezing, circumoral cyanosis occasional with frothy secretion from mouth for the 3 days. On examination, she looks well with moderate tachypnea and intercostal retraction. Chest reveals scatted rhonchi and crackles all over but good air entry.
Blood pressure 80/50 mmH
Heart rate 138 /min
Respiratory rate 48 /min
Temperature 36.6 °C
Oxygen saturation 91 %
Which of the following is the most likely diagnosis?
A. Bronchiolitis

B. Heart failure
C. Bronchial asthma
D. Gastrosophageal reflux

Answer: A
A 6-month-old infant with cough and fever diagnosed as bronchiolitis. On examination, she had mild distress with intercostal retraction and chest auscultation reveals diffuse crackles.
Blood pressure 92/67 mmHg
Heart rate 145 /min
Respiratory rate 56 /min
Temperature 36.6 °C
Oxygen saturation 89 %
Which of the following is the best management?
A. Admit for oxygen and hydration therapy
B. Send home with instruction when to come back
C. Admit for antibiotics and oxygen administration
D. Admit for bronchodilation and steroid management

Answer: A
A 2.5-year-old boy seen in the clinic with skin rash. There is history of mild abdominal pain but the child looks healthy otherwise (see image). Which of the following is the most likely diagnosis?
A. Measles
B. Chickenpox
C. Meningococcemia
D. Henoch-schnlein purpura

Answer: D
A 6-year-old child seen in the clinic with fever and sore throat. She has had contact with child developed flu-like symptoms. Throat exam showed mild hyperaemia. Rapid throat test was negative for streptococcal infection and throat swab sent for culture
Blood pressure 110/70 mmHg
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following is the best management at this stage?
A. Supportive treatment
B. Start antibiotics pending culture result
C. Administer 1 shot of ceftriaxone intramuscular
D. Start amoxicillin for 10 days regardless of culture result

Answer: A

A 9-month-old infant presents with history of fever, irritability and right ear rubbing. Ear exam showed air-fluid levels and bulging tympanic membrane.
Blood pressure 70/50 mmHg
Heart rate 110 /min
Respiratory rate 18 /min
Temperature 38.8 °C
Oxygen saturation 95 %
Which of the following is the most likely diagnosis?
A. Acute otitis media
B. Acute otitis externa

C. Chronic otitis media
D. Chronic otitis externa

Answer: A
Which of the following organism is the most causative agent in paediatrics acute otitis media?
A. Viral
B. Fungal
C. Bacterial
D. Mycoplasma

Answer: C
A 3-year-old child present to the clinic with skin lesion that is not itchy. No history of fever and no previous illness (see image). Which of the following is the most likely diagnosis?
A. Cellulitis
B. Folliculitis
C. Impetigo
D. Toxic shock syndrome

Answer: C
Which of the following interval (in days) defines fever of unknown origin in paediatrics?
A. 7
B. 14
C. 21
D. 28

Answer: B

A 4-year-old boy previously healthy came to the clinic with history of limping for the last 2 days. On physical examination, he is keeping his left leg flexed and refusing to move it or allowing anyone to touch it. Imaging is showing effusion of the left hip.
Blood pressure 90/61 mmH
Heart rate 126 /min
Respiratory rate 29 /min
Temperature 38.8 °C
Oxygen saturation 98 %
Which of the following is the most likely causative organism?
A. Enterobacter
B. H. Influenzae
C. Group B streptococcus
D. Staphylococcus aureus

Answer: D

A 3-month-old girl presented to the Emergency Room with symptoms of paroxysmal episodes of cough followed by whooping sound at the end. It is occasionally associated with perioral cyanosis and post- tussive vomiting. These symptoms have gotten worse when started to have fever of 39.5° C. History revealed she did not receive her 2-month vaccine. Which of the following is the most common expected complication?
A. Pneumonia
B. Dehydration
C. Otitis media
D. Pneumothorax

Answer: A
A 3-month-old girl presented to the Emergency Room with symptoms of paroxysmal episodes of cough followed by whooping sound at the end. It is occasionally associated with perioral cyanosis and post- tussive vomiting. History revealed that she did not receive her 2-month vaccine. Her mother has already given her 2 doses of antimicrobial treatment. What is the gold standard method to diagnose this condition?
A. Blood culture
B. Serologic testing
C. Respiratory culture
D. Nasopharyngeal polymerase chain reaction

Answer: D
A 3-month-old boy presented to the Emergency Room with history paroxysmal episodes of cough followed by whoopi sounds at the end of each episode. It is associated with perioral cyanosis and post- tussive vomiting. There is no history of diarrhea. This illness has been preceded by runny nose a few days ago. History revealed he did not receive his 2-month vaccine yet. On examination, the child looks unwell and tachypneic. Which of the following is the most likely diagnosis?
A. Pertussis
B. Adenovirus pneumonia
C. Mycoplasma pneumonia
D. Chlamydia trachomatis pneumonia

Answer: A
A 3-month-old boy presented to the Emergency Room with history paroxysmal episodes of staccato cough, inspiration between each cough. It is associated with perioral cyanosis and post-tussive vomiting. There is no history of diarrhea. This illness has been preceded by conjunctivitis few days ago. On examination, the child looks unwell and tachypneic with auscultatory crackles. He has normal white blood cell count with elevated eosinophils on laboratory evaluation. Which of the following is the most likely diagnosis?
A. Pertussis
B. Adenovirus pneumonia
C. Mycoplasma pneumonia
D. Chlamydia trachomatis pneumonia

Answer: D
A 3-month-old boy presented to the Emergency Room with history paroxysmal episodes of cough associated with an inspiratory whoop and possessive emesis. It is associated with fever, conjunctival redness and diarrhea. His immunization is up to date. Which of the following is the most likely diagnosis?
A. Pertussis
B. Adenovirus infection
C. Mycoplasma pneumonia
D. Chlamydia trachomatis pneumonia

Answer: B
A 9-month-old boy presents to the clinic with skin rash noticed by his parent over the past few weeks. His parents are known to have eczema before. Which of the following is the typical site of eczematous plaques at this age?
A. Upper back
B. Diaper area
C. Dorsum of the feet
D. Scalp, cheeks, and forehead

Answer: D
An 8-year-old boy presents to the clinic with skin rash noticed by his parent over the past few weeks. His parents are known to have eczema before. Which of the following is the typical site of eczematous plaques at this age?
A. Upper back
B. Dorsal feet
C. Flexural surface
D. Scalp, cheeks and forehead

Answer: C
A 14-year-old boy presents to the clinic with skin rash noticed by his parent over the past few weeks. His parents are known to have eczema before. Which of the following is the typical site of eczematous plaques at this age?
A. Popliteal fossa
B. Perioral region
C. Antecubital fossa
D. Scalp and forehead

Answer: B
A 6-years-old boy who has been treated since infancy for eczema is coming now to be evaluated for honey-colored crusted lesion in antecubital skin area bilaterally. He has no fever but skin infection is suspected. Which of the following is the most likely cause of the skin infection?
A. Coxsackie virus
B. Herpes simplex virus
C. Staphylococcus aureus
D. Group A streptococcus

Answer: C
A 6-year-old boy who has been treated since infancy for eczema is coming now to be evaluated for severely itching umbilicated vesico pustular skin rash that is grouped in "clusters of grapes" appearance overlying diffuse eczematous plaques. A superimposed skin infection is suspected. Which of the following is the most likely etiology?
A. Coxsackie virus
B. Herpes simplex virus
C. Staphylococcus aureus
D. Group A streptococcus

Answer: B
A 5-year-old boy who has been treated for eczema with increasing strength of topical steroid is presenting with worsening eczematous plaques on fleural surfaces of his elbows and knees. Which of the following is the best next step in the management?
A. Consider another diagnosis
B. Switch to topical calcineurin inhibitors
C. Referral to dermatologist to start narrow-band UV-B phototherapy
D. Inquire about parent compliance with the treatment and answer their concerns

Answer: D
A 10-year-old girl presented to the clinic with pruritic, eczematous plaques over her flexural region and xerosis. Good bathing practice and adequate hydration are recommended to the patient. Which of the following is considered a good bathing practice recommendation?
A. Emollients are not important after good prolonged bathing
B. Olive oil is superior to coconut oil in decreasing severity
C. Using strong soap cleansers is more effective than non-soap cleansers

D. Emollient choice is highly dependent on provider and patient predilection

Answer: D
A 5-year-old boy presents to the clinic with pruritic, eczematous plaques over his flexural region and xerosis. Good bathing practice and adequate hydration are recommended to the patient. Which of the following is considered a good bathing practice recommendation?
A. Using non-soap cleansers is preferred to avoid irritation
B. Emollients are not important after good prolonged bathing
C. Lotions have been proven to be more effective than creams
D. Olive oil is superior to coconut oil in decreasing severity

Answer: A

A 3-month-old girl is hospitalized for confirmed pertussis infection. She is started on macrolide therapy. She has 2 other siblings at home, 3 and 5 years old who are both up to date for their vaccinations. Which of the following is the most appropriate recommendation for her asymptomatic sibling?
A. Immediate booster vaccine with DaP
B. Start chemoprophylactic therapy with macrolide
C. Close observation for signs and symptoms of pertussis
D. Booster vaccine with DTaP if they are from high risk group

Answer: B

A 4-month-old girl with unilateral purulent effusion, decreased tympanic membrane mobility with pneumatic otoscopy, no fever, and no other symptoms. She is breastfeeding very well and does not use pacifier. She has no recent history of infections. Which of the following is the best next step in management?
A. Start amoxicillin treatment
B. Start amoxicillin-clavulanate
C. Observe with follow-up in 48hr
D. ENT referral for evaluation of tympanostomy tubes

Answer: A
A 19-month-old boy presented to the clinic with right lower eyelid swelling, erythema, and tenderness for 1 day. He has fever reaching up to 38.5° C with decreased appetite and activity. He was given diphenhydramine without any improvement of the eyelid swelling. His parent denied any visual disturbances, diplopia or any other swelling. Which of the following is the most likely diagnosis?
A. Cellulitis
B. Hypoproteinemia
C. Allergic reactions
D. Subperiosteal hematomas

Answer: A
A previously healthy 13-month-old girl has 2-day history of fever and increasing cough. She continued to be breastfed with no vomiting. Her immunizations are up to date. She is alert and mildly ill appearing. On examination, there is no grunting or chest retractions. There are crackles heard over the right lung base. The remainder of the examination findings are normal. She has no known allergies.
Blood pressure 110/70 mmHg Heart rate 146 /min Respiratory rate 45 /min Temperature 38.6 °C Oxygen saturation 95 %
Which of the following is the most likely pathogen?

A. Moraxella catarrhalis
B. Mycoplasma pneumonia
C. Streptococcus pneumonia
D. Haemophilus influenza type B

Answer: C
A 7-year-old girl is complaining of sore throat for the last 2 days. Today, she has developed fever of 38.7 °C when measured at home, which has been relieved by acetaminophen. On examination, she was found to have tender anterior cervical adenopathy and pharyngeal and tonsillar exudate. The rest of her physical exam is unremarkable.
Which of the following is best next step toward diagnosis?
A. Throat culture
B. Echocardiography
C. Rapid antigen detection test
D. Fine needle aspiration for cervical adenopathy

Answer: C

A 4-year-old girl presented to the clinic with history of fever for last 3 days associated with sore throat and decrease oral intake. She also has difficulty in swallowing solid food. On examination, she has palpable cervical lymph node that is about 2 cm in size. Which of the following is the most important step in evaluation?
A. Chest radiograph
B. Lateral neck radiograph
C. Contrast-enhanced CT neck
D. Fine needle aspiration of the lymph node

Answer: B
A 4-year-old girl presented to the clinic with history of fever for last 3 days associated with sore throat and decrease oral intake. She also has difficulty in swallowing solid food. Her vaccination is up to date. On examination, she has palpable cervical lymph node that is about 2 cm in size, tender and fluctuates. Which of the following is the most likely causative organism?
A. Pneumococcus
B. Streptococcus pyogenes
C. Staphylococcus epidermidis
D. Haemophilus influenzae type B

Answer: B

A 5-year-old boy is diagnosed with group A streptococcus tonsillitis and started on antibiotic treatment. He has a 2-year-old brother who is well and asymptomatic. Which of the following is the best approach for his brother?
A. Observation
B. Throat culture
C. Start antibiotic treatment
D. Throat swab for rapid antigen detection test.

Answer: A

A newborn baby boy is born to unbooked 25-year-old mother. The mother has developed febrile illness immediately after birth but she is otherwise stable. She does not have record for any of the transmissible infections. She asked if she can breastfed her baby. In which of the following conditions breastfeeding or expressed breast milk in the immediate postpartum period is acceptable?

A. Breastfeeding of HIV mother
B. Expressed breast milk of HIV mother
C. Breastfeeding in untreated active tuberculosis
D. Expressed breast milk of untreated active tuberculosis

Answer:D

A 2-week-old neonate is brought by his mother with complains of poor oral intake for the last 2 days. On examination, he has oral thrush and diaper dermatitis.
Which of the following is the most applicable management?
A. Oral anti-fungal for 7 days
B. Topical anti-fungal for 5 days
C. Systemic anti-fungal for 5 days
D. Oral and topical anti-fungal for 5 days

Answer: B
A 3-year-old child presented with fever, headache, difficulty swallowing and throat pain. On examination, there is intense inflammation of the tonsils with purulent exudate that is more membranous in appearance.
Blood pressure 110/70 mmHg
Heart rate 150 /min
Respiratory rate 28 /min
Temperature 38.8 °C
Oxygen saturation 96 %
Which of the following is the most likely causative organism?
A. Corona virus
B. Influenza virus
C. Epstein barr virus
D. Respiratory syncytial virus

Answer:C
A 7-month-old baby presented to the clinic with history of fever, irritability and disturb sleeping for 2 days. Ear examination showed perforation of the eardrum with pus visible in the external canal.
Blood pressure 110/70 mmHg
Heart rate 115 /min
Respiratory rate 22 /min
Temperature 38.8 °C
Oxygen saturation 97 %
Which of the following is the most likely diagnosis?
A. Acute otitis media
B. Serious otitis media
C. Chronic otitis media
D. Otitis media with effusion

Answer: A
A 7-month-old baby presented to the clinic with history of fever, irritability and disturb sleeping for 2 days. Ear examination showed perforation of the eardrum with pus visible in the external canal.
Blood pressure 110/70 mmg
Heart rate 115 /min
Respiratory rate 24 /min
Temperature 39 °C
Oxygen saturation 97 %
Which of the following is the most common causative organism?

A. Rotavirus
B. Rhinovirus
C. Adenovirus
D. Coronavirus

Answer: B

A 3-year-old child presented to Emergency Room with complains of fever, headache and lethargy. On examination, clear purpuric rash, sign of shock and positive Kernig sign present. Eye exam revealed the presence of papilledema. The parents are bedside and worried about child condition and prognosis. Which of the following complication is important for the parents?
A. Vision loss
B. Hearing loss
C. Renal failure
D. Retinal haemorrhage

Answer:B
which of the following represent a fetal complication from maternal infection with Parvovirus B19?
A. Usually asymptomatic
B. May lead to fetal hydrops and death
C. Fetal arrhythmia and limp anomalies
D. Sever fetal skin disease with eye involvement

Answer:B

During counseling an HIV positive mother who is pregnant and had several concerns regarding the outcome of her pregnancy. Her viral load is detectable although she is on treatment. What would be the best advice during counselling the mother?
A. The neonate should not receive any therapy after delivery
B. She can breastfeed normally if she continued on the therapy
C. She should deliver by cesarean section to decrease the chance of transmission
D. Therapy during pregnancy will not prevent the transmission of the virus to the baby

Answer: C

Which of the following is the most causative organism of bronchiolitis in infant?
A. Influenza
B. Adenovirus
C. Parainflunza
D. Respiratory syncytial virus

Answer: D

An 18-month-old child presents with stridor and barking cough for 2 days after flu-like symptoms. Examination reveals active well looking child and stable vital signs.
Blood pressure 100/70 mmHg
Heart rate 102 /min
Respiratory rate 20 /min
Temperature 37.6 °C
Oxygen saturation 95 %
Which of the following is the most likely diagnosis?
A. Viral croup
B. Epiglottis

C. Trachiaitis
D. Foreign body aspiration

Answer: A

A 4-month-old boy presents to the clinic with rash over the neck for 3 days. The rash is not itchy (see image). Which of the following is the most likely diagnosis?
A. Cellulitis
B. Candidal intertrigo
C. Erythema Multiform
D. Seborrheic dermatitis

Answer: B

An 18-month-old child presents to the clinic with unilateral eye swilling for 2 days. On examination, there is clear proposis, painful and limited ocular movement with reduce visual acuity. The mother said they missed 12 and 18 months vaccines. Which of the following is the most appropriate next step in management?
A. Arrange for 18 month vaccines administration
B. Brain MRI to roll out intraventricular hemorrhage
C. Urgent CT scan to assess the spread of the infection
D. Perform funduscopic exam to roll out inclusion body

Answer: C
An 18-month-old child presents to the clinic with unilateral eye swilling for 2 days. On examination, there is clear proposis, painful and limited ocular movement with reduce visual acuity. The mother said they missed 12 and 18 months vaccines. Which of the following is the most likely diagnosis?
A. Orbital myositis
B. Orbital cellulitis
C. Periorbital cellulitis
D. Thyroid-associated orbitopathy

Answer: B

A 2-year-old child presents to the clinic with skin lesions on the face, neck and hands. The mother said it was started as erythematous macules, which became vesicular then ruptured. On examination, ruptured vesicles with exudation that look like honey colored crust lesions. The mother mentioned that her child complain of atopic eczema since infancy.
Which of the following is the most likely diagnosis?
A. Boils
B. Impetigo
C. Atopic dermatitis
D. Scalded skin syndrome

Answer: B
A 2-year-old child presents to the clinic with skin lesions on the face, neck and hands. The mother said it was started as erythematous macules, which became vesicular then ruptured. On examination, ruptured vesicles with exudation that look like honey colored crust lesions. The mother mentioned that her child complain of atopic eczema since infancy. The mother is asking if she can allow her child to return to the nursery.
Which of the following is the most appropriate counseling should be given?
A. Lesions are not contagious and can go to nursery

B. Lesions are not contagious and can go to nursery after 5 days
C. Lesions are contagious and should not go to nursery for 3 days
D. Lesions are contagious and should not go to nursery until it become dry

Answer: D
A 2-year-old child presents to the clinic with 3-day history of vesicular lesions on the lips, gums and anterior surface of the tongue and hard palate. On examination, child looks dehydrated with dry mucus membrane the lesions look extensive, painful with mild bleeding. According to the mother, she is not eating or drinking well because of the pain.
Which of the following is the most likely diagnosis?
A. Impetigo
B. Gingivostomatitis
C. Herpetic whitlows
D. Eczema herpeticum

Answer: B
A 2-year-old child presents to the clinic with 3-day history of vesicular lesions on the lips, gums and anterior surface of the tongue and hard palate. On examination, the lesions look extensive, painful with mild bleeding. According to the mother, she is not eating or drinking well because of the pain. Which of the following is the most likely the causative organism?
A. Herpes simplex virus
B. Varicella zoster virus
C. Staphylococcal aureus
D. Streptococcus pneumonia

Answer: A
A 2-year-old child presents to the clinic with 3-day history of vesicular lesions on the lips, gums and anterior surface of the tongue and hard palate. On examination, child looks dehydrated and the lesions look extensive, painful with mild bleeding. According to the mother, she is not eating or drinking well because of the pain for the last 2 days.
Which of the following is the best management?
A. Acyclovir
B. Intravenous fluid and acyclovir
C. Intravenous fluid and topical antibiotics
D. Narrow spectrum systemic antibiotics and topical antibiotics

Answer: B
A 5-year-old child presents to the Emergency Room with severe sore throat, high-fever and increasing difficulty breathing over 8 hours. On examination, ill toxic looking child with drooling of saliva and sitting immobile with open mouth.
Blood pressure 110/70 mmHg
Heart rate 115 /min
Respiratory rate 32 /min
Temperature 38.8 °C
Oxygen saturation 95 %
Which of the following is the most appropriate immediate action should be taken?
A. Urgent referral to ENT for tonsillectomy
B. Complete sepsis work up, followed by antibiotics for 7 days
C. Send home on oral antibiotics for 7 days and referral to ENT
D. Urgent admission to ICU for intubation then sepsis work to follow

Answer: D

A 5-year-old child presents with severe sore throat, high fever and increasing difficulty breathing over 8 hours. On examination, ill and toxic looking child with drooling of saliva and sitting immobile with open mouth.
Blood pressure 110/70 mmHg
Heart rate 120 /min
Respiratory rate 28 /min
Temperature 38.6 °C
Oxygen saturation 95 %
Which of the following is the most likely diagnosis?
A. Bronchitis
B. Pharyngitis
C. Acute Croup
D. Acute epiglottitis

Answer: D

A 6-year-old child presents with a high-grade fever, malaise, conjunctivitis, coryza, and a cough. A rash of discrete pink macules recently began on the face and extends to the hands and feet. The child also had small, whitish spots inside the mouth before the rash appeared.
Which of the following is the most likely diagnosis?
A. Toxic shock syndrome
B. Gonococcal bacteremia
C. Measles
D. Rubella

Answer: C

Pulmonology

A 4-year-old boy presents with a rectal prolapse. His parents report that he has constipation and his stools are bulky and foul smelling. His growth chart indicated that he is below the 5th percentile in both height and weight. In the past, he has had 2 episodes of pneumonia. During investigation, he tested positive for the sweat chloride test.
His parents and a 1-year-old brother are healthy and have no clinical symptoms.
Which of the following would be the most appropriate investigation?
A. Sibling's sweat chioride test
B. Parents' Sweat chloride test
C. Sibling's DNA analysis for the cystic fibrosis transmembrane conductance regulator gene
D. Parents' DNA analysis for the cystic fibrosis transmembrane conductance regulator gene

Answer: C
A 3-year-old child of a parent who is recently diagnosed with pulmonary tuberculosis has a tuberculin skin test. After 72 hours, the size of local induration is 10 mm.
Which of the following is the most appropriate interpretation?
A. Negative
B. positive
C. Needed to be repeated
D. Test is not recommended anymore

Answer: B

A 5-week-old baby is brought to the clinic with low-grade fever and difficulty breathing. Her parents report that she had 2 episodes in which she appeared to have stopped breathing. Physical examination confirms subcostal retractions with Inspiration and diffuse bilateral wheeze.
Temperature 37.9 °C
Heart Rate 130 /min
Respiratory Rate 55 /min
Oxygen saturation 90%
Which of the following is the most appropriate treatment?
A. Inhaled corticosteroids
B. Ventilatory management
C. Broad spectrum antibiotics
D. Supportive care and hydration

Answer: D
A 10-year-old boy is admitted to the hospital due to acute exacerbation of asthma. He is currently medicated with montelukast once daily and has last been seen by his primary care physician over 6 months ago. He reports daily episodes of dry cough and wheezing that are exacerbated by exercise. After admission, his symptoms stabilized with the administration of inhaled albuterol.
Which of the following the most appropriate long-term maintenance therapy?
A. Oral steroids once daily, plus a long-acting beta-2 agonist as needed
B. Inhaled corticosteroid twice-daily, plus a short acting beta-2 agonist as needed
C. Inhaled long-acting beta-2 agonist, plus a short acting beta-2 agonist as needed
D. Leukotriene receptor antagonist once daily plus a short acting beta-2 agonist as needed A
Answer: B

A 3-year-old child admitted to the hospital to initiate antibiotic treatment for pneumonia. She has long history of sinusitis and lower respiratory tract infections. During the current admission, she underwent fibre optic bronchoscopy with sputum collection and both pseudomonas aeruginosa and

burkholderia cepacia are isolated.
Which of the following is the most likely diagnosis?
A. Asthma
B. Cystic fibrosis
C. Primary ciliary dyskinesia
D. Alpha1-Anttrypsin deficiency

Answer: B
A 3-year-old child is a known asthmatic who usually has 1 acute exacerbation each month. The acute symptoms are relieved by nebulizer therapy at the Emergency Department or the nearby health centre.
Which of the following should be included in the daily management?
A. Slow-release theophylline
B. Sodium cromoglycate
C. Inhaled budesonide
D. Oral steroids

Answer: C
A 22-month-old boy presents to the Emergency Department with a sudden onset of respiratory distress during a family picnic. He is febrile and has otherwise been well. Physical examination demonstrates unilateral wheezing in the right hemithorax.
Which of the following is the most likely to be seen in his chest X-ray?
A. Pneumothorax on the right side
B. Consolidation of the right lower lobe
C. Lower lobe collapse on the right side
D. Hyperinflation of the right lung with mediastinal shift to the left

Answer: D
A 2-month -old bay presents to the Paediatric Emergency Department with coughing, wheezing and rapid breathing for the last 2 days. The infant was born at 38 weeks' gestation; his birth weight is 3.5 kg and now weighs 5.2 kg. He had developed a runny nose, imitability and difficulties feeding without having to pause several times. His 3-year-old brother has a cold with fever and cough. Physical examination shows a prolonged expiratory time, nasal flaring and subcostal retractions. Cardiac auscultation results are normal and the spleen and liver are of normal size for the child's age.

Blood pressure 86/58 mmHg
Heart rate 146 /min
Respiratory rate 56 /min
Temperature 38.6° C
Oxygen saturation 96% on room air
Which of the following is the most appropriate next step in making a definitive diagnosis?
A. Chest X-ray
B. Throat swap
C. Bronchoscopy
D. Nasopharyngeal aspirate

Answer: D
A 1-year-old child is brought to the clinic with a high fever, cough and respiratory distress for 3 days. On auscultation of the chest, there are generalised fine crackles (see report).
Chest X-ray:
Showed puffy infiltrates with ill-defined margins throughout both lung fields.
Which of the following is the most likely diagnosis?
A. Bronchopneumonia

B. Acute epiglottitis
C. Bronchiectasis
D. Cystic fibrosis

Answer: A
A 20-month-old child is brought to the Emergency Department due to respiratory distress with stridor and fever. Her parents mention that 2 days ago she started rhinorrhoea and mild cough with progressive worsening of the cough that is now "barky".
Which of the following is the most likely diagnosis?
A. Tracheomalacia
B. Tracheal foreign body
C. Laryngotracheobronchitis
D. Acute infectious laryngitis

Answer: C
A 13-year-old boy comes to clinic complaining of fatigue, non-productive cough, low-grade fever and night sweats over the past 7 weeks. He also mentioned that he has had 2 episodes of hemoptysis.
Which of the following is the most likely diagnosis?
A. Tuberculosis
B. Bronchiectasis
C. Lung neoplasm
D. Mycoplasma pneumonia

Answer: A

A 5-year-old child with bronchial asthma seen in the clinic because of uncontrolled course of disease despite using Ventolin regularly. Mother has asthma and using similar inhaler but she expresses poor technique of Ventolin delivery to her child. Which of the following is the most likely reason for her child's problem?
A. Child neglect
B. Poor compliance
C. Improper technique
D. Need further evaluation

Answer: C
A 5-year-old child with bronchial asthma under control with inhalers. Mother concerned about the future of her child's asthma during adulthood. Which of the following is the most likely prognosis?
A. Chronic lung disease
B. Most of the children grow out of it
C. Asthmatic during adulthood
D. No correlation between childhood and adulthood asthma

Answer: B
A 3-year-old child with bronchial asthma on ventolin nebulizer at home as per needs. The mother expresses concern that the family cannot go out of the home most of the time because the nebulizer machine need electricity power source. Which of the following is the most appropriate counselling should be given?
A. Agree and support their decision as the priority is asthma control
B. Find someone to take care of asthmatic child if family goes out of the home
C. Educate with clear instructions inhaler usage and advice to stop nebulizer
D. Find out nebulizer machine with portability feature

Answer: C
What is the most effective treatment for moderate bronchial asthma severity?
A. Beta agonist agents
B. Beta agonist agents and steroids
C. Long acting beta agonist agents with steroids
D. Long acting beta agonist agents with anticholenergics

Answer: C
A 6-year-old boy presents to Emergency Room with acute asthma attack responding to initial management with improved respiratory distress. His vitals after stabilization are recorded. The decision made for hospitalization.
Blood pressure 110/70 mmHg
Heart rate 97 /min
Respiratory rate 20 /min
Temperature 36.6 °C
Oxygen saturation 88 %
Which of the following is the most likely reason for admission?
A. Hypoxia
B. Patient age
C. Hypotension
D. Asthma attack

Answer: A
A 2-year-old child diagnosed in the Emergency Room with pneumonia. The mother gave history of poor oral intake and vomiting for whatever orally introduced. Clinically, the child looks well with no distress. Blood pressure 88/56 mmH
Heart rate 110 /min
Respiratory rate 23 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following reasons justifies the child hospitalization?
A. Age of the child
B. Vital signs findings
C. Inability to take orally
D. Primary diagnosis of pneumonia

Answer: C
A 3-year-old child seen in the clinic with fever and cough for 5 days following flu-like prodrome. On examination, he had decrease air entry and crackles over right lung middle zone. Chest X- ray is done (see image and lab results).
Blood pressure 90/64 mmH
Heart rate 110 /min Respiratory rate 20 /min Temperature 36.6 °C
Oxygen saturation 97 %
Test Result Normal Values
Hb 12 165-195 q/L (Newborn) 112-165 g/L (Child)
Platelets count 328 150-400 x 109/L
Neutrophils 80 40-60%
WBC 22 4.5-13.5 x 109/L (2-10 years)
4.5-11.1 x 109/L (11 years-older adult)
Neutrophils 84% (2.4-7.3 × 109/L) 30%-77% (2-10 years) (2.7-6.5 x 109/L) 40%-75% (11 years-older adult) Lymphocytes 15% (2.8-9.3 x 109/L) 26%-56% (Birth) (3.7-10.7 × 109/L) 20%-64% (1-23 months)
Which of the following is the most appropriate management?
A. Reassurance

B. Bronchodilator and steroids
C. Admit for intravenous antibiotics
D. Discharge on amoxicillin for 7 days

Answer: D

A previously healthy 13-month-old girl comes to the clinic with 2-day history of fever and increasing cough. She continued to be breastfed with no vomiting. Her immunizations are up to date. She is alert and mildly ill appearing. On examination, there is no grunting or chest retraction. She has crackles heard over the right lung base. The remainder of the examination findings is normal. She has no known allergies. Bacterial community-acquired pneumonia is suspected.
Respiratory rate 36 /min
Temperature 38.6 °C
Oxygen saturation 96 %
Which of the following would indicate a need for hospitalization?
A. Age of 13 month
B. Crackles on auscultation
C. Respiratory rate of 36 breath per min
D. Inability to maintain hydration or oral intake

Answer: D

Which of the following asthmatic patients require hospital admission?
A. Heart rate more than 120 PPM
B. Whenever presented in acute attack
C. After 3 presentation to emergency room
D. Oxygen saturation below 92% after high dose therapy

Answer: D

A 4-year-old child presented with recurrent chest infection and poor growth. On examination, chest auscultation revealed inspiratory crepitation with expiratory wheeze and finger clubbing. Further exploration in the history revealed the presence of very offensive and greasy stools. Which of the following is expected to be further identified in the physical examination?
A. Nasal polyp
B. Gynecomastia
C. Inguinal hernia
D. Small joint swelling

Answer:A

A 4-year-old child with bronchial asthma on ventolin inhaler. The mother gave history of cough attack more than twice per week and she brought her child to the Emergency Room last month twice. Which of the following is the most appropriate next step in management?
A. Start inhaled steroids
B. Increase ventolin frequency
C. Continue same management
D. Admit to hospital for further workup

Answer: A

Which of the following is the most common mode of inheritance in cystic fibrosis?
A. Autosomal recessive

B. Autosomal dominant
C. X-linked recessive
D. Multifactorial

Answer: A
A 2-year-old child presents with cough and fever for 1 week duration following flu-like symptoms. On examination, well looking with decreased air entry and crackles in right lower lung on auscultation and just below this finding there is stony dullness on percussion.
Blood pressure 98/65 mmHg
Heart rate 105 /min
Respiratory rate 22 /min
Temperature 38.6 °C
Oxygen saturation 95 %
Which of the following is the expected finding in chest X-ray match child's chest finding?
A. Bronchopneumonia
B. Plural effusion
C. Pneumothorax
D. Lung collapse

Answer: B
A 3-month-old infant with noisy breath since birth change with positioning tends to disappears in prone position and become more obvious with supine position. Examination and growth parameter are normal. Which of the following is the most likely diagnosis?
A. Nasal atresia
B. Laryngomalacia
C. Subglottic stenosis
D. Vocal cord paralysis

Answer: B
A 3-month-old infant with noisy breath since birth change with positioning tends to disappears in prone position and become more obvious with supine position. Examination and growth parameter are normal. The mother concerned about this sound. Which of the following is the most appropriate counselling should be given?
A. Will worsen with time
B. Needs surgical correction
C. Normal developmental sounds
D. Will improve by her first birthday

Answer: D
An 8-month-old infant with recurrent wheeze and distress resistance to asthma management and required multiple hospitalizations. After detailed history and examination, his pediatrician suspects tracheomalacia. Which of the following has the highest diagnostic value?
A. Chest CT
B. Chest X-ray
C. Fluoroscopy
D. Bronchoscopy

Answer: D

Which of the following is the most common chronic disease in children?
A. Renal failure
B. Cystic fibrosis
C. Bronchial asthma

D. Diabetes mellitus

Answer: C
Which of the following is the most important elements in bronchial asthma management?
A. Control acute attacks
B. Annual influenza vaccination
C. Compliance with inhaled steroids
D. Family education and awareness

Answer: D
A 3-year-old child with suspected bronchial asthma disease. Which of the following has the highest diagnostic value at this age?
A. Brochoscopy and lung biopsy
B. Pulmonary function test and spirometry
C. Chest imaging and specific immunological blood test
D. Detailed history and physical examination and response to brochodilation

Answer: D
A 10-year-old boy with recurrent wheeze and cough responded partially to asthma management. He has poor weight gain and strong family history of bronchial asthma. Which of the following supports diagnosis of cystic fibrosis?
A. Partial response to brochodilation
B. Family history of bronchial asthma
C. Recurrent wheeze and cough
D. Poor weight gain

Answer: D

A 6-year-old boy presented with recurrent cough and wheeze, He is on ventolin and fluxotide inhaler, labelled as bronchial asthma sine age of 4. The mother is concerned about poor weight gain and worsening lung condition. His examination reveals thin child with weight less than 5th percentile and height at 25th percentile and has normal chest exam.
Blood pressure 110/70 mmHg Heart rate 76 /min Respiratory rate 18 /min Temperature 36.6 °C Oxygen saturation 95 %
Which of the following is the most appropriate counselling should be given?
A. Reassurance and dietitian referral
B. Need escalation in bronchial asthma control
C. Start antibiotics and follow up after one month
D. Need further work out to roll out bronchial asthma mimicker

Answer: D

A 4-year-old child presents with shortness of breath and cough following flu-like illness for 2 days. Examination reveals child in severe distress with increased effort of respiration with bilateral wheeze and prolonged expiatory phase. He had slimier attacks before that required hospitalization.
Blood pressure 100/65 mmHg Heart rate 124 /min Respiratory rate 33 /min Temperature 36.6 °C Oxygen saturation 82 %
which of the following is the most appropriate management?
A. Antibiotic and systematic steroid
B. Ventolin nebulization and antibiotics
C. Intravenous fluid and ventalin nebulization
D. Ventolin nebulization and systematic steroid

Answer: D

A 13-year-old boy comes to clinic complaining of fatigue, non-productive cough, low-grade fever and night sweats over the past 7 weeks. He also mentioned that he has had 2 episodes of hemoptysis. Which of the following is the most likely diagnosis?

A. Tuberculosis
B. Bronchiectasis
C. Lung neoplasm
D. Mycoplasma pneumonia

Answer: A

Gastroenterology

Which of the following is most appropriate management for Hirschsprung's disease?
A. Surgical resection
B. Dietary alteration
C. Antibiotic therapy
D. Immuno-supressive agents

Answer: A
A 3-week-old infant is brought to the Paediatrics Clinic. On examination, a right, non-tender scrotal mass is seen. It is non-reducible and not transparent to light.
Which of the following is the most likely diagnosis?
A. Orchitis
B. Hydrocele
C. Inguinal hernia
D. Testicular torsion

Answer: C

An 8-year-old boy presents to the Pediatrics Clinic with a history of recurring abdominal cramps and diarrhea over the last 3 months. His height and body weight had fallen from the 25th to the 3rd percentile during this same period. He has a decreased appetite and feels full after having eaten small quantities of food. Physical examination shows diffuse abdominal tenderness and a perineal fissure. There are no other abnormal findings (see reports)
Blood pressure 100/60 mmH
Heart rate 80 /min
Respiratory rate 20 /min
Temperature 38.0° C
Colonoscopy:
Non-contiguous lesions in the proximal colon and terminal ileum.
Biopsy proximal colon:
Transmural inflammation.
Which of the following is the most likely diagnosis?
A. Crohn's disease
B. Ulcerative colitis
C. Behçet's syndrome
D. Pseudomembranous colitis

Answer: A

A 10-year-old girl presents to the Paediatrics Clinic with a history of weight loss over the last 6 months. The bodyweight had fallen from the 50th to the 20th percentile. She has little appetite and complains of postprandial cramps and recurrent nausea. Physical examination confirms a bloated abdomen with diffuse tenderness on palpation and perianal skin tags (see reports).
Endoscopy:
Focal ulcerations separated by otherwise intact mucosa which is spread over different GI segments.
Biopsy:
Transmural inflammation.
What part of the gastrointestinal tube is most likely to be affected?
A. Jejunum
B. Ileum
C. Colon
D. Rectum

Answer: B

A 4-month-old presents with frequent post-prandial regurgitation. The symptoms have been present since birth but have worsened when the mother increased the formula feeding volumes. The patient's weight is less than it was at the three-month check-up. General examination is otherwise normal
Blood pressure 80/40 mmH
Heart rate 120 /min
Respiratory rate 32 /min
Temperature 36.6° C
Which of the following is the most likely cause?
A. Duplication of the gastric pylorus
B. Erosion of the lower esophagus
C. Infection of the gastric antrum
D. Decreased Lower Esophageal Sphincter pressure

Answer: D

A 6-year-old child is brought to the Paediatric Clinic. The parent reports intermittent fever, vomiting, diarrhea, and diffuse abdominal pain for the past week. The parent indicated that the child's eyes also appears to have a yellowish skin stain for the past 2 days. The liver is palpable 5 cm below costal margin and there is jaundice, scleral icterus, and mild abdominal tenderness on examination.
Blood pressure 110/70 mmlg
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 38.0° C
Which of the following is the most likely type of viral hepatitis?
A. A
B. B
C. C
D. D

Answer: A

An 11-month-old is brought to the clinic. The Parent reports that he has a 1-day history of passing bright red blood mixed with foul smelling, dark brownish-black stool. The child is pale. Coagulation tests are normal (see lab result).
Blood pressure 118/56 mmHg
Heart rate 160 /min
Respiratory rate 26 /min
Temperature 36.6° C
Test Result Normal Value
Hb 70 112-165 g/L
Which of the following investigations is most likely to confirm the source of bleeding?
A. Abdominal ultrasonography
B. Radioisotope imaging
C. Barium enema
D. Barium meal

Answer: D

A 10-year-old bay presents to the clinic complaining of abdominal pain, frequent diarrhea that is sometimes bloody, and pain usually accompanying defecation. His mother noted that he has lost weight over the last three months since the onset of the problem. The boy also has aches and pains of his joints. On examination, he looked ill, pale, with a soft abdomen, no tenderness or organomegaly (see lab results).
Blood pressure 110/70 mmH Heart rate 100 /min Respiratory rate 16 /min Temperature 38.6° C
Test Result Normal Values
Sodium 132 134-146 mmol/L
Potassium 3.2 3.5-5.1 mmol/L
Creatinine 31 27-62 mol/L
Blood urea nitrogen 5.6 1.8 to 6.4 mmol/L
Hb 82 112-165 g/L
C-reactive peptide 1.1 0-0.5 mg/di
Albumin 32 36-52 g/L
Which of the following treatments should be tried first?
A. Metronidazole
B. Corticosteroids
C. Gluten free diet
D. 5 amino salicylates

Answer: D
A 10-year-old boy presents to the clinic complaining of abdominal pain, frequent diarrhoea that is sometimes bloody, and pain usually accompanying defecation. His mother observed that he has lost weight over the last 3 months since the onset of the problem. The boy also has aches and pains of his joints. On examination, he looks ill, pale, with a soft abdomen, no tenderness or organomegaly (see lab resuits).
Blood pressure 110/70 mm
Heart rate 100 /min
Respiratory rate 16 /min
Temperature 38.6° C
Test Result Normal Values
Sodium 132 134-146 mmol/L
Potassium 3.2 3.5-5.1 mmol/L
Creatinine 31 27-62 mol/L
Blood urea nitrogen 5.6 1.8 to 6.4 mmol/L
H6 82 112-165 g/L
C-reactive peptide 1.1 0-0.5 mg/di
Albumin 32 36-52 g/L
Which of the following is the most likely diagnosis?
A. Celiac disease
B. Crohn's disease
C. Ulcerative colitis
D. Chronic dysentery

Answer: C
A 2-year-old boy presents with poor growth, abdominal distension and diarrhoea. On examination, he is at the 5th percentile on the growth chart. He looks pale, has abdominal distension, and wasting of the buttocks.
Which of the following is the most appropriate investigation?
A. Ant-endomysial antibodies
B. Complete blood count
C. Duodenal biopsy

D. Colonoscopy

Answer: A

Which of the following radiographic signs is associated with Duodenal obstruction in a neonate?
A. Kerley B lines
B. Bird's beak sign
C. Hampton's hump
D. Double bubble sign

Answer: D
A 2-year-old boy presents with poor growth, abdominal distension and diarrhea. On examination, he is at the 5th percentile on the growth chart. He looks pale, has abdominal distension, and wasting of the buttocks.
Which of the following is the most likely diagnosis?
A. Celiac disease
B. Cystc fibrosis
C. Crohn's disease
D. Toddler diarrhoea

Answer: A
A 6-week-old infant begins to exhibit forceful vomiting after each meal. The infant was born at term to a 23 year-old primigravida whose pregnancy was uncomplicated and ended with a normal vaginal delivery. No congenital anomalies are noted upon examination of the infant. Which of the following is the most likely diagnosis?
A. Congenital duodenal atresia
B. Necrotzing enterocolits
C. Hirschsprung's disease
D. Pyloric stenosis

Answer: D
A 10-year-old patient presents with a 3-day history of yellow discoloration of the sclera, anorexia and abdominal pain (see lab results).
Test Result Normal Values
Indirect bilirubin 140 <19 mol/L
Total bilirubin 260 <20.5 mol/L
Albumin 40 36-52 g/L
Total Proteins 76 56-80 g/L
Alanine aminotransferase 878 10-35 U/L
Aspartate aminotransferase 1005 12-40 IU/L
Which of the following is the most likely diagnosis?
A. Gilbert's syndrome
B. Infective hepatitis
C. Obstructive jaundice
D. Acute pancreatitis

Answer: B

A 7-year-old child presents with recurrent attacks of abdominal pain for the last 2 weeks, decreased appetite and general weakness. On examination, she looks ill. Abdominal palpation detected a tender mass located in the right lower quadrant; it was confirmed by rectal examination. Complete blood count shows leukocytosis with left shift

Blood pressure 90/60 min
Heart rate 100 /min
Temperature 37.5° C
Which of the following investigations is most likely to confirm the diagnosis?
A. Abdominal ultrasound
B. Plain abdomen X ray
C. Abdominal CT scan
D. Barium enema

Answer: C
A 10-year-old girl is referred to the gastroenterologist after complaining about intense upper abdominal pain for the last 2 months. The pain is independent of meals and there is no vomiting or reflux. She has a normal stool and appetite. The patient's father has a history of peptic ulcer. Laboratory findings are normal. The C13-Urea-breath test is positive. The gastroenterologist performs a gastroscopy and takes a biopsy for microbiological testing.
Which bacterium is most likely to be found in the biopsy?
A. Helicobacter pylori
B. Shigella dysenteriae
C. Campylobacter jejuni
D. Salmonella enteritidis

Answer: A

A 12-year-old patient presents with recent onset of fever and abdominal pain. Just before arrival, the patient vomited. Pain began in the peri-umbilical area. After several hours, the pain localized to the right lower abdominal quadrant. The patent stated that any food offered would be refused. The patent is alert. Physical examination confirms guarding and rebound
tenderness in the right lower abdominal quadrant and there are positive psoas and obturator signs (see lab result).
Blood pressure 110/70 mmHg
Heart rate 120/min
Respiratory rate 22 /min
Temperature 38.2° C
Test Result Normal Value
WBC 15.8 4-10.5 x 109/L
Which of the following is the most likely pathophysiology?
A. Crohn's disease
B. Pin-worm infestation
C. Intussusception of the ileum
D. Obstruction of appendiceal lumen

Answer: D
A 2-week-old infant presents with projectIle, non-bilious vomiting that frequently contains clotted milk. After vomiting, the child still seems to be hungry. On examination, there is mild dehydration. A firm, non-tender and mobile hard 1-2 cm in diameter mass is palpable in the right upper abdominal quadrant at the lateral edge of the rectus abdominus muscle (see report).
Heart Rate 95 /min Respiratory Rate 25 /min Temperature 36.7 °C
Abdominal X-ray:
Shows "single bubble" sign.
Which of the following is the most likely cause?
A. Gastroenteritis
B. Pyloric stenosis
C. Reyes syndrome

D. Gastric mucosal atrophy

Answer: B

A 4-week-old infant is brought to the clinic with progressive, projectile, non-bilious vomiting with a palpable 'olive in right upper quadrant.
Which of the following is the most appropriate first investigation?
A. PH monitoring
B. Scintigraphy
C. Sonography
D. Endoscopy

Answer: C
A 2-month-old boy is brought to the clinic with non-bilious and progressive vomiting. The symptoms have developed 2 weeks prior and the episodes of vomiting have been increasing in frequency. He is now vomiting after every feeding. His weight fell from the 50th percentile to the 25th percentile on the growth chart. The fontanels are sunken and skin turgor was reduced. Physical examination shows a small, firm and movable mass located between the umbilicus and the sternum.
Which of the following is the most likely laboratory result?
A. Elevated sodium level
B. A decreased hematocrit
C. Elevated C-reactive protein
D. Hypochloremic metabolic alkalosis

Answer: D
A 2-month-old baby is brought to the clinic for evaluation of spitting up immediately after every meal. On examination, his abdomen is soft with no hepatosplenomegaly, stool is Negative for occult blood, and he is at the 50th percentile for height and weight (see lab results).
Test Results Normal Values
Hb 150 112-165 g/L
WBC 6.1 4-10.5 x 109/L
Platelets count 250 150-400 x 109/L
Urinalysis:
Protein Absent Absent
Leukocytes 0 0-3 per high power field
Erythrocytes 1 0-2 per high power field
Which of the following is the most appropriate next step in management?
A. Oral proton pump inhibitor
B. Reassure the parents
C. Surgical referral
D. Abdominal CT

Answer: B
A 20-day-old infant has non-fading yellow discoloration of the skin and pale stool that began on the third day after birth. A week ago, the infant was diagnosed with normal physiological jaundice and advised phototherapy (see lab results).
Test Result Normal Value
Total bilirubin 150 < 19 umol/L
Direct bilirubin 98 <10 mol/L
Which of the following is the most likely diagnosis?
A. ABO incompatibility
B. Rh incompatibility
C. G6PD deficiency

D. Biliary atresia

Answer: D
A 2-week-old newborn is brought to the clinic. The mother reports that her baby passes a small amount of unformed stool after each feed. The child is on breastfeed since birth.
Which of the following is the most appropriate management?
A. Prescription of lactose free milk
B. Oral rehydration solution
C. Reassure the mother
D. Infant formula milk

Answer: C

A 12-year-old boy referred to the hospital for treatment of pneumonia. Detailed history reveals intermittent dysphagia; retrosternal pain and regurgitation (see report).
Barium swallow:
Dilatation of the esophagus, lack of peristalsis, and many synchronous contractions. The lower end of esophagus shows gradual narrowing.
Which of the following is the most likely diagnosis?
A. Achalasia cardia
B. Oesophageal malignancy
C. Diffuse esophageal spasm
D. Gastrosophageal reflux disease

Answer: A
A 15-month-old infant has been ill for the past 12 hours with apparent abdominal discomfort, emesis, and the passing of bloody mucous per rectum. The infant is slightly pale. On examination, found to have an increased bowel sounds. The abdominal film shows an obstructive pattern.
Which of the following is the most important step of management?
A. Bowel rest and IV hydration until resolution
B. Intravenous antibiotics
C. Immediate surgery
D. Barium enema

Answer: D
Which of the following is the most appropriate management of an asymptomatic umbilical hernia in a 4-year-old?
A. Reassurance
B. Local cauterization
C. Surgical correction
D. Elastic support to the mid-abdomen

Answer: C

A 5-week-old infant is brought to the clinic with a 3-week history of intermittent vomiting. After vomiting, the infant becomes very hungry and wants to feed again. Physical examination shows no alterations apart from a 2.50 kg weight loss (see lab results).
Test Results Normal Values
Sodium 147 134-146 mol/L
Potassium 2.9 3.5-5.1 mmol/L
Bicarbonate 32 21-28 mmol/L
Chloride 89 97-108 mmol/L
Which of the following is the most likely diagnosis?

A. Gastric volvulus
B. Cyclic vomiting syndrome
C. Hypertrophic pyloric stenosis
D. Intestinal malrotation with volvulus

Answer: C
A 4-day-old infant is seen by a pediatrician in the newborn nursery, for bilious vomiting and abdominal distention. The newborn has been feeding poorly and is difficult to calm. The newborn passed meconium shortly after birth and has passed several seedy, yellow stools each day since then. The physical examination confirmed abdominal distention.
Which of the following is the most likely diagnosis?
A. Allergic reaction to formula
B. Hirschsprung's disease
C. Meconium ileus
D. Midgut volvulus

Answer: D

A 2-year-old African child is skinny with obvious muscle wasting, loss of body fat, and no edema.
Which of the following malnutrition does this child have?
A. Muscle wasting syndrome
B. Kwashiorkor
C. Marasmus
D. Pica

Answer: C
Which of the following diets could cause kwashiorkor?
A. High-protein and high-carbohydrate
B. High-protein and low-carbohydrate
C. Low-protein and high-carbohydrate
D.Low-protein and low-carbohydrate

Answer: C
A young boy developed gingival hypertrophy with enlarged, bluish purple friable gums. Which of the following vitamin deficiencies is the most likely the cause?
A. A
B. C
C. D
D. E

Answer: B

A 3-month-old breastfed child becomes anorexic and develops generalized edema with some degree of aphonia. The infant is diagnosed with infantile beriberi.
What vitamin deficiency does this infant have?
A. Vitamin B1
B. Vitamin E
C. Vitamin C
D. Niacin

Answer: A

A 7-year-old child developed attacks of post-meal nausea, abdominal distension, and central abdominal pain and cramps that usually followed by watery diarrhea. On examination, the child looked well, with a slightly distended abdomen.
Blood pressure 90/70 mmH
Heart rate 98 /min
Respiratory rate 18 /min
Temperature 36.6° C
Which of the following is the most likely diagnosis?
A. Chronic giardia lamblia infection
B. Chronic salmonella infection
C. Inflammatory bowel disease
D. Carbohydrate intolerance

Answer: D
A preterm newborn presents with abdominal distention and showed air lucency in the bowel wall
Which of the following is the most likely diagnosis?
A. Infection
B. Malrotaton
C. Meconium ileus
D.Oesophageal atresia

Answer: D
A 2-month-old baby is brought to the clinic. The mother reports her baby had been crying so much for the last 2 weeks and all her efforts to calm him had failed. The child was delivered normally at 36 weeks' gestation. He has been breast fed since birth, but the mother recently introduced an evening bottle to help calm the baby. On examination, he is crying and irritable. He is at the 50th percentile on the growth curve, and apart from abdominal distension there are no abnormal physical signs.
Which of the following is the most likely diagnosis?
A.Infant colic
B. Infant constipation
C.Infant regurgitation
D.Cow's milk intolerance

Answer: A

A 9-day-old infant presentes to the clinic. The infant was born at 40 weeks' gestation by spontaneous vaginal delivery and had an uneventful birth and nursery course. The infant is breastfed and nursing well. On examination, the infant is mildly jaundiced with a yellow colouration to the face down to the mid-chest. The extremities do not appear jaundiced. The examination is otherwise normal.
Which of the following is the most likely diagnosis?
A. Deficiency of glucuronyl transferase
B. Resolution of birth trauma
C. Breast milk jaundice
D. Hepatitis

Answer: C

A 12-month-old girl presents to the Emergency Department with cramping, colicky abdominal pain progressively developing over the past 12 hours. She has had 2 episodes of vomiting. On physical examination, the child is irritable, and there is abdominal tenderness. Vital signs are normal. During the examination, the child passed a bloody stool with mucus. Lab testing and ultrasound shows leukocytosis and a doughnut-shaped image, respectively.
Which of the following is the most appropriate step in management?
A. Watchful waiting

B. Emergency surgery
C. Radiographic reduction
D. Intravenous normal saline

Answer: C

A 6-month-old child regularly regurgitates a large portion of his feeds. A pH probe study shows significant periods of low esophageal pH. The child has normal growth and no other significant past medical history.
Which of the following is the most appropriate management?
A. Oral antacids
B. Close follow-up
C. Oesophageal manometry
D. Surgery with fundoplicaton

Answer: B
A 13-year-old girl has a 10-month history of diarrhea, postprandial periumbilical abdominal pain, and fever with weight loss. She has had several episodes of blood in her stools. Which of the following is the most likely diagnosis?
A. Peptic ulcer
B. Biliary lithiasis
C. Crohn's disease
D. Chronic pancreatts

Answer: C
An 18-month-old Caucasian boy is evaluated in the clinic for failure to thrive, poor appetite, abdominal distention, diarrhea and irritability. He had been well until 12 months of age when he was weaned from milk to regular foods. Since then, his growth curve is flattened.
Which of the following is the most likely diagnosis?
A. Giardiasis
B. Cystc fibrosis
C. Celiac disease
D. Whipple disease

Answer: C
A 12-year-old boy presents to the clinic with intermittent abdominal distention, abdominal colic and excessive flatulence. He first started noticing these symptoms when he moved into his father's house and his stepmother insisted on milk at dinner every night. He is otherwise healthy, has normal growth, no weight loss, and no travel history.
Which of the following tests has the highest diagnostic value?
A. Serum lactose levels
B. Esophageal manometry
C. Upper gastrointestinal endoscopy
D. Hydrogen breath test after oral lactose

Answer: D

A 7-year-old boy presented with recurring upper abdominal pain and occasional vomiting after meals for the last 6 months. The child has not gained any weight since the beginning of the symptoms and has a loss of appetite. The father has a history of peptic ulcer. Physical examination confirms epigastric tenderness at palpation. A gastroscopy is performed and a biopsy is taken for microbiological testing. A triple treatment with amoxicillin, clarithromycin and a proton pump inhibitor is recommend. Laboratory findings are normal (see report).

13C-Urea-breath test: positive.
Which of the following is the most appropriate way to monitor the response to treatment?
A. Serologic testing
B. Clinical manifestation
C. 13C-Urea-breath test
D. Biopsy for microbiological testng

Answer: C

A 7-year-old boy presents with recurring upper abdominal pain and occasional vomiting after meals over the last 6 months. The child has not gained any weight since the beginning of the symptoms and has a loss of appetite. The father has a history of peptic ulcer. Physical examination confirms epigastric tenderness on palpation. A gastroscopy is performed and a biopsy is taken for microbiological testing. Treatment with metronidazole and a proton pump inhibitor is recommended. Laboratory findings are normal (see report).
13C-Urea-breath test:
positive.
Which additional medication should be added to the treatment?
A. Clarithromycin
B. Tetracycline
C. Prednisone
D. Sucralfate

Answer: A
A 6-week-old infant is admitted to the hospital with jaundice (see lab results).
Test Results Normal Values
Total bilirubin 65 <20.5 mol/L
Direct bilirubin 52 <5 mol/L
Which of the following is the most likely diagnosis?
A. Choledochal cyst
B. Gilbert syndrome
C. ABO incompatibility
D. Crigler-najar syndrome

Answer: A

A 10-year-old boy presents to the clinic with a progressive yellow discoloration of his sclera and skin as well as abdominal distention. examination reveals jaundice and a Kayser-Fleischer ring (see lab results).
Test Result Normal Values
Total bilirubin 81 <20.5 mol/L
Aspartate aminotransferase 65 12-40 IU/L
Aspartate aminotransferase 105 12-40 IU/L
Ceruloplasmin 132 200-540 pg/L
Which of the following is the most appropriate treatment?
A. Ethylene diamine tetra acetic acid
B. British ant-Lewisite
C. Desferrioxamine
D. Penicillamine

Answer: D

A 7-month-old baby presents the Emergency Department with irritability, and intermittent crying during which he pulls up his legs, followed by periods with no discomfort for the last four hours. He had started to vomit just before his parents brought him. On examination, the child looked ill and lethargic, with a distended abdomen, and a palpable mass in the right upper quadrant. Heart rate 140 /min
Respiratory rate 28/min
Temperature 38.6° C

Which of the following is the most appropriate investigation?
A. Contrast barium enema
B. Abdominal ultrasound
C. Plain Abdomen X-ray
D.Abdominal CT scan

Answer: A
A 7-month-old baby presents to the Emergency Department with irritability, and intermittent crying during which he pulls up his legs, followed by periods with no discomfort for the last 4 hours. He had started to vomit just before his parents brought him. On examination, the child looked ill and lethargic, with a distended abdomen, and a palpable mass in the right upper quadrant.
Heart rate 140 /min
Respiratory rate 28 /min
Temperature 38.6° C
Which of the following is the most likely diagnosis?
A. Pancreatitis
B. Appendicitis
C. Gastroenteritis
D. Intussusceptions

Answer: D
A 1-month-old baby is brought to the clinic with signs of constipation, abdominal distension, and bilious vomiting. The parents explain that constipation has been a problem since the baby's birth. Which of the following tests has the highest diagnostic value?
A. Plain X-ray
B. Rectal biopsy
C. Anal manometry
D. US abdomen

Answer: B

A 10-day-old full term boy presents with abdominal distension for 1 week. It is associated with greenish vomiting and difficulty in passing stool. His stools are loose in constancy. On per-rectal examination, the rectum is empty and he passed large amount of stool and gases after the figure was removed (see image).
Which of the following is most likely diagnosis?
A. Pyloric stenosis
B. Intussusception
C. Incarcerated hernia
D. Hirschsprung's disease

Answer: D

A 2-week-old girl presents with persistent vomiting since birth. She vomits about 1 hour after feeding. She passes stool 3 times per day. The physical examination shows a small non-tender mass

in the epigastrium.
Blood pressure 95/55 mmHg
Heart rate 135 /min
Respiratory rate 35 /min
Temperature 37.1° C
Which of the following is the most appropriate investigation?
A. Barium swallow
B. Abdominal X-ray
C. Abdominal CT scan
D. Abdominal ultrasound

Answer: D
A 3-year-old boy presents to the clinic with blood in his diapers for 2 days. There is no associated abdominal pain or constipation. He had similar presentation 3 months ago. The physical examination shows pale conjunctiva and dry oral mucosa.
Blood pressure 130/75 mmH
Heart rate 124 /min
Respiratory rate 35 /min
Temperature 37.4° C
Which of the following is the most likely diagnosis?
A. Juvenile polyp
B. Intussusceptions
C. Hirschsprung disease
D. Meckel's diverticulum

Answer: D

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Blood pressure 130/75 mmHg
Heart rate 124 /min
Respiratory rate 35 /min
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which of the following is the most likely diagnosis?
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B. Intussusceptions
C. Hirschsprung disease
D. Meckel's diverticulum

Answer: D
A 2-week-old girl presents with persistent vomiting since birth. She vomits about 1 hour after feeding. She passes stool 3 times per day. The physical examination shows a small non-tender mass in the epigastrium.
Blood pressure 95/55 mmH
Heart rate 135 /min
Respiratory rate 35 /min
Temperature 37.1° C
Which of the following is the most appropriate investigation?
A. Barium swallow
B. Abdominal X-ray
C. Abdominal CT scan
D. Abdominal ultrasound

Answer: D

A 9-year-old boy presented to the Emergency Room with severe abdominal pain for 8 hours. Which of the following is the most common surgical emergency in such patient?
A. Appendicitis
B. Cholecystitis
C. Intussusception
D. Perforated ulcer

Answer: A
An 8-year-old girl presented to the Emergency Room with severe abdominal pain for the last 6 hours. The pain is constant and not relieved by acetaminophen. She cannot stand up straight because of the pain. Physical Examination reveals a mild grade fever with severe abdominal tenderness over the right lower quadrant and rebound tenderness
Which of the following imaging study is the most appropriate to confirm the diagnosis?
A. CT abdomen
B. US abdomen
C. MRI abdomen
D. Magnetic resonance cholangiopancreatography

Answer: A
A 2-year-old girl presented to the Emergency Room with intermittent, severe abdominal pain and unusual appearance of her stool "currant jelly". On examination, she looks well and alert with sausage-shaped abdominal mass. Which of the following is the diagnostic method of choice in this condition?
A. CT abdomen
B. MRI abdomen
C. Contrast enema
D. Ultrasound abdomen

Answer: D
A 2-year-old girl presented to the Emergency Room with intermittent, severe abdominal pain and unusual appearance of her stool "currant jelly”. On examination, she looks well and alert with sausage-shaped abdominal mass. Ultrasound abdomen revealed "target-shaped" finding. Which of the following is the best option in treatment?
A. Emergency surgical correction
B. Keep her NPO and start total parental nutrition
C. Warmed saline enemas under ultrasound visualization
D. Insert nasogastric tube to gravity drain and keep her NPO

Answer: C

A 4-month-old girl brought to well-baby visit. Her mother has complained of frequent episodes of spitting up after each feeding. These episodes have gotten worse over the past few weeks.
There is no history of fever, cough or diarrhea. The giri has been exclusively breastfed for good time every 3-4 hours. On physical examination, her growth parameters are at 90th percentile for age. The rest of her exams are normal. Which of the following is the next step in the management?
A. Ph probe
B. Reassurance
C. Upper endoscopy
D. Ultrasound abdomen

Answer: B
A 1-month-old boy presented to the Emergency Room with history of forceful, nonbilious vomiting. The mother is very concerned as he is her first baby and was delivered by caesarean section. She stated that he is always hungry no matters how much he feeds. Physical examination showed a palpable "olive" in the abdominal right upper quadrant. Which if the following is the likely finding in his initial laboratory evaluation?
A. Hyperkalemia
B. Hyponatremia
C. Hypochloremia
D. Metabolic acidosis

Answer: C
A 1-month-old boy presented to the Emergency Room with history of forceful, nonbilious vomiting. The mother is very concerned as he is her first baby and was delivered by caesarean section. She stated that he is always hungry no matters how much he feeds. Physical examination showed a palpable "olive" in the abdominal right upper quadrant. Which of the following is the most appropriate initial imaging study?
A. CT abdomen
B. MRI abdomen
C. Upper GI series
D. Ultrasound abdomen

Answer: D
A 12-year old girl is complaining of nausea and vomiting when going on flight longer than 1 hour. Which of the following vomiting treatments is the best choice in her condition?
A. Ginger
B. Ondansetron
C. Granisetron
D. Diphenhydramine

Answer: D
A 1-month-old boy presented to the Emergency Room with history of forceful, nonbilious vomiting. The mother is very concerned as he is her first baby and was delivered by caesarean section. He has no fever or diarrhea but seems hungry all the time despite her best efforts feeding him. On examination, he looks dehydrated and his weight did not change in comparison to his birthweight. His abdominal palpation revealed an olive on the right upper quadrant. Which of the following is the most appropriate immediate next step in the management?
A. Upper GI series
B. Prescribe promethazine
C. Abdominal ultrasonography
D. Correction of dehydration and electrolyte imbalance

Answer: D

A 2-month-old boy is being evaluated in Well Baby Clinic. He is the first child for a young mother who is concerned about his fussiness and spitting up after feeds and in between feeds. He is otherwise healthy. His growth parameters are normal. Which of the following is the most likely cause of his condition?
A. Gastroenteritis
B. Eosinophilic esophagitis
C. Gastrosophageal reflux disease
D. Physiologic gastrosophageal reflux

Answer: D
A 2-month-old boy is being evaluated in Well Baby Clinic. He is the first child for a young mother who is concerned about his fussiness and spitting up after feeds and in between feeds. He is taking cow-milk based formula and otherwise healthy. His growth parameters are normal. His physical examination is normal What would be your next step in his management?
A. Observation
B. Pyloric ultrasonography
C. 24-hour impedance pH study test
D. Reflux precautions and reassurance

Answer: D

A 2-month-old boy is being evaluated in Well Baby Clinic. He is the first child for a young mother who is concerned about his fussiness and spitting up after feeds and in between feeds. He is taking cow-milk based formula and otherwise healthy. His growth parameters are normal. His physical examination is normal. Which of the following is an indication for further evaluation?
A. Male gender
B. Failure to thrive
C. Spitting up while asleep
D. Spitting up between feeds

Answer: B
A 6-year-boy diagnosed previously with gastrosophageal disease. He is on proton pump inhibitor for long time is complaining of dysphagia and epigastria discomfort. Which of the following is important history clue for Eosinophilic Esophagitis?
A. Overweight
B. History of snoring
C. Chewing food excessively
D. Worsening of epigastric pain at night

Answer: C

An 8-year-old boy has been complaining of recurrent abdominal pain for the past several months. Which of the following would prompt an immediate search for an organic explanation?
A. Pain is around his umbilicus
B. It last for more than 10 minutes
C. Occurs primarily around awakening
D. Occurs primarily around sleep time

Answer: B
A 7-year-old boy has had frequent periumbilical pain and nausea in the moming. He occasionally vomits. The symptoms started just after he started going to school. School phobia and separation anxiety are suspected. Which of the following is the most appropriate in management?
A. Start domperidone
B. Start amitriptyline
C. Keeping the child out of school for at least 3 months
D. Increasing the child's participation in age-appropriate activities

Answer: D
A 13-year-old boy is complaining of recurrent abdominal pain for the last year. The pain takes about 5 to 7 days to improve and it is usually associated increased frequency of stooling. In these occasions,

his stool is watery and explosive. No history of vomiting, constipation or blood passage. Irritable bowel syndrome is suspected. Which of the following is the most appropriate in management?
A. Avoid peppermint oil
B. Avoid lentil ingestion
C. Increase Fruit Ingestion
D. Increase Honey ingestion

Answer: B

A 6-month-old infant presents to the Emergency Room with sudden apnea and color change. The mother gave history of recurrent vomiting and multiple chest infection. On examination, wasted and underweight infant, with failure to thrive and dystonic neck posturing while crying. Which of the following contribute to the disease?
A. Hypertrophy of the pyloric muscles
B. Hypertrophy of the esophageal muscles
C. Inappropriate relaxation of the upper esophageal sphincter
D. Inappropriate relaxation of the lower esophageal sphincter

Answer: D

A 6-month-old infant presents to the Emergency Room with sudden apnea and color change. The mother gave history of recurrent vomiting, and multiple chest infection. On examination, he looks failing to thrive with dystonic neck posturing while crying.
Which of the following is the most likely diagnosis?
A. Pyloric stenosis
B. Cow's milk allergy
C. Whooping cough
D. Gastrosophageal reflux

Answer: D; Sandifer syndrome associated with GERD and Hiatal Hernia
A 3-week-old neonate presents to the Emergency Room with persistent projectile vomiting. The mother mentioned that he is immediately show sign of hunger after vomiting, but he is not gaining weight as expected.
Which of the following finding associated with this presentation?
A. Hypochloraemic metabolic alkalosis
B. Hyperchloraemic metabolic acidosis
C. Hypochloraemic metabolic acidosis
D. Hyperchloraemic metabolic alkalosis

Answer: A
A 4-week-old neonate presents to the Emergency Room with persistent projectile vomiting. The mother mentioned that he is immediately show sign of hunger after vomiting, but he is not gaining weight as expected (see lab results)
Test Result Normal Values
Sodium 130 134-146 mmol/L
Potassium 2.8 3.5-5.1 mmol/L
Chloride 88 97-108 mmol/L
Bicarbonate 26 21-28 mmol/L
Which of the following is the most likely diagnosis?
A. Intussusception
B. Pyloric stenosis
C. Intestinal obstruction

D. Gastrosophageal reflux

Answer: B
Which of the following common cause of vomiting in infants' age group?
A. Coellac disease
B. Diabetic ketoacidosis
C. Urinary tract infection
D. Cyclical vomiting syndrome

Answer: C
What is the most appropriate counseling should be given regarding infantile colic?
A. It occurs in 80% of babies
B. Usually resolve by 6 weeks of age
C. Associated with excessive passage of flatus
D. Gripe water has proven evidence to control it

Answer: C
A 5-year-old girl presents to the Emergency Room with history of severe abdominal pain, which is central and colicky, and repeated vomiting. The pain is getting worse by movement. On examination, there is generalized guarding in the abdomen, with more tendemess toward the right iliac fossa (see lab results). Blood pressure 110/70 mmHg
Heart rate 135 /min
Respiratory rate 22 /min Temperature 38 °C
Oxygen saturation 95 %
Test Result Normal Values
RBC 5 4.8-7.1 x 1012/L (Newbor) 4.6-4.8 x 1012/L (Child)
Hb 132 165-195 g/L (Newborn) 112-165 g/L (Child)
Platelets count 320 150-400 x 109/L
ESR 12 2-10 mm/h
WBC 15.2 4.5-13.5 x 109/L (2-10 years)
4.5-11.1 x 109/L (11 years-older adult)
Which of the following is the most appropriate counseling regarding this condition?
A. Occurs in 2% of individual
B. Its uncommon below 3 years old
C. A sausage shaped mass often palpable
D. Passage of redcurrant jelly stool confirmed the diagnosis

Answer: B
Which of the following is the most appropriate counseling for celiac disease?
A. No diagnostic test is available for celiac disease
B. Intestinal biopsy is required to confirm the diagnosis
C. Diagnosis can be confirmed by positive serological test
D. Improvement after usage of gluten-free diet can work as diagnostic test

Answer: B
A 14-month-old child presents to the Emergency Room with history of paroxysmal, sever colicky abdominal pain, which comes and goes in bouts. He had vomiting and suddenly pass red currant jelly stool. Which of the following is the most appropriate counseling for this presentation?
A. Shock is a common complication
B. Recurrence is common after surgical correction
C. Treatment requires immediate surgical intervention
D. Reduction by therapeutic enemas is carried out in case of peritonitis

Answer: A

A 4-year-old girl presents to the clinic with recurrent abdominal pain in addition to headaches She has poor appetite and refuse feeding because the pain. The attacks of the abdominal pain are midline associated with vomiting and facial pallor. There is a family history of severe headache in 2 sisters.Which of the following is the most likely diagnosis?
A. Abdominal migraine
B. Chronic constipation
C. Irritable bowel syndrome
D. Inflammatory bowel disease

Answer: A

Which of the following describes the role of antii-diarrheal drugs in the management of gastroenteritis?
A. They are effective
B. Can be associated with side effects
C. Can replace oral rehydration in severe cases
D. May shorten the excretion of bacteria in stool

Answer: B
A 12-month-old child presents to the clinic following an episode of gastroenteritis as diarrhea returns back after introduction of normal diet. The mother said that she used the oral rehydration solution as prescribed and followed the advices received from the pediatrician. On examination, the child looks mildly dehydrated with normal activity. Which of the following is the most appropriate counseling should be given?
A. Change to gluten free formula
B. Change to cow's milk free formula
C. Use oral rehydration solution for 5 days followed by introduction of normal diet
D. Use oral rehydration solution for 24 hours followed by introduction of normal diet

Answer: D
A 12-month-old child presents to the clinic following an episode of gastroenteritis as diarrhea returns back after introduction of normal diet. The mother said that she used the oral rehydration solution as prescribed and followed the advices received from the pediatrician. On examination, the child looks mildly dehydrated with normal activity. Which of the following can explain the new presentation?
A. Temporary lactose intolerance
B. Temporary lactase intolerance
C. Temporary glucose intolerance
D. Temporary sucrose intolerance

Answer: A

A 3-year-old girl found during health campaign in poor area to have generalized edema, sever wasting, hyperkeratosis, distended abdomen and sparse hair (see lab results).
Blood pressure 85/45 mmH
Heart rate 80 /min
Respiratory rate 25 /min Temperature 36.0 °C
Test Result Normal Values

Sodium 132 134-146 mmol/L Potassium 3.2 3.5-5.1 mmol/L Magnesium 0.4 0.7-1.0 mmol/L Glucose 3.1 2.8-5 mmol/L (Newborn) 3.3 -5.5 mmol/L (Child)
Albumin 20 36-52 g/L
Which of the following contributes to the finding?
A. Vitamin B12 deficiency
B. Vitamin D deficiency
C. Sever protein malnutrition
D. Sever carbohydrate malnutrition

Answer: C
A 3-year-old girl found during health campaign in poor area to have generalized edema, sever wasting, hyperkeratosis, distended abdomen and sparse hair (see lab results).
Blood pressure 85/45 mmH
Heart rate 85 /min
Temperature 36.0 °C
Test Result Normal Values
Sodium 132 134-146 mmol/L
Potassium 3.0 3.5-5.1 mmol/L
Magnesium 0.4 0.7-1.0 mmol/L
Glucose 3.1 2.8-5 mmol/L (Newborn)
3.3 -5.5 mmol/L (Child)
Calcium total 2.1 2.2-2.7 mmol/L
Phosphate 1.2 1.3-2.3 mmol/L
Albumin 18 36-52 g/L
Which of the following is the most likely diagnosis?
A. Rickets
B. Marasmus
C. Kwashiorkor
D. Coeliac disease

Answer: C
A 3-year-old boy known case of celiac disease, presented to the clinic with frontal bossing, bowlegs and harrison sulcus. He missed the follow up in the clinic for the last 6 month. Which of the following can help in confirming the diagnosis?
A. X-ray of the wrist joint shows cupping
B. Flaky-paint skin rash with hyperkeratosis
C. Ultrasound abdomen shows enlarger liver
D. Low plasma albumin, potassium and magnesium

Answer: A
A 3-year-old boy with celiac disease presents to the clinic with frontal bossing, bowlegs and harrison sulcus. He missed the follow up in the clinic for the last 6 month. Which of the following is the best management?
A. Daily administration of Vitamin D3
B. Daily administration of Vitamin A
C. Daily administration of Vitamin B12
D. Daily administration of Vitamin C

Answer: A

Neonatology

A 4-month-old child presents to the Paediatric Outpatient Clinic with a 2-day history of constipation, lethargy, and fever. The child was born to a 25 year-old primigravida and is being breastfed. On examination, the child has flattened facies, an inability to smile or gaze away when a light is shined directly in the eye.
Blood pressure 110/70 mmH
Heart rate 140 /min
Respiratory rate 26 /min
Temperature 38.2° C
Which of the following is the most likely diagnosis?
A. Hirschsprung's disease
B. Infantile botulism
C. Hypothyroidism
D. Amoebic colitis

Answer: B

What is the most important measure to protect neonates against infection in a nursery?
A. Thorough hand washing before and after exams
B. Compulsory use of shoe covers and gowns
C. Wearing sterile gloves before entry
D. Use of prophylactic antibiotics

Answer: A

A 3-week-old boy is brought to the Emergency Department because of fever, poor feeding, watery stools. and abdominal distention. He was a product of a full-term pregnancy with normal spontaneous vaginal delivery. He has history of intrauterine growth restriction. He has full sepsis and meningitis workup which came back to be negative.
Which of the following features is associated with a lower risk for developing necrotizing enterocolitis?
A. Being male gender
B. Being full-term baby
C. Age less than 4 weeks
D. Negative sepsis and meningitis work up

Answer: B
A male neonate born by cesarean section at 31 weeks' gestation with birth weight 1300g. He has developed apnea, high gastric residual levels, and bloody stools. Necrotizing enterocolitis is suspected. which of the following features is associated with a higher risk for developing necrotizing enterocolitis?
A. Male gender
B. Bloody stool
C. Cesarean section delivery
D. Birth weight of less than 1500g

Answer: D

A 3-day-old neonate start to develop skin lesions look like erythematous macules on an erythematous base, over the upper chest and the back (see lab results).
Blood pressure 85/55 mmHg Heart rate 110 /min Respiratory rate 18 /min Temperature 36.6 °C

Oxygen saturation 99 %
Test Result Normal Values
RBC 5 4.8-7.1 x 1012/L (Newbor)
4.6-4.8 x 1012/L (Child)
Hb 145 165-195 g/L (Newborn)
112-165 g/L (Child)
Platelets count 250 150-400 x 109/L
WBC 9 9.1-30.1 x 109/L (Birth)
6.1-17.5 x 109/L (1-23 months)
Which of the following is the best management approach?
A. Topical antibiotics for 5 days
B. Consult dermatology for skin biopsy
C. Reassurance as it is a benign condition
D. Full sepsis work up and oral antibiotics for 7 days

Answer: C
Which of the following is the most causative factor for retinopathy of prematurity?
A. Acidosis
B. Hyperoxemia
C. Apnea of prematurity
D. Gram negative sepsis

Answer: B

A newborn delivered normally and discharged with the mother next day. What is the recommended sensory screening should be performed?
A. Hearing
B. Vision
C. Hearing and vision
D. Depends on clinical exam

Answer: A

Respiratory distress syndrome is common disease encountered in newborn delivered prematurely. Which of the following is the most basic lung part affected?
A. Epithelial cell producing surfactant
B. Connective tissue protein
C. Alveolar space structure
D. Bronchial tissue enzyme

Answer: A

The illnesses and problems children encounter are highly age dependent. Which of the following represent an infant age group?
A. Less than 4 weeks
B. Less than 6 month
C. Less than 1 year
D. Less than 2 years

Answer: C
During normal newborn exam, an absent red reflex in the eye and cardiac mummer are discovered. After interviewing the mother, she gave a history of viral infection during the first trimester.
Which of the following infection can cause this?

A. Rubella
B. Syphilis
C. Toxoplasmosis
D. Cytomegalovirus

Answer: A

A 2-hour-old baby delivered by emergency cesarean section because of placental abruption, and required resuscitation with intubation and cardiac message (see lab results).
Test Result Normal Values
ABG HCO3- 7 13-22 mmol/L
ABG PCO2 65 27-40 mmHg
pH 6.9 7.36-7.45
ABG PO2 5 8-24 mmHg (Newbor)
55-80 mmHg (Child)
Base access -18 -2 to 2 mmol/L
Which of the following proven to reduce death and sever disability within 6 hours?
A. Fluid restriction
B. Mild hypothermia
C. Respiratory support
D. Treatment of hypotension

Answer: B; Prevents Perinatal Asphyxia
Newborn delivered with ventouse due to malposition. While assessment in the nursery, a hematoma from bleeding below the periosteum confined within the margins of the skull sutures noted. Which of the following is the most likely diagnosis?
A. Chignon
B. Cephalohematoma
C. Caput succedaneum
D. Subaponeurotic hematoma

Answer: B
A 30-minute-old, preterm at 27-week, delivered by emergency cesarean section for maternal reason have tachypnea, labored breathing and chest wall recession.
Blood pressure 55135 mmH
Heart rate 135 /min
Respiratory rate 65 /min
Temperature 36.2 °C
Oxygen saturation 86 %
Which of the following is the most likely diagnosis?
A. Pneumothorax
B. Apnea of prematurity
C. Respiratory distress syndrome
D. Transient tachypnea of newborn

Answer: C
Necrotizing enterocolitis is a serious illness affecting preterm infants in the first few weeks of life
Which of the following is true in regard this condition?
A. Its associated with bacterial invasion of ischemic bowel wall
B. Treatment required antibiotics but rarely parenteral nutrition
C. Has low morbidity and mortality compared with inguinal hemia
D. Preterm infants fed cow's milk formula have low risk to develop it

Answer: A
A 5-day-old girl brought to Emergency Room with lethargy and poor feeding. On examination, she was irritable, deeply jaundiced and dehydrated. The mother gave history of previous babies' required admission and exchange transfusion (see lab results).
Blood pressure 90/55 mmHg, Heart rate 95 /min, Respiratory rate 18 /min
Temperature 36.6 °C
Oxygen saturation 94 % Weight 3 kg
Test Result Normal Values
Hb 95 165-195 g/L (Newborn) 112-165 g/L (Child) Reticulocyte 3 0.2-1.2 %
Total bilirubin 360 <274 mol/L (Preterm (3-5) days) <205 mol/L (Term (3-5) days)
<21 mol/L (Older infant pre-term)
<20.5 mol/L (Older infant term)
Which of the following is important to be obtained in the history?
A. Apgar score
B. Birth weight
C. Type of delivery
D. Maternal blood group

Answer: D
A term baby delivered by cesarean section, started to be tachypnic with nasal flaring and expiratory grunting. Chest X-ray showed fluid in the horizontal fissure
Blood pressure 60/35 mmHg
Heart rate 140 /min
Respiratory rate 65 /min
Temperature 36.6 °C
Oxygen saturation 92 %
Which of the following is the most likely diagnosis?
A. Pneumonia
B. Meconium aspiration
C. Respiratory distress syndrome
D. Transient tachypnea of the newborn

Answer: D
A 6-hour-old newborn delivered by cesarean section because of sever fetal deceleration and meconium stained fluid. He required intubation and ventilation with iontropic support. Few hours later, he started to have low saturation with increase oxygen requirement. The pre ducal saturation is 92 while the post ductal saturation is 81. Chest X-ray done and showed no pneumothorax. Which of the following is the next step in the management?
A. Surfactant lavage
B. Diuretics infusion
C. Inhaled Nitric Oxide
D. Chest tube insertion

Answer: C
A term infant delivered with antenatal diagnosis of diaphragmatic hernia. She required vigorous resuscitation in the delivery room. Which of the following is the first step in managing this infant after stabilization?
A. Starting antibiotics
B. Chest tube insertion
C. Immediate surgical repair
D. Large nasogastric tube insertion

Answer: D

In colonized mother with Group B Streptococcal infections, which of the following is risk factor for infection in the newborn?
A. Preterm baby
B. Twin pregnancy
C. Gestational diabetes
D. Rupture of membranes for 6 hours

Answer: A; D would be a correct answer if more than 18 Hrs
In managing infant of diabetic mother who required high level of glucose infusion concentration up to 20% to control the hypoglycemia.
Which of the following route is preferred to give the glucose infusion?
A. Orogastric tube
B. Nasogastric tube
C. Central venous catheter
D. Peripheral venous catheter

Answer: C

Cardiology

A 14-month-old asymptomatic boy has a grade 2 pansystolic murmur detected on a routine consultation. Echocardiography reveals a 2 mm ventricular septal defect in the muscular portion of the septum.
Which of the following is the most appropriate management?
A. Surgical repair
B. Watchful waiting
C. Afterload reducing agents
D. Interventional cardiac catheterization

Answer: B
A 4-month-old boy with down syndrome exhibits poor weight gain, tachypnea, a low-pitched grade 2 systolic murmur and a fixed splitting of S2. Chest X-ray reveals cardiomegaly and increased pulmonary vascularity. Electrocardiography documents biventricular hypertrophy and a superior frontal plane QRS axis.
Which of the following is the most likely diagnosis?
A. Coarctation of the aorta
B. Patent ductus arteriosus
C. Ventricular septal defect
D. Atrioventricular septal defect

Answer: D
A 4-week-old baby presents with tachypnea, tachycardia, differential cyanosis and irritability. Which of the following is the most common accompanying feature?
A Cough
B. Ascites
C. Difficulty feeding
D. Jugular venous distension

Answer: C
A 6-month-old girl presents with difficulty feeding and the parents report that her skin becomes bluish during feeding and crying. Cardiac auscultation reveals a harsh systolic ejection murmur and a loud, single S2.
Which of the following is the most likely diagnosis?
A. Aortic stenosis
B. Tetralogy of fallot
C. Ventricular septal defect
D. Transposition of the great arteries

Answer: B
A newborn child presents with cyanosis, tachypnea, irritability and difficulty feeding. Physical examination findings are a parasternal heave and a loud, single S2 on cardiac auscultation. No murmurs were heard.
Which of the following is essential in the early management?
A. NSAIDs
B. Diuretics
C. Dopamine
D. Prostaglandin E1

Answer: D
A 5-year-old girl is found to have a loud pan-systolic murmur at the lower left sternal border. Which of the following is the most likely diagnosis?
A. Atrial septal defect
B. Patent ductus arteriosus
C. Ventricular septal defect
D. Pulmonary valve stenosis

Answer: C
Which of the following ECG abnormalities is an expected finding in ostium secundum atrial septal defect?
A. Delta waves
B. Left axis deviaton
C. Right bundle branch block
D. Left ventricular hypertrophy

Answer: C
Which of the following is characteristic of the cardiac auscultation of a child with an atrial septal defect?
A. S3 gallop
B. Fixed, widely split S2
C. Continuous murmur
D. Systolic ejection dick

Answer: B
A 3-year-old asymptomatic child is examined prior to elective oral surgery. A grade 2/6 low- pitched continuous murmur is heard at the upper right steal border when the child is sitting; the murmur disappears when the child is supine.
Which of the following is the most appropriate acton?
A. Reassure, murmur is innocent
B. Obtain a cardiac consultation
C. Antibiotic prophylaxis prior to oral surgery

D. Postpone oral surgery and re-evaluate the child

Answer: A
Which of the following findings in a newborn requires immediate treatment?
A. An erupted tooth
B. Breast enlargement
C. Shallow sacral dimple
D. Absent femoral pulses

Answer: D

A 13-year-old boy presents for a physical check-up. He has had normal growth and development and has been able to do the usual activities for his age without difficulty. His older brother had had a sudden cardiac death while walking to work. The boy's examination is normal. Which of the following must be excluded?
A. Bicuspid aortc valve
B. Patent ductus arteriosus
C. Ventricular septal defect
D. Hypertrophic cardiomyopathy

Answer: D
Which of the following patents with congenital heart disease would be expected to exhibit cyanosis?
A. 6-year-old girl with coarctation of the aorta
B. 18-month-old boy with a muscular VSD
C. Term newborn with truncus arteriosus
D. 10-year-old boy with a congenital ASD

Answer: C

An infant born to a 36-year-old mother presents with poor feeding, sweating over the forehead, and repeated chest infections. The child has an upward slant of the eyes, delayed mile stones, short stubby fingers, clinodactyly of the little finger, and a single transverse palmer crease. The child is in respiratory distress.
Blood pressure 65/45 mmHg
Heart rate 120 /min
Respiratory rate 45 /min
Temperature 36.8° C
Which of the following is the most likely cardiac anomaly?
A. Tetralogy of fallot
B. Patent ductus arteriosus
C. Endocardial cushion defect
D. Transposition of great arteries

Answer: C
Which of the following congenital heart diseases is associated with the lowest risk of infective endocarditis?
A. Patent ductus arteriosus
B. Ventricular septal disease
C. Subvalvular aortic stenosis (discrete)
D. Atrial septal defect (ostium secundum)

Answer: D

Which of the following is a component of the Tetralogy of Fallot?

A. Ventricular septal defect
B. Left ventricular hypertrophy
C. Atrial septal defect
D. Tricuspid stenosis

Answer: A
A 9-month-old infant has a congenital heart lesion but the parents do not know the specific diagnosis. On examination, it was found that the infant has central and peripheral cyanosis. Which of the following is the most likely diagnosis?
A. Patent ductus arteriosus
B. Coarctation of the aorta
C. Tetralogy of fallot
D. Atrial septal defect

Answer: C
A 10-year-old boy is diagnosed with acute rheumatic fever with cardiac involvement. For how many years is rheumatic fever prophylaxis recommended for this child?
A. 1
B. 5
C. 10
D. 15

Answer: B
A 9-year-old presents with weakness, fatigue, and hypertension in upper limbs. examination confirms delayed pulses in lower extremities.
Which of the following is the most likely aetology?
A. Atrial septal defect
B. Mitral valve prolapse
C. Coarctation of the aorta
D. Ventricular septal defect

Answer: C

Which of the following has the greatest risk of developing endocarditis in patients with congenital heart disease?
A. Patent ductus arteriosus
B. Ventricular septal defect
C. Atrial septal defect
D. Tetralogy of fallot

Answer: B
A 12-day new-born girl is brought to Emergency Department. Her mother complaining that the girl turns blue when she cries. After examination and investigation, the pediatrician explains to the mother that she had a congenital heart disease called Fallot's tetralogy.
Which of the following is a part of Fallot tetralogy?
A. Pulmonary stenosis
B. Aortic valve stenosis
C. Interatrial septal defect
D. Hypertrophy of left ventricle

Answer: A
An 8-year-old child is brought to the clinic for a routine physical examination. During the exam, the physician notes a cardiovascular murmur that is high pitched and heard best at the midsternal border and late systolic in timing. The pattern is crescendo-decrescendo in nature. Which of the following is the most likely type of murmur?
A. Systolic regurgitation
B. Systolic ejection
C. Physiologic
D. Innocent

Answer: B

A 5-year-old overweight girl has systolic blood pressure above the 95th percentile and has a family history of hypertension. Physical examination reveals a strong radial pulses and weak femoral pulses. Which of the following is the most likely diagnosis?
A. Pheochromocytoma
B. Renal artery stenosis
C. Essential hypertension
D. Coarctation of the aorta

Answer: D
A 12-year-old boy comes to the clinic with symptoms of unnatural tiredness and exertional dyspnoea. He has a past history of myopia. On physical examination, he has scoliosis and a pectus excavatum (see report).
Height above the 90th percentile
Weight below the 25th percentile
Echocardiography:
Mitral valve prolapse and dilatation of the ascending aorta. Which of the following is the most likely diagnosis?
A. Aortic dissection
B. Marfan Syndrome
C. Congenital syphilis
D. CHARGE associaton

Answer: B

A 6-year-old boy presents for his routine preschool check-up. The cardiovascular examination reveals a systolic murmur grade Ill, with no thrill, located at the lower sternal border, it is vibratory in character and accentuated when the boy is in a supine position.
Which of the following is the most likely diagnosis?
A. Still murmur
B. Aortic stenosis
C. Atrial septal defect
D. Ventricular septal defect

Answer: A
A 14-year-old-boy presents for a routine paediatrics exam.
There is no history of any major illness. Physical examination confirms an irregular pulse with no other abnormal findings (see image).
Blood pressure 115/70 mmHg
Heart rate 80 /min
Respiratory rate 20 /min
Temperature 36.8° C

Oxygen saturation 95% on room air
Height 175 cm (> 90th percentile)
Weight 45 kg (25th percentile)
Which of the following is most likely prognosis?
A. Syncopal episodes
B. Need of pacemaker
C. Normal development
D. Myocardial dysfunction

Answer: C
A 21-day-old newborn recently diagnosed with a patent ductus arteriosus undergoes surgery to ligate the duct.
Which of the following nerves would be at risk?
A. Left vagus
B. Left phrenic
C. Left recurrent laryngeal
D. Left Superficial cardiac plexus

Answer: C
Which of the following syndromes is associated with coarctation of the aorta?
A. Down
B. Turner
C. Edward
D. Patau

Answer: B

A 9-year-old boy seen in pre-sport screening clinic found to have fixed widely split second heart sound with grade 2 ejection systolic murmur at upper left steal border. ECG was normal.
Blood pressure 110/70 mmH
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 36.6 °C
Oxygen saturation 95 %
Which of the following is the most appropriate next action?
A. Treadmill exercise test
B. Start diuretics and captopril
C. Reassurance and discharge from clinic
D. Cardiology referral for catheterization

Answer: A

Which of the following is the most common congenital heart lesion?
A. Persistent ductus arteriosus
B. Ventricular septal defect
C. Atrial septal defect
D. Tetralogy of fallot

Answer: B
A 24-hour-old neonate found to have abnormal heart rhythm during admission exam. ECG done showed complete heart block. ECHO showed small persistent ductus arteriosus (see lab results).
Test Result Normal Values

RBC 5 4.8-7.1 x 1012/L (Newborn) 4.6-4.8 x 1012/L (Child)
Hb 165 165-195 g/L (Newborn) 112-165 g/L (Child)
HCT 0.45 0.44-0.64 (Newborn)
0.32-0.42 (Child)
Platelets count 65 150-400 x 109/L
WBC 11 9.1-30.1 x 109/L (Birth)
6.1-17.5 x 109/L (1-23 months)
Which of the following maternal disorders contributes to the finding?
A. Hypothyrodism
B. Diabetes mellitus
C. Rubella infection
D. Systemic lupus erythematosus

Answer: D
A 2-year-old child presents to the Emergency Department with high-grade fever, persistent vomiting, diarrhea and cough for 3 days. On examination, a murmur with decrease air entry over the right lung. For the last 4 visit to the pediatrician, the patient had a normal exam including cardiac (see lab results). Blood pressure 110/70 mmHg
Heart rate 125 /min
Respiratory rate 22 /min Temperature 38.5 °C
Oxygen saturation 97 %
Test Result Normal Values
RBC 4.5 4.8-7.1 x 1012/L (Newborn) 4.6-4.8 x 1012/L (Child)
Hb 95 165-195 g/L (Newborn) 112-165 g/L (Child)
MCH 20 28-33 pg/cell
MCV 65 80-95 f1
Platelets count 200 150-400 × 109/L
WBC 6 4.5-13.5 x 109/L (2-10 vears)
4.5-11.1 X 109/L (11 years-older adult)
Which of the following is the best next step in management for the heart murmur?
A. Urgent referral to cardiology
B. Immediate ECHO and ECG
C. Admission in cardiac ICU for monitoring and intervention
D. Re-examine the heart murmur after correction of the underlying illness

Answer: D
A 2-year-old child presents to the Emergency Department with high-grade fever, persistent vomiting, diarrhea and cough for 3 days. On examination, a murmur with decrease air entry over the right lung. For the last 4 visit to the pediatrician, the patient had a normal exam including cardiac (see lab results).
Blood pressure 110/70 mmg
Heart rate 110 /min
Respiratory rate 22 /min
Temperature 38.5 °C
Oxygen saturation 97 %
Test Result Normal Values
RBC 4.2 4.8-7.1 x 1012/L (Newborn)
4.6-4.8 x 1012/L (Child)
Hb 92 165-195 g/L (Newborn)
112-165 g/L (Child)
MCH 20 28-33 pg/cell
MCV 65 80-95 fl
Platelets count 350 150-400 x 109/L

Which of the following is the most likely diagnosis for the heart murmur?

A. Aortic stenosis
B. Innocent murmur
C. Pulmonary regurgitation
D. Ventricular septal defect

Answer: B
An 8-week-old infant presents to the clinic with wheezing. coughing, poor feeding and poor weight gain for the previous 2 weeks. Before this, she had been well. Her routine neonatal examination had been normal apart from pansystolic murmur at lower left sternal edge. On examination, the liver was enlarged below costal margin.
Blood pressure 110/50 mmg
Heart rate 95 /min
Respiratory rate 32 /min
Temperature 36.6 °C
Oxygen saturation 92 %
Which of the following is the most likely diagnosis?
A. Cystic fibrosis
B. Bronchial asthma
C. Atrial septal defect
D. Ventricular septal defect

Answer: D
An 8-week-old infant presents to the clinic with fever, wheezing. coughing, poor feeding and poor weight gain for the previous 2 weeks. Before this, she had been well. Her routine neonatal examination had been normal apart from pansystolic murmur at lower left sternal edge. On examination, the liver was enlarged below costal margin (see report).
Blood pressure 95/50 mmHg
Heart rate 85 /min
Respiratory rate 34 /min
Temperature 38 °C
Oxygen saturation 92 %
Chest X-Ray: Showed cardiomegaly and increase pulmonary vascular marking with bilateral patches in the lung. Which of the following is the most appropriate next step in management?
A. Referral for surgical intervention
B. Medical treatment with diuretics
C. Admit to ICU for diagnostic catheterization
D. Full course of IV antibiotics then re-evaluate

Answer: B
Which of the following represent the best management approach for significant atrial septal defect (ASD)?
A. Secundum ASD require surgical correction
B. Treatment usually undertaken at 3-5 years of age
C. Treatment usually undertaken within 8 weeks form birth
D. No need to treat significant ASD as it will close by age of 5 years

Answer: B
An 18-month-old child diagnosed with congenital heart disease, presented to Emergency Room with severe cyanosis and hypercyantic spells. The mother mentioned that he was irritable and inconsolable crying for the last 1 hours and the cyanosis is getting worse. On examination, he looks cyanotice centrally, with loud harsh ejection systolic murmur heard at left sternal border (see report).

Blood pressure 100/60 mmHg Heart rate 95 /min Respiratory rate 28 /min Temperature 37.8 °C Oxygen saturation 85 %
Chest X-Ray: Showed small heart with decrease pulmonary vascular marking. Which of the following is to most appropriate management?
A. Sedation and pain relief
B. Intravenous diuretics bolus
C. Immediate cardiac catheterization
D. IV antibiotics and total parenteral nutrition

Answer: A; TOF

A 2-day-old neonate presents to the Emergency Department with sudden sever cyanosis. On examination, he looks cyanotic, lethargic; with cardiac exam reveals no murmur and normal pulses (see report).
Chest X-Ray:
Showed egg on side appearance of the cardiac shadow and increased pulmonary vascular marking.
Which of the following is the most likely diagnosis?
A. Tetralogy of fallot
B. Coarctation of aorta
C. Transposition of great arteries
D. Persistence ductus arteriosus

Answer: C
A 14-month-old presents to Primary Clinic with recurrent syncopal attack, associated with worsening heart failure manifestation and poor growth. On examination, he has crackles all over the chest with palpitation and abnormal cardiac rhythm (see report).
ECG: Showed narrow complex tachycardia of 250-300 beats/min.
Which of the following is the most likely diagnosis?
A. Atrial flutter
B. Atrial fibrillation
C. Ventricular tachycardia
D. Supraventricular tachycardia

Answer: D

Rheumatology

Which of the following are most consistent with the diagnosis of acute rheumatic fever?
A. Fever, Roth spots and ischemic stroke
B. Malar facial rash, joint aches and fever
C. Chorea and multiple painful joint swelling
D. Fever, cough and purulent sputum production

Answer: C
Which of the following represents the main treatment for symptomatic acute rheumatic fever?
A. Intravenous ampicillin every 6 hours
B. High dose oral acetyl salicylic acid
C. Daily intravenous hydrocortisone
D. Monthly intramuscular penicillin

Answer: B

Which of the following echocardiographic findings is consistent with the diagnosis of acute rheumatic fever?
A. Mitral valve stenosis
B. Aortic valve stenosis
C. Mitral valve regurgitation
D. Pulmonary valve regurgitation

Answer: C
A 12-year-old boy comes to the clinic with multiple painful swollen joints with reduced range of motion. examination shows multiple non-tender subcutaneous nodules.
Which of the following would most probably confirm the diagnosis?
A. Elevated ESR
B. positive blood culture
C. High C-reactive protein
D. Elevated antistreptolysin O titer

Answer: D
Which of the following is the main treatment of Kawasaki disease?
A. Aspirin
B. Ibuprofen
C. Paracetamol
D. Intravenous Immunoglobulin

Answer: D

A 5-year-old boy presents with fever and a rash on his lower limbs, 2 weeks after developing a sore throat. There is also a history of abdominal pain and painful knees and ankles. Physical examination confirms palpable purpuric lesions on the lower limbs that do not blanch on pressure (see lab result).
Test Result Normal Value
ESR 92 2-10 mm/h
Which of the following is the most likely diagnosis?
A. Vitamin C deficiency
B. Factor VII deficiency
C. Hypersensitivity vasculitis
D. Drug hypersensitivity reaction

Answer: C

A 4-year-old boy presents with a 1-day history of abdominal pain and vomiting. He was afebrile and has no diarrhea. He has bilateral knee pain, and there was some tenderness of the knee joints, but no effusions. Within 24 hours, he developed a rash on his legs and buttocks, which is petechial and purpuric, and his platelet count was normal.
Which of the following is the most likely diagnosis?
A. Systemic lupus erythematosus
B. Hemolytic uremic syndrome
C. Henoch-schönlein purpura
D. Kawasaki disease

Answer: C

An 8-year-old girl is brought to the Paediatric Clinic with pain in the left wrist and right knee. These symptoms have occurred several times in the past 6 months with asymptomatic periods in between. Usually, the pain is worse in the morning and it improves during the day. She is otherwise healthy. On physical examination, there is swelling and a decreased range of motion of both joints.
Which of the following is the most likely diagnosis?
A. Septic arthritis
B. Reactive arthritis
C. Acute rheumatic fever
D. Juvenile idiopathic arthritis

Answer: D

A 2-year-old girl with a high, relapsing fever of 9-day duration associated with marked irritability. a non-purulent conjunctivitis, red cracked lips, swollen and erythematous hands and feet, and a maculopapular erythematous rash on the trunk and extremities.
Which of the following is the most likely diagnosis?
A. Epstein bar viral infection
B. Kawasaki disease
C. Rubella infecton
D. Measles

Answer: B

A 9-year-old child presents with a fever and sore throat associated with bilateral knee pain. The laboratory investigation shows a high sedimentation rate.
Which of the following is the most likely diagnosis?
A. Juvenile rheumatoid arthritis
B. Osgood-schlatter disease
C. Infectious mononucleosis
D. Acute rheumatic fever

Answer: D

A 3-year-old boy presents to the Emergency Room for the second time with non-resolving high-grade fever (39.5 °C). Fever has started 6 days ago and was prescribed antipyretic, which did not help much in relieving the fever. Instead, He has developed bilateral non-exudative conjunctivitis.

His immunizations are up-to-date. On examination, he has diffusely erythematous oropharynx, red cracked lips, and a strawberry tongue. He has non-exudative congested throat with cervical lymph node that measure around 1.5 cm.
Which of the following is the most likely diagnosis?
A. Mumps
B. Measles
C. Enterovirus
D. Kawasaki disease

Answer: D
A 3-year-old boy presents to the Emergency Room for the second time with non-resolving high-grade fever (39.5 °C). Fever has started 6 days ago and was prescribed antipyretic when he was evaluated initially 3 days ago. However, Antipyretic did not help much in relieving the fever. Instead, He has developed develop bilateral non-exudative conjunctivitis. His immunizations are up-to-date. On examination, he has diffusely erythematous oropharynx, red cracked lips, and a strawberry tongue. He
has non-exudative congested throat with cervical lymph node that measured around 1.5 cm. IVIG is ordered. Which of the following may indicate a poor response to IVIG in such patient?
A. Anemia
B. Neutropenia
C. Elevated CRP
D. Hypernatremia

Answer: C
A 4-year-old boy is diagnosed with Kawasaki Disease on day 8 of illness. He is admitted to the hospital for further management. Which of the following is the most appropriate initial treatment regimen?
A. Aspirin and IVIG
B. Aspirin and IV corticosteroid
C. IVIG and cyclosporine
D. Aspirin and infliximab

Answer: A
A 4-year-old boy is diagnosed with Kawasaki Disease on day 6 of illness. He is admitted to the hospital for further management. He has received aspirin and IVIG without any complications. His fever has subsided and his conjunctivitis has significantly improved. Which of the following modalities is the best to evaluate his coronary arteries?
A. Echocardiography
B. Cardiac catheterization
C. Magnetic resonance angiography
D. Computed tomographic angiography

Answer:A

A 7-year-old girl presents to the clinic with fever and skin rash for 1 day. She has also pain and swelling in her both knees and wrists. The family gave a history of fever and sore throat two weeks ago which has been recovered completely. On examination, she has pain, swelling and limited movement in both knees and wrists. She has erythema marginatum rash. The remaining exam is unremarkable. She is started on antibiotic and anti-inflammatory medications.
Which of the following is the best next step in management?
A. Echocardiography
B. Musculoskeletal MRI
C. Pulse methylprednisolone

D. Intravenous immunoglobulin

Answer: A

A 4-year-old girl presents with morning limp and swelling of her right knee that has been worsening for the last 2 months. On physical examination, she clearly has decreased range of motion as well as swelling and warmth of right knee. She has negative antinuclear antibody (ANA). When should she be screened for uveitis by an ophthalmologist?
A. Every 3 months
B. Every 6 months
C. Every 12 months
D. Not recommended if her ANA is negative

Answer: B
A 4-year-old girl presents with morning limp and swelling of her both knees that has been worsening for the last 2 months. She has no other symptoms. On physical examination, she clearly has decreased range of motion as well as swelling and warmth of both knees. She has normal laboratory investigation except a positive antinuclear antibody (ANA). Juvenile idiopathic arthritis (JIA) suspected.
Which of the following is the best classification of her disease?
A. Polyarticular JIA
B. Oligoarthritis JIA
C. Systemic-onset JIA
D. Undifferentiated JIA

Answer: B
A 2-year-old boy brought to the Emergency Room for the second time with non-resolving high grade fever 39.5 °C. Fever started 6 days ago and was prescribed antipyretic when he was evaluated initially 3 days ago. However, antipyretic did not help much in relieving the fever. Kawasaki disease suspected. Which of the following is one of the five principal clinical features of Kawasaki disease?
A. Arthritis
B. Myocarditis
C. Anterior uveitis by slit lamp examination
D. Bilateral bulbar conjunctival injection without exudate

Answer: D
A 6-year-old girl presents to the clinic complaining of morning neck stiffness and pain in her fingers joints to the degree that she needs help in buttoning herself. She has flat erythema over the malar eminences that are sparing the nasolabial folds. One examination, she had painless mouth ulcer. On her finger's
joints. there is clear swelling and tenderness. Which of the following will make her fulfil 4 of the 11 American College of Rheumatology criteria for the classification of Systemic Lupus Erythematosus?
A. Weight loss
B. Erythema marginatum
C. Positive rheumatoid factor
D. Hemolytic anemia with reticulocytosis

Answer: D

A 12-year-old girl who is known to have systemic lupus erythematosus (SLE). Her disease is controlled on immunosuppressive medication. She is complaining of mild grade fever and dysuria. Her examination is unremarkable. The urinalysis is suggestive of urinary tract infection. Which of the following should be avoided in her condition as it may cause SLE flare?

A. Cephalexin
B. Amoxicillin
C. Nitrofurontoin
D. Sulfamethoxazole

Answer: D

which of the following considered as minor manifestation of Jones criteria for diagnosis of rheumatic fever?
A. Pancarditis
B. Polyarthralgia
C. Sydenham chorea
D. Subcutaneous nodules

Answer: B
A 2-year-old child presented to the clinic with high-fever of 5-day duration. Examination showed ill child with mild conjunctivitis, a rash, cervical lymphadenopathy, and cracked red lips (see lab results),
Blood pressure 110/70 mmHg
Heart rate 115 /min
Respiratory rate 22 /min Temperature 39.5 °C
Test Result Normal Values C-reactive peptide 135 0-0.5 mg/di RBC 4 4.8-7.1 x 1012/L (Newborn)
4.6-4.8 x 1012/L (Child)
Hb 115 165-195 g/L (Newborn) 112-165 g/L (Child)
Platelets count 550 150-400 x 109/L ESR 125 2-10 mm/h
WBC 15 4.5-13.5 x 109/L. (2-10 years)
4.5-11.1 x 109/L. (11 years-older adult)
Which of the following is the best management?
A. Aspirin
B. Ribavirin
C. Vitamin A
D. Acyclovir

Answer: A
A 2-year-old child presented to the clinic with high-fever of 5-day duration. Examination showed ill child, with mild conjunctivitis, a rash, cervical lymphadenopathy and cracked red lips (see lab results),
Blood pressure 110/70 mmHg
Heart rate 120 /min
Respiratory rate 24 /min Temperature 39.2 °C
Test Result Normal Values
RBC 5 4.8-7.1 x 1012/L (Newborn) 4.6-4.8 x 1012/L (Child)
Hb 115 165-195 g/L (Newborn) 112-165 g/L (Child)
Platelets count 550 150-400 x 109/L ESR 125 2-10 mm/h
WBC 18 4.5-13.5 x 109/L (2-10 vears)
4.5-11.1 x 109/L (11 years-older adult)
C-reactive peptide 135 0-0.5 mg/dl
Which of the following is the most likely diagnosis?
A. Mumps
B. Rubella
C. Measles
D. Kawasaki disease

Answer: D

Hematology/ Oncology

A 4-year-old boy presents with fever and abdominal pain after a fall on his right side a day ago. His mother noticed that his abdomen appears distended today, particularly on the right side. examination shows a pale and hypertensive boy.
Which of the following is the most likely diagnosis?
A. Liver contusion
B. Pyelonephritis
C. Neuroblastoma
D. Wilm's tumour

Answer: C

A 5-day-old born at home to a 25 year-old primigravida at full term developed bruising over the thigh. examination is normal, except for a few bruises on the left thigh. Weight and height are at the 95 percentile for age (see lab results).
Test Result Normal Values
Prothrombin time 21 12-14 s
APTT 66 30-40 s
Platelet 280 150-450
Which of the following is the most likely diagnosis?
A. Idiopathic thrombocytopenic purpura
B. Neonatal hemorrhagic disease
C. Deficiency of factor X
D. Haemophilia

Answer:B

A 2-week-old newborn presents to the Outpatient Department with mild jaundice of skin and conjunctiva that had begun 6 days ago. The baby was born full-term following a normal pregnancy and uncomplicated birth. Her birth weight was 3.45 kg. The girl is exclusively breastfed and gaining weight properly. No pathologic findings are evident on physical examination. Ultrasound examination of liver and biliary systems is normal (see lab results).
Test Result Normal Values
Indirect bilirubin 20 <19 mol/L
Direct bilirubin 1 <10 molL
Alanine aminotransferase 25.8 13-45 U/L
Aspartate aminotransferase 23.7 12-40 U/.
Which of the following is most appropriate initial management?
A. Observation
B. Phototherapy
C. Exchange transfusion
D. Phenobarbital treatment

Answer: A

A 5-day-old infant has significant nasal and rectal bleeding. He was delivered by a midwife at home. The pregnancy was normal and the baby had good Apgar scores. He is breastfeeding well.
Which of the following vitamin deficiencies is the most likely the cause?
A. B1
B. C
C. D
D. K

Answer: D

Which of the following is the most common intra-abdominal tumour in children?
A. Rhabdomyosarcoma
B. Hepatoma
C. Wilm's
D. Ewing

Answer: C
A 3-year-old boy presents to the Outpatient Department with the complaint of lethargy and failure to thrive (see lab results.)
Test Result Normal Values
RBC 4.6 4.6-4.8 x 1012/L.
Hb 88 112-165 g/L
Reticulocyte 5 0.2-1.2 %
MCH 24 28-33 pg/cell
MCV 65 80-95 f
Which of the following is the most appropriate management?
A. Blood transfusion
B. Intramuscular iron
C. Oral ferrous sulfate
D. Iron fortified cereal

Answer: C
A 1-year-old infant presents with massive hepatosplenomegaly, skin bruises, and swelling in the lateral aspect of the neck.
Which of the following is the most appropriate investigation?
A. Bone marrow aspiration
B. Lumbar puncture
C. EBV serology
D. MRI thorax

Answer: A

A 29-year-old woman gave birth to a boy who is severely jaundiced. She already has two children who were normal at birth. The woman has A-Negative blood group and the newborn tested to be O-positive.
Which of the following is the most likely cause?
A. Maternal anti-RhD antibodies causing placental damage
B. Fetal anti-RhD antibodies causing agglutination of fetal RBCs
C. Fetal anti-RhD antibodies causing agglutination of maternal RBCs
D. Maternal anti-RhD antibodies causing agglutination of fetal RBCs

Answer: D
A 5-year-old girl presents to the clinic with pallor and irritability. On examination, she has mild jaundice and her spleen was enlarged by 4 cm (see lab results).
Test Result Normal Values
RBC 4.4 4.6-4.8 x 1012/L
Hb 82 112-165 g/L
Reticulocyte 9 0.2-1.2 %
MCH 39 28-33 pg/cell
MCV 80 80-95 fl

Peripheral blood smear
Test Result
Microspherocytes ++
Anisocytosis Mild
Which of the following is the most likely diagnosis?
A. Biliary obstructon
B. Sickle cell anemia
C. Hereditary spherocytosis
D. Autoimmune haemolytc anemia

Answer: C
A 2-year-old boy is brought to the Paediatric Emergency Department. The mother reports that the boy has become increasingly irritable and feeble over the last 2 days. He has passed very little urine and his skin has become pale. The mother had consulted the paediatrician 2 weeks before because the baby had suffered from bloody diarrhea, vomiting and high fever. Symptomatic treatment with acetaminophen and probiotics had been started. Physical examination shows mild petechiae on the lower extremities. Urinalysis reveals microhematuria and mild proteinuria (see lab results).
Test Result Normal Values
Hb 73 112-165 g/L
INR 1.1 0.9-1.2
WBC 48.6 4.5-10.5 x 109/L
Platelets count 95 150-400 x 109/L
MCV 86 80-95 fl
MCH 30 28-33 pg/cell
RBC count 3.3 4.6-4.8 x 1012/L.
Prothrombin time 12 12-14 sec
APTT 34 30-40 s

Which of the following is the most likely diagnosis?
A. HUS
B. ITP
C. TIP
D. DIC

Answer: A

A 12-year-old girl presents with bleeding gums while brushing her teeth. This had started 1 week ago after a viral upper respiratory tract infection. On physical examination, the only finding are petechiae on her extremities and abdomen. There is no lymphadenopathy, or hepatosplenomegaly (see lab results and report).
Test Result Normal Values
Hb 130 112-165 g/L
HCT 0.39 0.32-0.42
MCV 88 80-95 f1
WBC 8 4.5-10.5 x 109/L
Platelets count 20 150-400 x 109/L
Bone Marrow:Increased megakaryocytes.

Which of the following is the most appropriate management?
A. Azathioprine
B. Prednisolone
C. Splenectomy

D. Cyclophosphamide

Answer: D

An 8-year-old girl presents with fever, many bruises over the entire body, and pain in both legs. examination confirms pallor, ecchymoses and petechiae on the face, trunk and extremities (see lab results).
Test Result Normal Values
Hb 63 112-165 g/L
WBC 28 4.5-10.5 x 109/L
Platelets count 29 150-400 x 109/L
Which of the following would be the most appropriate diagnostic test?
A. Bone marrow aspiration
B. Coagulation profile
C. Hb electrophoresis
D. CT scan liver

Answer: A
A 5-year-old developed acute abdominal pain in the upper right quadrant, following an upper respiratory tract infection. On examination, there are normal bowel sounds and no rebound tenderness (see lab results and report).
Blood pressure 110/70 mmHg
Heart rate 110 /min Respiratory rate 24 /min
Temperature 38.9° C
Test Result Normal Values
Hb 72 112-165 g/L
WBC 12 4.5-10.5 x 109/L
Reticulocyte 7.2 0.2-1.2 %
Peripheral smear:
Normochromic normocytic red blood cells with target cells and red cell inclusion bodies. Which of the following is the most likely diagnosis?
A. Gl infarction
B. Lead poisoning
C. Sickle cell disease
D. Acute appendicitis

Answer: C
A 2-year-old child presents with painful swelling on the dorsal surfaces of the hands, feet, fingers, and toes (see lab results and report).
Blood pressure 110/70 mmHg
Heart rate 100 /min
Respiratory rate 22 /min Temperature 38.9° C
Test Result Normal Values
Hb 77 112-165 g/L
Reticulocyte 3.9 0.2-1.2 %
Blood smear:
Several elongated and crescent shaped red blood cells.
Which of the following is the most appropriate on-going care?
A. Vitamin B12
B. Corticosteroids
C. Iron supplement
D. Penicillin and immunisation

Answer: D
A 5-year-old child presents with bleeding from the nose and gums and easy bruising of the extremities for 3 days. Prior to this presentation, the child had a viral upper respiratory tract infection.
Which of the following is the most likely diagnosis?
A. Idiopathic thrombocytopenic purpura
B. Henoch-schönlein purpura
C. Meningococcemia
D. Haemophilia A

Answer: A
A 3-year-old child presents for a routine physical examination. The parent is concerned that the child appears pale. The child's diet includes a lot of milk but very little red meat. Haemoglobin is 80 g/L with hypochromic and microcytic indices.
Which of the following is the most appropriate management?
A. Daily multivitamin with iron
B. Trial of iron and follow-up
C. Bone marrow biopsy
D. Blood transfusion

Answer: B
Which of the following vitamins is administered to newborns to prevent haemorrhagic disease?
A. A
B. B
C. C
D. K

Answer: D
A 6-year-old child is brought to the Emergency Department with chest pain and signs of respiratory distress. The child has a minor upper respiratory tract infection, which suddenly worsened with temperature elevation. The physical examination found an increased pulmonic and systolic ejection murmurs (see lab results and report).
Test Result Normal Values
Hь 65 112-165 g/L
WBC 18 4.5-10.5 x 109/L.
Reticulocyte 12 0.2-1.2 %
Chest X-ray:
Shows lobar infiltrates.
Which of the following is the most likely diagnosis?
A. Sickle cell disease
B. Splenic sequestration
C. Myocardial infarction
D. Pneumococcal pneumonia

Answer: A
A 2-year-old boy presents with irritability, lack of attention, and an abnormal craving for chalk to eat. He was born full term by vaginal delivery and has had appropriate developmental milestones. He was breached and weaning started at the age of 9-months. The examination is normal except for pale conjunctiva (see lab results).
Test Result Normal Values
Hb 76 112-165 g/L
HCT 0.22 0.32-0.42

WBC 5 4.5-10.5 x 109/L
Platelets count 250 150-400 x 109/L
MCV 65 80-95 f1
MCH 22 28-33 pg/cell
MCHC 248 260-340 g/L
Reticulocyte 0.9 0.2-1.2 %
Total iron binding capacity 701 17.9-71.6 mol.
Which of the following is the most suitable treatment?
A. Folic acid
B. Vitamin B12
C. Iron supplementation
D. Packed cell transfusion

Answer: C
A 7-year-old boy is brought to the clinic with the complaints of mild abdominal pain, constipation and aggression. The mother told the physician that he is used to eating tissue paper and chipped off wall paint (see lab results and report).
Test Result Normal Values
RBC 3.9 4.6-4.8 x 1012/L
Hb 74 112-165 g/L
MCH 23 28-33 pg/cell
MCV 70 80-95 fl
Peripheral blood smear:
Microcytic hypochromic.
Which of the following is the most likely type of anaemia?
A. Aplastic
B. Pernicious
C. Haemolytic
D. Iron deficiency

Answer: D
A 16-year-old boy presents with 6-day fever and abdominal pain. He looks pale and toxic and had a tender and mildly enlarged spleen.
What is the most sensitive culture to be recommended?
A. Stool or urine culture
B. A single blood culture
C. Multiple blood cultures
D. Bone marrow aspirate culture

Answer: D

Which of the following conditions can lead to hyposplenism?
A. G6PD deficiency
B. Sickle cell disease
C. Vitamin B12 deficiency
D. Hereditary spherocytosis

Answer: B
A 12-year-old girl presents with bleeding gums while brushing her teeth. This had started 1 week ago after a viral upper respiratory tract infection. On physical examination, the only finding are petechiae on her extremities and abdomen. There is no lymphadenopathy, or hepatosplenomegaly (see lab results).
Test Result Normal Values

Hb 130 112-165 g/L
HCT 0.39 0.32-0.42
MCV 88 80-95 f1
WBC 8 4.5-10.5 x 109/L
Platelets count 20 150-400 x 109/L
Which of the following findings is expected in this patent's bone marrow?
A. Hypoplasia
B. Reduced iron stores
C. Megaloblastic changes
D. Increased megakaryocytes

Answer: D
A 5-year-old boy presents with sudden and severe bone pain in his hands and feet. On examination, he appears dehydrated (see lab results).
Blood pressure 110/60 mmHg
Heart rate 125 min
Respiratory rate 28 /min Temperature 38.8° C rectal
Test Result Normal Values
Hb 108 112-165 g/L
HCT 0.30 0.32-0.42
MCV 76 80-95 fl
Which of the following is responsible for this clinical presentaton?
A. Aplastic crisis
B. Sequestration crisis
C. Vaso-occlusive crisis
D. Sickle chest syndrome

Answer: C
A 1-year-old boy is diagnosed with factor VII deficiency with level of less than 1%. Which of the following investigations will be affected?
A. Bleeding time
B. Platelet count
C. Prothrombin time
D. Activated partial thromboplastin time

Answer: C
A 1-year-old bay presents with prolonged bleeding after circumcision (see lab results).
Test Result Normal Values
Bleeding time 1.6 up to 4 min
Prothrombin time 40 12-14 sec
APTT 120 30-40 sec
Platelets count 160 150-400 x 109/L
Which factor deficiency is the most likely cause?
A. VIII
B. VII
C. IX
D. X

Answer: D
Which of the following signs are more likely to be associated with haemophilia?
A. Epistaxis
B. Gum bleeding
C. Haemarthrosis

D. Petechial haemorrhages

Answer: C
A 15-year-old patient presents for a routine check-up (see lab results).
Test Result Normal Values
RBC 5.3 4.4-4.8 x 1012/L
WBC 68 4.5-10.5 x 109/L
Hb 98 112-165 g/L
MCV 69 80-95 fl
MCHC 261 260-340 g/L
Reticulocyte 1.0 0.2-1.2%
Which of the following is the most likely diagnosis?
A. Sickle cell anaemia
B. B-thalassemia trait
C. Iron deficiency anaemia disease
D. Anaemia of chronic medical disease

Answer: B
A 5-day-old infant was delivered normally at home attended by a midwife. He had never been to a hospital, but now noted to have bleeding from the umbilical cord that seems increasing in severity. He has been breached since after delivery. The mother had a normal perinatal period, but was diagnosed with epilepsy 10 years ago, and has been taking phenobarbital (see lab results).
Test Result Normal Values
Bleeding time 3.8 up to 4 min
Prothrombin time 21 12-14 sec
APTT 46 30-40 sec
Which of the following is the most appropriate treatment?
A. Corticosteroids
B. Platelet transfusion
C. Fresh frozen plasma
D. Parenteral 1 mg vitamin K1

Answer: D
A 1-year-old boy is being investigated for pallor, jaundice, lethargy and delayed growth. On examination, found to be short for his age, with prominent forehead and splenomegaly (see lab results).
Test Result Normal Values
HB 40 112-165
RDW 14 11.5-14.5%
MCV 70 80-95f
Reticulocyte 2.4 0.2-1.2%
WBC 5.5 4.5-10.5 x 109/L
RBC 5.8 4.6-4.8 x 1012/L
HB electrophoresis
Hb A2 30 2-3%
Hb F 50 1-2%
Which of the following is the most likely diagnosis?
A. Iron deficiency anemia
B. B- Thalassemia
C. sickle cell disease
D. a- Thalassemia

Answer: B

A preschool child presents for his routine preschool-entry examination (see lab results).
Test Result Normal Values
HB 90 112-165
RDW 20 11.5-16%
Platelets count 490 150-400 x 109/L
MCV 68 80-95 fl
MCH 26 28-33 pg/cell
MCHC 241 260-340 g/L
Reticulocyte 1.1 0.2-1.2 %
WBC 7.5 4.5-10.5 x 109/L
RBC 5 4.6-4.8 x 1012/L
Which of the following is the most likely diagnosis?
A. Aplastic anaemia
B. Thalassemia minor
C. Iron deficiency anaemia
D. Hereditary spherocytosis

Answer: C
A2-year-old child with sickle cell disease had an upper respiratory tract infection, which was followed by shortness of breath, and pallor. On examination, looked pale, sick and had a splenomegaly (see lab results).
Blood pressure 90/60 mmHg
Heart rate 110 /min
Respiratory rate 25 min
temperature 38.6° C

Test Result Normal Values
Hb 34 112-165 g/L
WBC 3.1 4.5-10.5 x 109/L
Lymphocytes 52 20-40 %
Neutrophils 58 40-60%
Platelets count 166 150-400 x 109/L
Reticulocyte 0.1 0.2-1.2 %
ESR 25 2-10 mm/h
Which of the following is the best immediate management?
A. Splenectomy
B. Fluids and analgesia
C. Intravenous antibiotics
D. Packed red cell transfusion

Answer: D
A 53-day-old boy is referred to the Paediatric Clinic for investigation of anaemia. He was born full term and he is a known case of G6PD deficiency. On examination, both height and weight are in the 10th percentile, the rest of the examination is normal (see lab results).
Test Result Normal Values
Hb 70 112-165 g/L
Platelet count 350 150-400 x 109/L
Total bilirubin 25.1 <20.5 mol/L
Urinalysis Result Normal Values
Color Clear Clear or light yellow
Leukocytes 0 0-3 per high power field

Erythrocytes 0 0-2 per high power field
Which of the following is the best management?
A. Blood transfusion
B. Iron supplement
C. Reassurance
D. Folic acid

Answer: C
A 53-day-old boy is referred to the Paediatric Clinic for investigation of anemia. He was born at full term and he is a known case of G6PD deficiency. On examination, both height and weight are in the 10th percentile, the rest of the examination is normal (see lab results).
Test Result Normal Values
Hb 70 112-165 g/L
Platelet count 350 150-400 x 109/L
Total bilirubin 25.1 <20.5 mol/L
Urinalysis Result Normal Values
Color Clear Clear or light yellow
Leukocytes 0 0-3 per high power field
Erythrocytes 0 0-2 per high power field
Which of the following is the most likely diagnosis?
A. Haemolysis
B. Iron deficiency
C. Physiological jaundice
D. Anaemia of chronic disease

Answer: C
A 6-year-old girl presents to the Emergency Department with a history of feeling unwell and tired over the last 10 hours. On examination, she looks pale, had a palpable liver and her spleen is 6 cm below the costal margin (see lab results).
Blood pressure 93/55 mmH
Heart rate 169 /min
Respiratory rate 28 /min
Temperature 37.5° C
Test Result Normal Values
Hb 52 112-165 g/L
Platelet count 102 150-400 × 109/L
Sickle cell screen positive Nil
Which of the following is the most appropriate treatment?
A. Blood transfusion
B. Immunoglobulin
C. Clarithromycin
D. Prenisolone

Answer: A
A 6-year-old girl presents to the Emergency Department with a history of feeling unwell and tired over the last 10 hours. On examination, she looks pale, had a palpable liver and her spleen is 6 cm below the costal margin (see lab results).
Blood pressure 93/55 mmHg
Heart rate 169 /min
Respiratory rate 28 /min Temperature 3t.5° C
Test Result Normal Values
Hb 52 112-165 g/L
Platelet count 102 150-400 × 109/L

Sickle cell screen positive Nil
Which of the following is the most likely diagnosis?
A. Sequestraton crisis
B. Evans syndrome
C. Bacterial sepsis
D. Aplastc crisis

Answer: A

A 6-year-old girl presents to the Emergency Department with a history of feeling unwell and tired over the last 10 hours, she had 2 previous similar attacks. On examination, she looks pale, has a palpable liver and her spleen is 6 cm below the costal margin (see lab results).
Blood pressure 93/55 mmHg
Heart rate 169 /min
Respiratory rate 28 /min
Temperature 37.5° C
Test Result Normal Values
Hb 52 112-165 g/L
Platelet count 102 150-400 x 109/L
Sickle cell screen positive Nil
Which of the following is the most appropriate management?
A. Regular blood transfusions
B. Hydrocortisone
C. Splenectomy
D. Hydroxyurea

Answer: C
A 2-year-old boy presents to the Paediatric Clinic for a routine examination. He was born at full term with a birth weight of 3450 grams. The mother breastfeed him, for a year and since then he has been fed on whole formula milk with a balanced diet. His weight gain and development have always been normal for his age. At a routine check with one year, a haemoglobin level of 105 g/L was recorded and oral administration of ferrous sulphate drops was started. 6 months later the haemoglobin level was 103 g/L and lead to an increase of the daily dose of ferrous sulphate. Another 3 months later there was no significant rise in the level again. The mother reports having Continued giving the ferrous sulphate drops since then. The paediatrician orders a blood cell count (see lab results).
Test Result Normal Values
Hb 91 112-165 g/L
RDW 13.1 11.5-14.5 pm
WBC 10.4 4.5-10.5 x 109/L
MCV 62 80-95 fl
Reticulocyte 0.8 0.2-1.2 %
Platelet count 280 150-400 x 109/L
Which of the following is the most appropriate next step in definitive diagnosis?
A. Serum Vitamin B12 level
B. Bone marrow aspiration
C. Serum iron and ferritin level
D. Haemoglobin electrophoresis

Answer: D

A 2-year-old girt presents with low energy and failure to thrive. The child was born at 38 weeks' gestation with a birth weight of 3.3 kg. The actual body weight was 10 kg (3rd percentile). The mother reports poor appetite and little interest in food. The family follows a strict vegetarian diet. A

physical examination confirms pallor of the skin and the mucous membranes. There are cracks in the skin at the corner of the mouth and a peculiar appearance of the nails on the finger and toes (see image and lab results).
Heart rate 110/min
Respiratory rate 36 /min
Temperature 36.8° C
Oxygen saturation 95% on room air
Test Result Normal Values
Hb 90 112-165 g/L
HCT 0.29 0.32-0.42
Platelets count 470 150-400 x 109/L
MCV 78 80-95 f
MCH 25 28-33 pg/cell
Which of the following is the most appropriate treatment?
A. Transfusion of packed red blood cells
B. Erythropoiesis-stimulating agents
C. Oral ferrous sulfate
D. Parenteral iron

Answer: C
A newborn presents with jaundice 12 hours after birth. At 36 hours after birth, the jaundice became worse and blood smear was performed (see image and lab results).
Test Results Normal Values
Hb 91 112-165 g/L
Reticulocyte 7 0.2-1.2 %
Total bilirubin 2800 <205 mol/L
Which of the following is the most appropriate next step in the investigation?
A. Osmotic fragility test
B. Pyruvate kinase assay
C. Haemoglobin electrophoresis
D. Measurement of glucose-6-phosphate dehydrogenase activity

Answer: A
A 1-year-old boy is noted to have a microcytic anaemia, on routine screening. A haemoglobin electrophoresis was performed that showed and increased fraction of haemoglobin A2. Which of the following is the most likely diagnosis?
A. B-thalassemia
B. Iron deficiency
C. Sickle cell anaemia
D. Folic acid deficiency

Answer: A
A 2-year-old girt presents with diarrheal stools and weakness over the last 4 days. There is no family history of any serious illness. Physical examination confirms pallor of the skin and mucous membranes and a scleral icterus. The spleen can be palpated 2 cm below the costal margin (see image and lab results).
Heart rate 100 /min Respiratory rate 28 /min Temperature 38.6° C
Test Result Normal Values
Hb 55 112-165 g/L
WBC 15 4.5-10.5 x 109/L
Platelets count 200 150-400 x 109/L
Reticulocyte 7.5 0.2-1.2%
RBC 2 4.6-4.8 x 1012/L

Test Result
Direct Coombs test positive
Indirect Coombs test positive
Which of the following is the most likely diagnosis?
A. Spherocytosis
B. Fanconi anemia
C. Autoimmune hemolytc anemia
D. Glucose-6-phosphate dehydrogenase deficiency

Answer: C
An 18 month-old girt presents to the Paediatric Clinic with irritability and decreased appetite. The baby was bor at 38 weeks' gestation with a birth weight of 2.8 kg. She was breastfed for 9 months and then changed to a cow's milk-based formula. The infant is drinking about 1 liter of formula per day together with solid foods. The mother reports that the stool has been loose and smelly over the last 3 days. Physical examination confirms pallor of the skin and mucous membranes, and systolic heart murmur (see lab results).
Heart rate 120 /min
Respiratory rate 30 /min Temperature 37.3° C
Oxygen saturation 95% on room air
Test Result Normal Values
Hb 90 112-165 g/L
RBC 3.0 4.6-4.8 x 1012/L
Which of the following is the most likely underlying cause?
A. Gastrointestinal infection
B. Cow's milk consumption
C. Bone marrow defect
D. Low birth weight

Answer: B
A 12-month-old boy is brought to the clinic with a complaint of decreased appetite. The child was born at 36 weeks' gestation and has been fed with cow's milk formula since the age of 9 months. The body weight is 13.6 kg (> 97th percentile) and the length is 75 cm (50th percentile). Physical examination confirms pallor of the skin and buccal mucous membranes. There is also a systolic heart murmur and a palpable spleen (see image and lab results).
Test Result Normal Values
Hb 90 112-165 g/L
RBC 3.0 4.6-4.8 x 1012/L
What additional laboratory finding is most likely to be significantly lower than the normal value?
A. WBC
B. MCV
C. Platelets
D. Reticulocyte count

Answer: B
A 12-month-old boy is brought to the clinic with a complaint of decreased appetite. The child was born at 36 weeks' gestation and has been fed with cow's milk formula since the age of 9 months. The body weight is 13.6 kg (> 97th percentile) and the length is 75 cm (50th percentile). Physical examination confirms pallor of the skin and buccal mucous membranes. There is also a systolic heart murmur and a palpable spleen (see image and lab results). Test Result Normal Values
Hb 90 112-165 g/L
WBC 10 4.5-10.5 x 109/L
Platelets count 500 150-400 x 109/L

MCV 70 80-95 f
Reticulocyte 1.2 0.2%-1.2%
RBC 3.0 4.6-4.8 x 1012/L
Which of the following components need to be replaced?
A. Iron
B. Folic Acid
C. Vitamin B12
D. Erythrocytes

Answer: A
A 5-year-old boy presents with sudden and severe bone pain of his hands and feet, and on examination he appears dehydrated.
Blood pressure 105/60 mmHg
Heart rate 100 /min
Respiratory rate 22 /min
Temperature 38.8°C rectal
Which of the following is the most likely diagnosis?
A. Sickle cell disease
B. Alpha thalassemia
C. Thalassemia maior
D. Iron deficiency anaemia

Answer: A

A 2-year-old child presents with bruising and generalized petechiae, mostly on the legs. The child has always been healthy. The parents report that he had an upper respiratory infection 2 weeks before that resolved without medication (see lab results).
Test Results Normal Values
Hb 131 112-165 g/L
HCT 0.39 0.32-0.42
WBC 6.8 4.5-10.5 x 109/L
Lymphocytes 22 20-40 %
Monocytes 2 2-8 %
Neutrophils 58 40-60%
Platelets count 15 150-400 x 109/L
Which of the following is the most appropriate management?
A. Splenectomy
B. Platelet transfusion
C. Factor VIII transfusion
D. Intravenous y-globulin

Answer: D

A 3-year-ald boy presents with painless abdominal distension, he has had occasional attacks of abdominal pain, vomiting, and bloody urine. On examination, there is a smooth firm mass in the right loin, an absent iris and undescended testes (see lab results).
Urinalysis Result Normal Values
Color Clear Clear or light yellow
Appearance Turbid Clear
Specific gravity 1.025 1.001-1.030
pH 7.5 4.3-8
Leukocytes 5 0-3 per high power field
Erythrocytes 15 0-2 per high power field

Color Clear Clear or light yellow
Which of the following is the most likely diagnosis?
A. Hydronephrosis
B. Nephroblastoma
C. Urinary tract infection
D. Polycystic kidney disease

Answer:B

A 6-year-old girl presents to the Emergency Department with a history of feeling unwell and redness over the last 10 hours. On examination, she looks pale, had a palpable liver and her spleen was 6 cm below the costal margin (see lab results).
Blood pressure 93/55 mmH
Heart rate 169 /min
Respiratory rate 28 /min
Temperature 37.5° C
Test Result Normal Values
Hb 58 112-165 g/L
Platelet count 98 150-400 x 109/L
Sickle cell screen positive
Which of the following is the most useful investigation?
A. Bone marrow aspiration
B. Abdominal ultrasound
C. Antinuclear antibody
D. Reticulocyte count

Answer: D

A 2-year-old girl is seen in the clinic for pallor and tiredness. On examination, she looks healthy with normal vital signs; growth parameter and physical exam apart from conjunctival pallor (see lab results). Blood pressure 90/70 mmHg
Heart rate 97 /min
Respiratory rate 18 /min Temperature 36.6 °C
Oxygen saturation 95 %
Test Result Normal Values
RBC 4.5 4.8-7.1 x 1012/L (Newborn) 4.6-4.8 x 1012/L (Child)
Hb 87 165-195 g/L (Newborn) 112-165 g/L (Child)
HCT 0.32 0.44-0.64 (Newborn) 0.32-0.42 (Child)
MCH 25 28-33 pg/cell
MCV 74 80-95 fl
Reticulocyte 1.1 0.2-1.2 %
Platelets count 350 150-400 x 109/L
Iron 7 7.2-17.9 mol/L (Infant)
9-21.5 umol/L (Child)
Total iron binding capacity 94 17.9-71.6 mol/L Ferritin 12 20-200 pg/L
Which of the following is the most likely diagnosis?
A. Macrocytic anemia
B. Thalassemia trait
C. Sickle cell disease
D. Iron deficiency anemia

Answer: D

A 9-month-old infant is suspected to have iron deficiency anemia. CBC and iron profile ordered pending the results. which of the following indexes is expected to be high?
A. MCV
B. Ferritin
C. Hemoglobin
D. Iron binding capacity

Answer: D
A 4-month-old infant seen in the clinic with CBC result. The mother concern about the findings (see lab results).
Test Result Normal Values
Hb 9 165-195 g/L (Newborn)
112-165 g/L (Child)
MCH 25 28-33 pg/cell
MCV 77 80-95 fl
Reticulocyte 1.1 0.2-1.2 %
Platelets count 330 150-400 x 109/L
WBC 10 9.1-30.1 × 109/L (Birth)
6.1-17.5 x 109/L (1-23 months)
Which of the following is the most likely diagnosis?
A. Physiologic anemia
B. Thalassemia trait
C. Sickle cell anemia
D. Lead poisoning

Answer: A

A 4-year-old boy presented to the clinic with history of decreased appetite and activity over the past month. He lost significant weight since his last outpatient visit. Which of the following is the most common cause of cancer in such patient?
A. Leukemia
B. Lymphoma
C. Neuroblastoma
D. Retinoblastoma

Answer: A
A 4-year-old girl is brought to the clinic because her mother had felt a mass in the abdomen last night when she was bathing her. She has been otherwise well with no weight loss, fever, or bone pain. Her blood pressure is 130/80 mmH. Physical examination shows a well-thriving kid with a palpable firm mass in the right flank (see lab results).
Test Result Normal Values
Color Clear clear or light yellow
Appearance Clear clear
pH 6.5 4.3-8
Specific gravity 1020 1001-1030
Protein trace Absent
Glucose absent Absent
Ketones absent Absent
Urobilinogen absent No more than 6 mg/day
Nitrite absent Absent
Erythrocytes 40 0-2 per high power field
Hemoglobin +3 Absent
which of the following is the most likely diagnosis?

A. Leukemia
B. Lymphoma
C. Wilms tumour
D. Neuroblastoma

Answer:C
A 4-year-old girl is brought to the clinic because her mother had felt a mass in the abdomen last night when she was bathing her. She has been otherwise well with no weight loss, fever, or bone pain. Her blood pressure is 130/80 mmH. Physical examination shows a well-thriving kid with a palpable firm mass in the right flank (see lab results).
Test Result Normal Values
Color Clear clear or light yellow
pH 7.0 4.3-8
Specific gravity 1010 1001-1030
Protein Absent Absent
Glucose Absent Absent
Ketones Absent Absent
Nitrite Absent Absent
Hemoglobin +++ Absent
Leukocytes Absent 0-3 per high power field
Erythrocytes 50 0-2 per high power field
Which of the following is the most appropriate next step in her diagnostic work-up?
A. Plain abdominal radiography
B. Abdominal computed tomography
C. Abdominal magnetic resonance imaging
D. Plain abdominal radiography and abdominal ultrasonography

Answer:D

Which of the following is the most suggestive of hemolysis as a cause for neonatal hyperbilirubinemia?
A. Low reticulocyte count
B. Negative direct Coombs test result
C. Increased conjugated bilirubin level
D. Increased unconjugated bilirubin level

Answer: C

A 13-year-old boy who has palpable lymph node in his neck. One of his close relatives has been diagnosed recently with lymphoma and his family is very worry about it.
Which of the following may influence the physician to consider lymph node biopsy?
A. Associated with hepatosplenomegaly
B. Size more than 1 cm
C. Tender on palpation
D. Absence of fever

Answer: A
A 13-year-old boy who has palpable lymph node in his neck. One of his close relatives has been diagnosed recently with lymphoma and his family are very worry about it. Which of the following is most suggestive of a malignant rather than an infectious cause of lymphadenopathy?
A. Age of less than 5 years
B. Cervical Lymphadenopathy
C. Duration of less than 4 weeks

D. Supraclavicular lymphadenopathy

Answer: D
An 11-year-old girl brought to the clinic for physical exam because one of her cousins has been diagnosed with lymphoma recently. She is worried about palpable lymph node in her neck. Which of the following lymph nodes locations would be most concerning for malignancy?
A. Cervical lymphadenopathy
B. Inguinal lymphadenopathy
C. Submandibular lymphadenopathy
D. Supraciavicular lymphadenopathy

Answer: D

A 5-year-old child presented with bruising and skin rash that developed over 24 hours. She had had upper respiratory tract infection a week ago. On examination, she appeared well but had purpuric skin rash with some bruises on legs. No lymphadenopathy or hepatosplenomegaly (see lab results).
Test Result Normal Values
RBC 4.7 4.8-7.1 x 1012/L (Newborn)
4.6-4.8 x 1012/L (Child)
Hb 115 165-195 g/L (Newborn)
112-165 g/L (Child)
MCH 30 28-33 pg/cell
MCV 85 80-95 f1
Platelets count 17 150-400 x 109/L
ESR 3 2-10 mm/h
Clotting time 5 5-10 min
Bleeding time 3 up to 4 min
APT 35 30-40 sec
INR 1.1 0.9.1.2
Prothrombin time 13 12-14 sec
WBC 6 4.5-13.5 x 109/L (2-10 years)
4.5-11.1 x 109/L (11 years-older adult)
Color clear clear or light yellow
Nitrite absent Absent
Hemoglobin absent Absent
Erythrocytes 0 0-2 per high power field
Which of the following is most likely the diagnosis?
A. Wiskott-aldrich syndrome
B. Bernard-souller syndrome
C. Immune thrombocytopenic purpura
D. Thrombotic thrombocytopenic purpura

Answer: C
A 2-year-old girl brought to Primary Clinic because the mother noticed she is pale. She had history of drinking 3 glasses of cow's milk daily, very picky eater and refusing to eat meat. The mother gave history of similar presentation on 2 of her brother (see lab results).
Test Result Normal Values
RBC 3 4.8-7.1 x 1012/L (Newborn) 4.6-4.8 x 1012/L (Child)
Hb 4 165-195 g/L (Newborn) 112-165 g/L (Child)
MCH 18 28-33 pg/cell
MCV 62 80-95 f1
Reticulocyte 3 0.2-1.2 %
Platelets count 480 150-400 x 109/L Ferritin 9 20-200 g/L

Which of the following is the most likely diagnosis?
A. Sickle cell disease
B. Alpha thalassemia trait
C. Iron deficiency anemia
D. Anemia of chronic disease

Answer: A
A 2-year-old girl brought to Primary Clinic because the mother noticed she is pale. History of drinking 3 pints of cow's milk daily and is a very picky eater. The mother gave history of similar presentation on 2 of her brother (see lab results).
Test Result Normal Values
RBC 3 4.8-7.1 x 1012/L (Newborn) 4.6-4.8 x 1012/L (Child)
Hb 4 165-195 g/L (Newborn) 112-165 g/L (Child)
MCH 18 28-33 pg/cell
MCV 62 80-95 fl
Reticulocyte 3 0.2-1.2 %
Platelets count 480 150-400 x 109/L Ferritin 9 20-200 ug/L
Which of the following is the most likely diagnosis?
A. Sickle cell disease
B. Alpha thalassemia trait
C. Iron deficiency anemia
D. Anemia of chronic disease

Answer: C
A 2-years-old boy presents to the clinic with complain of sudden jaundice, fever for the last 3 days and passage of dark urine. Family history is positive for one brother who has severe neonatal jaundice required exchange transfusion, and they are originated from Mediterranean region (see lab results).
Blood pressure 110/70 mmHg Heart rate 120 /min Respiratory rate 22 /min Temperature 38.5 'C
Test Result Normal Values
RBC 4 4.8-7.1 x 1012/L (Newborn)
4.6-4.8 x 1012/L (Child)
Hb 9 165-195 g/L (Newborn)
112-165 g/L (Child)
Reticulocyte 3 0.2-1.2 %
Platelets count 300 150-400 x 109/L
WBC 10 4.5-13.5 x 109/L (2-10 years)
4.5-11.1 x 109/L (11 years-older adult)
Which of the following medications should be avoided in the management?
A. Aspirin
B. Amoxicillin
C. Ibuprofen
D. Acetaminophen

Answer: A; G6PD
Which of the following leukemia is the most common in pediatric age group?
A. Acute myeloid
B. Chronic myeloid
C. Acute lymphoblastic
D. Acute non-lymphoblastic

Answer:C

A 3-year-old boy is brought to the clinic by his mother because he was not eating well, and had distended abdomen. Recently he appeared reluctant to walk and lost weight. On examination, he looks pale, with hypertension and distended abdomen. MRI abdomen, confirmed the presence of large upper abdominal mass extending toward the midline

Which of the following is the most likely diagnosis?

A. Wilms tumor

B. Neuroblastoma

C. Hodgkin lymphoma

D. Non-Hodgkin lymphoma

Answer: B

Endocrinology

A 14-year-old boy has developed unilateral gynecomastia which makes him embarrassed. Which of the following is the most appropriate advice?
A. Reassurance that most cases resolve within a few years
B. Explaining that majority of men have this condition to a certain degree
C. Using a compression bandage at night
D. Decreased intake of soy protein

Answer: A
An 11-year-old patient presents with increased thirst, hunger and urinary frequency. The patient has had nothing to eat or drink for 3 hours before the clinic visit. examination is unremarkable (see lab result).
Blood pressure 110/70 mmH
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 36.6° C
Test Result Normal Value
Random Glucose 25.5 3.9-5.5 mol/L
Which of the following is the most likely cause?
A. Elevated renin
B. Elevated insulin
C. Decreased insulin
D. Absence of glucagon

Answer: C
A 13-year-old girl comes to the clinic with her mother. The mother is anxious about her small stature. The girl had been born at full term with a birth weight of 3.5 kg. The subsequent physical development has been normal. There is no family history of any significant health problems. There is no sign of breast development or pubic hair growth.
Height 152 cm (25th percentile)
Weighs 37 kg (10th percentile)
Which of the following is the most likely cause?
A. Chromosomal
B. Constitutional
C. Psychological
D. Hormonal

Answer: B
A 15-year-old boy is diagnosed with diabetic ketoacidosis.
Which of the following types of insulin is used in acute management?
A. NPH
B. Aspart
C. Regular
D. Ultralente

Answer: C
A 12-year-old girl presents for a routine physical exam. She is found to have a thyroid nodule about 4 cm in diameter. There is no pain, hoarseness or local symptoms. Serum TSH is normal, and fine needle aspiration was inadequate.
Which of the following is the most appropriate next step in management?
A. Thyroid scan
B. Clinical follow up

C. Total thyroidectomy
D. Repeat fine needle aspiration

Answer: D
A 12-year-old girl is diagnosed with type 1 diabetes 3 years earlier presents for a routine examination. She is on a basal bolus regime which she manages well. The total daily insulin dose is 32 units. Physical examination shows no abnormal findings. There are no signs of puberty (see lab result).
Weight 40 kg (<50th percentile)
Height 156 cm (t5th percentile)
Test Result Normal Value
Glucose 6.0 3.5-5.5 mmol/L
Which of the following screening tests should be performed annually?
A. Ophthalmoscopy
B. Test for celiac disease
C. Growth hormone test
D. Ultrasound of kidneys

Answer: A

A 3-week-old girl presents with repeated vomiting over the last 4 days. The child was born full- term with a weight of 3.3 kg and weighs now 3 kg. There is no family history of any significant illness. Physical examination confirms reduced skin turgor and weak muscle tone. The external genitalia showed an abnormally enlarged clitoris. The pediatrician initiates intravenous treatment with isotonic sodium chloride solution (see lab results).
Test Result Normal Values
pH 7.25 7.36-t.45
Sodium 120 134-146 mmol/L
Potassium 5.8 3.5-5.1 mmol/L
Glucose 2.9 3.3 -5.5 mmol/L
Which of the following intravenous treatment is most appropriate?
A. Cefotaxime
B. Vasopressin
C. Sodium bicarbonate
D. Hydrocortisone

Answer: D
A 10-year-old girl is admitted to the Paediatric ICU after a car accident with extensive head trauma. 12 hours after being admitted she develops increased urine output, 100 ml/h (see lab results).
Temperature 36.9°C
Heart rate 102/min
Blood pressure 120/70 mmHg
Respiratory rate 18/min
Test Result Normal Value
Serum osmolality 326 280-300 mmol/kg
24 Urine Analysis Result Normal Value
Osmolality 90 280-910 mOsm/Kg H20
Which of the following is the most likely diagnosis?
A. Central diabetes insipidus
B. Nephrogenic diabetes insipidus
C. Excessive intravenous fluid administration
D. Syndrome of Inappropriate Antdiuretc Hormone Secretion

Answer: A
A 2-week-old baby is brought to the clinic by the mother. She is concerned that her baby has little appetite, only passes stool once in 48 h and is always very sleepy and sluggish. On physical examination, the child has a distended abdomen with an umbilical hernia and mildly jaundiced.
Which of the following is the most appropriate next step in investigation?
A. Neonatal metabolic screen
B. TSH and T4 of the newborn
C. TSH and T4 levels of the mother
D. Thyroid autoantibodies in the mother

Answer: A
A 16-year-old adolescent girl complains of galactorrhea for 3 months. Her menarche was at age 13 years and her menstrual cycles have been regular until two months ago. On examination, she is at Tanner stage 2.
Which of the following is the most appropriate hormone to measure first?
A. Prolactin
B. Oestrogen
C. Progesterone
D. Luteinizing hormone

Answer: A
In which of the following conditions would you expect a bone age greater than the chronologic age?
A. Hypothyroidism
B. Chronic renal failure
C. Familial short stature
D. Congenital adrenal hyperplasia

Answer: D
A 15-year-old girl presents to the clinic with her mother. The mother is concerned because she has never menstruated. On examination, the child has a short stature, short neck with a low hairline and hypertension. Her parents' stature is below average.
Which of the following is the most likely diagnosis?
A. Hypothyroidism
B. Hypopituitarism
C. Turner syndrome
D. Familial short stature

Answer: C
A 12-year-old obese boy is being observed in the Paediatric Outpatient Clinic as he has short stature, purple striae over his abdomen and hips, and hypertension.
Which of the following is the most important initial investigation?
A. MRI of the brain
B. Adrenal scintigraphy
C. MRI of the adrenal glands
D. Morning and evening cortisol levels

Answer: D
Which of the following is usually the first sign of puberty in boys?
A. Acne
B. Pubic hair
C. Penile enlargement
D. Testicular enlargement

Answer: D
A 2-week-old boy is brought tp the Paediatrics Clinic for a routine exam. The child was born at 41 weeks' gestation and has been exclusively breastfed. Physical examination confirms large fontanels and an umbilical hernia. The muscle tone is reduced. The skin feels cool and dry and there is generalized jaundice. The infant's tongue appeared large for his mouth.
Which of the following disorders is most likely to be found?
A. Congenital adrenal hyperplasia
B. Congenital hypothyroidism
C. Biotnidase deficiency
D. Galactosemia

Answer: B
A 7-year-old girl is brought to the Paediatric Outpatient Clinic because she is already showing breast development, appearance of pubic hair and acne. She is otherwise asymptomatic. Which of the following is the most likely diagnosis?
A. Ovarian tumour
B. Central precocious puberty
C. Gonadotropin-releasing tumour
D. Lesion in central nervous system

Answer: B
A 3-year-old girl is brought to the clinic with weight loss of 1 kg over the last 2 weeks in spite of increased appetite and thirst. The parents report that the child has been tired and moody lately. The frequency of wet diapers had increased significantly. Physical examination confirms dry mucous membranes and a generally reduced skin tonus. The genito-anal zone is reddened and inflamed (see lab result).
Blood pressure 90/60 mmHg Heart rate 70 /min Respiratory rate 60 /min Temperature 36.8° C
Test Result Normal Values
Glucose 25 3.5-5.5 mmol/L
Which of the following diagnostic tests is most important?
A. Glycosylated hemoglobin
B. Full blood cell count
C. HLA DR3 antigen
D. Urine dip stck

Answer: D

A 12-year-old boy with diabetes mellitus type 1 has been having recurrent attacks of hypoglycaemia for the last 2 weeks. They occur at different times of the day. He is compliant with his insulin dose and follows his dietary advice since diagnosis 2 months ago. On physical examination, he is normal.
Blood pressure 125/85 mmHg Heart rate 85 /min Respiratory rate 16 /min Temperature 37.2° C
Which of the following is the most likely explanaton?
A. Brittle diabetes
B. Dawn phenomenon
C. Honey-moon period
D. Somogyi phenomenon

Answer: C
A 14-year-old girl has been diagnosed with Turner's syndrome. Which of the following disorders is most likely to have?
A. Hypothyroidism
B. Addison's disease
C. Diabetes mellitus

D. Hyperprolactinemia

Answer: A
A 5-year-old girl started growing pubic hair and an adult body odour 8 months ago. examination shows her height is at the 75th percentile, weight is at the 95th percentile, and there was no acne or clitoromegaly.
Which of the following is the most important initial screening study?
A. Dehydroepiandrosterone sulphate
B. Follicle stimulating hormone
C. 17-hydroxyprogesterone
D. Testosterone

Answer: B
Which of the following hormonal assay is routinely recommended for newborns?
A. Prolactin level
B. Testosterone level
C. Insulin growth factor
D. Thyroid function test

Answer: D
A 9-year-old boy is brought to the clinic. The mother is concerned that her son is not eating well since early childhood. He is being teased for being small by his peers in the school. The mother has small body built and is short. In a systematic review, he is asymptomatic. Physical examination is normal (see lab results and report).
Blood pressure 130/85 mmH Heart rate 75/min Respiratory rate 16 /min Temperature 37.2° C
Weight 25 kg
Height 120 cm
Test Result Normal Values
Thyroid stimulating hormone 3.2 0.4 - 5.0 U/L
Follicle stimulating hormone 1.4 1.0-1.6 IU/L
Luteinizing hormone 0.1 0.05-0.5 lU/L
Growth hormone 4.9 2-10 pg/L
Free Testosterone 38.2 10.4-41.6 mol/L
Insulin like growth factor - 1 18 67-405 ng/ml
X-ray of left wrist and hand:
The bone age is as a 7-year-old child.
Which of the following is the most appropriate management?
A. High caloric diet
B. Growth hormone
C. Octreotide acetate
D. Behavioural therapy

Answer: B
A 9-year-old boy is brought to the clinic. The mother is concerned that her child is short. She appears to be short. in a systemic review, he is asymptomatic. Physical examination is normal (see lab results and report).
Blood pressure 130/85 mmhg
Heart rate 75/min
Respiratory rate 16 /min
Temperature 37.2° C
Weight 25 kg
Height 120 cm
Test Result Normal Values

Thyroid stimulating hormone 3.2 0.4 - 5.0 U/L
Follicle stimulating hormone 1.4 1.0-1.6 IU/L
Luteinizing hormone 0.1 0.05-0.5 IU/L
Growth hormone 4.9 2-10 pg/L
Free Testosterone 38.2 10.4-41.6 mol/L
Insulin like growth factor - 1 18 67-405 ng/ml
X-ray of left wrist and hand:
The bone age is as a 6-year-old child.
Which of the following is the most likely diagnosis?
A. Failure to thrive
B. Familial short stature
C. Growth hormone deficiency

Answer: C

An 8-year-old boy presents for follow up. He was diagnosed with diabetes mellitus type 1 a month ago. He is compliant with his insulin dose. Physical examination is normal.
Blood pressure 115/75 mmH
Heart rate 88 /min
Respiratory rate 16 /min
Temperature 37.2° C

When should an ophthalmology evaluation performed?
A. Immediately
B. After 1 year
C. After 3 years
D. After 5 years

Answer: D

A 9-year-old boy is brought for a check-up. Parents are worried because he is short. On examination, his height is below the 5th percentile for his age but increasing steadily. The parents ask if he will always be short in the future.
What is the most important factor to answer the question?
A. Sibling heights
B. Parental height
C. Weight for height and age
D. Upper to lower segment rato

Answer: B

A 15-year-old boy is diagnosed with diabetes mellitus type 1 and prescribed insulin. He and his family were educated about the use and recognition of adverse effects of insulin.
Which side effect would be most important to receive education on?
A. Hypoglycemia
B. Lipodystrophy
C. Hyperlipidemia
D. Hypersensitivity

Answer: A

An 8-year-old is brought to the clinic by his parents. The patient is overweight and looks much older than other children in the same age group. Weight and height are greater than the 95th percentile for the child's age, and body mass index is 30 Kg/m2. The rest of the examination is normal.
Which of the following is the most appropriate management?
A. Observation and revaluation in 12 months
B. Referral for surgical intervention
C. Prescription obesity medication
D. Lifestyle modification

Answer: D

A 2-year-old boy presents to the Pediatric Outpatient Clinic due to muscular weakness and bone pains. His legs are curved outwards. The child belongs to a strict vegetarian family. He has not been given vitamin supplements and is minimally exposed to the sun.
Which of the following vitamins should be prescribed?
A. A
B. D
C. E
D. K

Answer: B

A 12-year-old boy presents to the clinic with his mother. She is concerned that he is the shortest boy in the class. His mother's height is 155 cm. and father's height is 178 cm. The boy's height
Is 145 cm (between 5th - 10th percentile) and his bone age is 10 years.
What will be the patent's target height in cm?
A. 135-150
B. 151-165
C. 166-180
D. 181-195

Answer: C
A 19-year-old girl presents with a short stature, wide-spaced nipples, and primary amenorrhea. What is the most likely karyotype?
A. 45X
B. 46XX
C. 47XXX
D. 47XX+18

Answer: A
Which of the following findings would support a diagnosis of Turner's syndrome?
A. A palpable liver 2 cm below costal margin
B. Redundant skin folds at the nape of neck
C. Hypertonia and ankle clonus
D. Single palmer creases

Answer: B
A 13-year-old girl presents to the clinic for an assessment of her short stature. On examination, the patent has a height and weight below the 5th percentile, a webbed neck, lack of breast bud development, a high-arched palate, and a low-set posterior hairline.
Which of the following is the most likely diagnosis?
A. Trisomy 21
B. Turner's syndrome

C. Fragile X syndrome
D.Noonan's syndrome

Answer: B

A 7-year-old girl presents to the Paediatric Clinic with her mother, who is worried that her daughter has developed growth of pubic hair. She has had a normal physical development. Her school performance is acceptable and she is socially well integrated with her peers. Physical examination reveals pubic hair on the mons and the labia majora, but there is no axillary or facial hair. The clitoris is of normal size for that age. There is no sign of masculinization and no breast development. examination of the heart, lungs and abdomen shows normal results.
Blood pressure 95/55 mmHg
Heart rate 80 /min
Respiratory rate 26 /min
Temperature 36.8° C
Height 90 cm (75th percentile)
Weight 23 kg (50th percentile)
Which of the following is the most likely diagnosis?
A. Congenital adrenal hyperplasia
B. Premature adrenarche
C. Turner syndrome
D. Normal puberty

Answer: B
A 14-year-old girl presents to the Paediatric Clinic with her mother, who is worried because her daughter has not started menstruating yet. The girl has always been short of stature and there has been no pubertal growth spurt. Physical examination confirms that the girl has low hairline, prominent and low-set ears and a webbed, short neck. The lower legs are X-shaped and the chest is broad. There is no sign of breast development or other signs of sexual maturation. The mother reports that the adolescent has difficulties focusing on classwork and remembering what she has learned.
Height 150 cm (5th percentile)
Weighs 65 kg (90th percentile)
Which of the following is the most likely diagnosis?
A. Constitutional short stature
B. Endocrine short stature
C. Normal growth stature
D. Syndromic short stature

Answer: D
A 13-year-old twins, one boy and one girl present with their parent to the clinic. The parent is concerned because one of the children appears to have entered puberty while the other has not.
Which of the following should be explained about girls' puberty compared with boys?
A. 2-3 years earlier
B. 6-12 months earlier
C. 1 to 2 years after boys
D. Almost same time as boys

Answer: A

An 11-month-old child presents with failure to thrive, frequent large volumes of urine, excessive thirst, and 3 episodes of dehydration not associated with vomiting or diarrhoea. Several boys in his family showed similar symptoms.
Which of the following is the most likely diagnosis?

A. Diabetes mellitus
B. Diabetes insipidus
C. Water intoxication
D. Nephrotic syndrome

Answer: B

Which of the following criteria is utilized to diagnose hypertension in a 12-year-old girl?
A. Blood pressure > 120/80 mmH
B. Blood pressure > 140/90 mmHg
C. >90th percentile for the age and sex
D. >95th percentile for the age and sex

Answer: D

A 3-year-old girl presents to the screening clinic found to have central obesity and short stature. On examination, she has moon face with hump on her back.
Blood pressure 130/70 mmHg
Heart rate 76 /min
Respiratory rate 18 /min
Temperature 36.6 °C
Which of the following has highest diagnostic value?
A. Renal ultrasound
B. Height-weight chart
C. Thyroid function test
D. ACTH stimulation test

Answer: D
A 5-year-old child managed in Pediatrics Intensive Care Unit for diabetic ketoacidosis with insulin infusion and fluid correction (see lab result).
Test Result Normal Values
Sodium 138 134-146 mmol/L
Potassium 2.8 3.5-5.1 mmol/L
Chloride 101 97-108 mmol/L
Bicarbonate 22 21-28 mmol/L
Which of the following is the most likely explanation of low potassium?
A. Vomiting
B. Insulin therapy
C. Diuretics therapy
D. Bicarbonate therapy

Answer: B
A 4-month-old infant is brought to Emergency Room with complain of excessive sweating, pallor and altered level of conscious. The mother said he is not eating normally for the last 2 days and has recurrent vomiting (see lab results).
Test Result Normal Values
Blood urea nitrogen 2 0.7-6.7 mmol/L. (Newborn) 1.8 to 6.4 mmol/L (Child)
Creatinine 34 27-88 mol/L (Newborn)
27-62 pmol/L (Child)
Glucose 1.8 2.8-5 mmol/L (Newborn)
3.3 -5.5 mmol/L (Child)
Which of the following is the best initial management?
A. Intravenous infusion of glucose 2 ml/kg of 10% dextrose

B. Intravenous infusion of glucose 2 mi/kg of 12.5% dextrose
C. Intravenous infusion of glucose 4 ml/kg of 12.5% dextrose
D. Intravenous infusion of glucose 10 ml/kg of 10% dextrose

Answer: A
A 2-month-old infant presented to the clinic with history of prolonged jaundice, constipation, large tongue and mottled dry skin. There is a positive family history of autoimmune disease (see lab results).
Test Result Normal Values
ACTH 10 2-11 pmol/L
Insulin fasting 15 10-104 pmol/L
Thyroid stimulating hormone 22 1-25m IU/L (1-4 days)
1.7-9 mlU/L (2-20 weeks)
0.7-6.4 mlU/L (21 weeks-14 years)
Thyroxine T4 (serum) 55 125-215 mol/L (1-3 days)
100-200 mol/L (1-12 months)
94-193 nmol/L (1-5 years)
80-150 mol/L (5-15 years)
Triodothyronine T3 0.6 1.54-11.4 mol/L (1-3 days)
1.6-4 nmol/L (1-12 months)
1.2-3.2 mol/L (1-5 years)
1.2-3.2 mol/L (5-15 years)
Which of the following is the most appropriate step in management?
A. Oral thyroxine for 1 month
B. Life-long thyroxine replacement
C. Repeat the blood test after 1 month
D. Prednisolone replacement for 6 weeks

Answer:B
A 9-year-old boy presented with history of recurrent vomiting, lethargy, brown pigmentation of gum and skin crease with growth failure (see lab results).
Test Result Normal Values
Sodium 130 134-146 mmol/L
Potassium 6 3.5-5.1 mmol/L
Calcium total 3 2.2-2.7 mmol/L
Blood urea nitrogen 11 0.7-6.7 mmol/L (Newborn)
1.8 to 6.4 mmol/L (Child)
Creatinine 80 27-88 mol/L (Newborn)
27-62 mol/L (Child)
Prolactin 800 < 652 pmol/L (Male)
< 870 pmol/L (Female)
Cortisol 8 a,m. 66 138-635 mol/L
which of the following is the best management option?
A. Daily hydrocotrisone
B. Daily thyroxine
C. Daily vitamin D
D. Daily vitamin B6

Answer: A
An 8-year-old child presented to Emergency Room with constipation, anorexia, lethargy and behavioural abnormality. Bony erosion of the phalanges was seen on wrist radiography (see lab results).
Test Result Normal Values

Blood urea nitrogen 3.2 0.7-6.7 mmol/L. (Newborn) 1.8 to 6.4 mmol/L. (Child)
Creatinine 45 27-88 umol/L. (Newborn) 27-62 umol/L (Child)
Glucose 4 2.8-5 mmol/L (Newborn) 3.3 -5.5 mmol/L (Child)
Alkaline phosphatase 200 150-420 U/L (Infant) 100-320 U/L (Child)
Parathyroid hormone (Intact PTH levels) 6.2 1.1-5.3 pmol/L
Parathyroid hormone (N-terminal fraction) 3.1 0.8-2.5 pmol/L
Parathyroid hormone (C-terminal fraction) 11 0-35.8 pmol/L
Calcium total 5.3 2.2-2.7 mmol/L
Phosphate 1.2 1.3-2.3 mol/L
Magnesium 0.9 0.7-1.0 mol/L
Which of the following is the best management approach?
A. Rehydration and statins
B. Diuretics and rehydration
C. Bisphosphonates, diuretics and statins
D. Rehydration, diuretics and bisphosphonates

Answer: D
Which of the following is the most appropriate approach to manage nutritional rickets?
A. Daily administration of vitamin D1
B. Daily administration of vitamin D2
C. Daily administration of vitamin D3
D. Daily administration of vitamin D4

Answer: C

An 8-year-old boy being evaluated for short stature that includes detailed history and careful examination. Which of the following is the most important bedside test of highest diagnostic value?
A. Dipstick urinalysis
B. Portable spirometry
C. Arm circumference
D. Mid-parental target height

Answer: D
A 13-year-old girl screened for sport participation found to have delayed puberty and short stature. On examination, she was found to have webbed neck and wide shield-like chest. Which of the following is the most likely diagnosis?
A. Noonan syndrome
B. Turner syndrome
C. Growth hormone deficiency
D. Constitutional short stature

Answer: B
A 13-year-old girl screened for delayed puberty and short stature. On examination, she found to have webbed neck and wide shield-like chest. Which of the following tests has the highest diagnostics value?
A. Pelvic ultrasound
B. Bone age imaging
C. Growth hormone level
D. Chromosomal analysis

Answer: D
A 3-week-old boy infant presents to Emergency Room with history of vomiting and lethargy. He is hypoactive with mottled skin and no focus of infection. Perinatal history was unremarkable

(see lab results).
Blood pressure 50/30 mmHg
Heart rate 180 /min Respiratory rate 56 /min Temperature 36.6 °C Oxygen saturation 95 %
Test Result Normal Values
Sodium 122 134-146 mol/L
Potassium 6.2 3.5-5.1 mmol/L
Chloride 100 97-108 mmol/L
Bicarbonate 21 21-28 mmol/L
Calcium total 2.4 2.2-2.7 mmol/L
Calcium ionised 1.5 1.2-1.3 mmol/L
Blood urea nitrogen 3 0.7-6.7 mmol/L (Newborn) 1.8 to 6.4 mmol/L (Child)
Creatinine 44 27-88 mol/L (Newborn) 27-62 umol/L (Child)
Glucose 1.2 2.8-5 mmol/L (Newborn) 3.3 -5.5 mmol/L (Child)
Which of the following is the most likely definitive treatment?
A. Antibiotics
B. Fluid and dextose
C. Dextrose and intralipid
D. Cortisone and hormonal therapy

Answer: D
A 9-year-old child presents to Emergency Department with hypovolemic shock. Parents gave history of recent brain surgery and the child has polyurea and increased thirst since then (see lab results).
Blood pressure 70/40 mmg
Heart rate 136 /min Respiratory rate 18 /min Temperature 36.6 °C
Oxygen saturation 95 %
Test Result Normal Values ABG HCO3- 20 13-22 mmol/L ABG PCO2 35 27-40 mmHg pH 7.37 7.36-7.45
ABG PO2 88 8-24 mmHg (Newborn) 55-80 mmg (Child)
Base access -6 -2 to 2 mmol/L
Sodium 177 134-146 mol/L Potassium 4 3.5-5.1 mol/L
Chloride 140 97-108 mmol/L
Bicarbonate 20 21-28 mmol/L
Blood urea nitrogen 8 0.7-6.7 mmol/L (Newborn) 1.8 to 6.4 mmol/L (Child)
Creatinine 60 27-88 mol/L. (Newborn)
27-62 mol/L. (Child)
Glucose 4.5 2.8-5 mmol/L (Newborn)
3.3 -5.5 mmol/L (Child)
Color colorless clear or light yellow
Appearance Clear clear
Specific gravity 1003 1001-1030
Ketones absent Absent
Which of the following is the most likely diagnosis?
A. Diabetic ketoacidosis
B. Diabetes insipidus
C. Renal tubelur acidosis
D. Syndrome of inappropriate antidiuretic hormone secretion

Answer: B
A 9-year-old child presents to Emergency Department with hypovolemic shock. Parents gave history of recent brain surgery and child has polyurea and increased thirst since then (see lab results).
Blood pressure 70/50 mmH
Heart rate 122 /min Respiratory rate 18 /min Temperature 36.6 °C
Oxygen saturation 95 %

Test Result Normal Values ABG HCO3- 22 13-22 mmol/L ABG PCO2 36 27-40 mmHg pH 7.37 7.36-7.45
Sodium 167 134-146 mmol/L
Potassium 4.5 3.5-5.1 mmol/L
Chloride 122 97-108 mmol/L Bicarbonate 22 21-28 mmol/L
Color colorless clear or light yellow
Appearance Clear clear
Specific gravity 1004 1001-1030
Ketones absent Absent
Which of the following is the most appropriate treatment?
A. Desmopressin replacement
B. Insulin replacement regime
C. Free water replacement
D. Fluid and caloric control

Answer: A

A 7-year-old presents to Emergency Department with vomiting and abdominal pain for 1 day. The child has had history of polydipsia, polyuria and weight loss for the last 3 weeks. Examination reveals dehydration and hypoactivity (see lab results).
Blood pressure 90/50 mmH
Heart rate 110 /min Respiratory rate 24 /min Temperature 38.6 °C Oxygen saturation 95 % Weight 23 kg
Test Result Normal Values ABG HCO3- 5 13-22 mmol/L ABG PCO2 22 27-40 mmHg pH 7.12 7.36-7.45
ABG PO2 100 8-24 mmHg (Newborn) 55-80 mmHg (Child)
Base access -17 -2 to 2 mmol/L
Color light yellow clear or light yellow Appearance clear clear
pH 6 4.3-8
Specific gravity 1025 1001-1030
Protein +1 Absent
Glucose +4 Absent
Ketones +4 Absent
Nitrite absent Absent
Leukocytes 3 0-3 per high power field
Which of the following is the most likely diagnosis?
A. Gastritis
B. Food poisoning
C. Diabetic ketoacidosis
D. Urinary tract infection

Answer: C

Which of the following considered a complication of obesity in childhood?
A. Primary enuresis
B. Hypertension
C. Type 1 diabetes
D. Hyperventilation syndrome

Answer: B

Which of the following phases contribute the most for the final height of the child?
A. Fetal phase
B. Infantile phase

C. Childhood phase
D. Pubertal phase

Answer: C
A child presents to the clinic for pubertal assessment. He had pubic hair filling out towards adult distribution, and darkening of scrotal skin. Which stage of puberty he reached according to Tanner staging?
A. I
B. III
C. IV
D. V

Answer: C
An 18-month-old girl presents with asymmetrical breast enlargement. There was no pubic hair growth or body odor and her height was in mid parental range. Her bone age was mildly advanced. Which of the following is the best next step in management?
A. ACTH test
B. MRI brain
C. CT chest
D. Pelvic ultrasound

Answer: D
A 2-year-old girl is brought by her mother with breast enlargement. There was no pubic hair growth or body dour and her height was in mid parental range. Which of the following is the most likely diagnosis?
A. Precocious puberty
B. Premature pubarche
C. Central precocious puberty
D. Premature breast development

Answer: D
A 24-hour-old neonate with ambiguous genitalia. 17-hydroxyprogesterone was abnormally high (see lab results).
Test Result Normal Values
Sodium 126 134-146 mmol/L
Potassium 7.1 3.5-5.1 mmol/L
Bicarbonate 15 21-28 mmol/L
Glucose 2.6 2.8-5 mmol/L (Newborn) 3.3 -5.5 mmol/L (Child)
which of the following is the correct management for this case?
A. Hydrocortisone and saline bolus
B. Saline bolus and glucose infusion
C. Antibiotics, hydrocortisone and glucose infusion
D. Hydrocortisone, saline bolus and glucose infusion

Answer: D
A 24-hour-old neonate with ambiguous genitalia. 17-hydroxyprogesterone was abnormally high (see lab resuits).
Test Result Normal Values
Sodium 127 134-146 mmol/L
Potassium 6.8 3.5-5.1 mmol/L Bicarbonate 15 21-28 mmol/L Glucose 2.5 2.8-5 mmol/L. (Newborn) 3.3 -5.5 mmol/L (Child)
Which of the following is the inheritance of this disease?
A. Multifactorial

B. X-linked dominant
C. Autosomal dominant
D. Autosomal recessive

Answer: D
A 24-hour-old neonate with ambiguous genitalia. 17-hydroxyprogesterone was abnormally high (see lab results).
Test Result Normal Values
Sodium 128 134-146 mmol/L
Potassium 6.8 3.5-5. 1 mmol/L Bicarbonate 16 21-28 mmol/L Glucose 2.5 2.8-5 mmol/L (Newborn) 3.3 -5.5 mmol/L (Child)
Which of the following is the most likely diagnosis?
A. 21-hydroxylase deficiency
B. 17-hydroxylase deficiency
C. 11-beta hydroxylase deficiency
D. 36-hydroxysteroid dehydrogenase deficiency

Answer:A

Nephrology

A 3-day-old infant is brought to the Paediatrics Clinic with refusal to feed, vomiting and inactivity. The infant was born at term with no perinatal problems. The infant's urine smells like burned sugar.
Which of the following is the most likely diagnosis?
A. Maple syrup urine disease
B. Urinary tract infecton
C. Urea cycle disorder
D. Phenylketonuria

Answer: A

A 4-year-old boy with nephritic syndrome presents with vomiting, abdominal pain and diarrhoea. On examination, his abdomen is distended and there is an evidence of generalised and rebound tenderness.
Which of the following is the most likely diagnosis?
A. Peritonitis
B. Gastritis
C. Pancreatitis
D. Appendicitis

Answer: A

Which of the following investigations is the most important to perform on a child with enuresis?
A. Urinalysis
B. Urine culture
C. Complete blood count
D. Intravenous pyelography

Answer: A
A 6-year-old child has asymptomatic haematuria. There is no significant past medical history and the physical examination is unremarkable (see lab result).
Urinalysis Result Normal Value
Erythrocytes 15 0-2 per HPF
Which of the following should the initial evaluation include?
A. Repeat urinalysis
B. Urine cytology
C. Renal biopsy
D. Cystoscopy

Answer: A
A 6-year-old boy complains of passing Coca cola-coloured urine for the last week. On examination, he had a congested throat, and anterior cervical lymphadenopathy (see lab results).
Blood pressure 155/100 mmHg
Heart rate 125 /min
Respiratory rate 20/min
Temperature 38.5° C

Urinalysis Result Normal Values
Colour Brown Clear or light yellow

Appearance Turbid Clear
Leukocytes 3-5 0-3 per high power field
Erythrocytes 20-30 0-2 per high power field
Which of the following is the most likely diagnosis?
A. Acute cysts
B. IgA nephropathy
C. Alports syndrome
D. Acute glomerulonephritis

Answer: D
A 7-year-old boy complains of passing tea-coloured urine for the last 3 days. On examination, there is a congested throat, with no other abnormal findings (see lab results).
Blood pressure 140/98 mmHg
Heart rate 120 /min
Respiratory rate 20/min
Temperature 37.5° C
Urinalysis Result Normal Values
Colour Brown tea Clear or light yellow
Appearance Turbid Clear
Leukocytes 3-5 0-3 per high power field
Erythrocytes 20-30 0-2 per high power field
Which of the following is the most appropriate investigation?
A. Urine culture and sensitivity
B. Abdominal ultrasound
C. Plain abdomen X-ray

Answer:
A 3-year-old girl is brought to the Emergency Department with fever and dark urine. Her parents state that she had been constipated. On examination, she looks unwell and irritable with abdominal tenderness, but no palpable liver or spleen (see lab results).
Blood pressure 125/86 mmHg
Heart rate 125 min
Respiratory rate 29 /min
Temperature 39.9°C
Test Result Normal Values
WBC 16.5 4.5-10.5 x 109/L
Hb 81 112-160 g/L
Platelets count 325 150-400 x 109/L Urinalysis
Test Result Normal Values
Leukocytes 16 0-3 per high power field
Erythrocytes 18 0-2 per high power field
Protein ++ Absent
Which of the foilowing is the most appropriate management?
A. Frusemide
B. Ceftrixone
C. Prednisone
D. Salt free diet.

Answer: B
A 3-year-old girl is brought to the Emergency Department with fever and dark urine. Her parents state that she had been constipated. On examination, she looks unwell and irritable with abdominal tenderness, but no palpable liver or spleen (see lab results).
Blood pressure 125/86 mmHg

Heart rate 125 /min
Respiratory rate 29/min Temperature 39.9° C
Hb 81 112-160 g/L
Platelets count 325 150-400 x 109/L Urinalysis
Test Result Normal Values
Leukocytes 16 0-3 per high power field
Erythrocytes 18 0-2 per high power field
Protein ++ Absent
Which of the following is the most likely diagnosis?
A. Hypertensive nephropathy
B. Acute glomerulonephritis
C. Acute pyelonephritis
D. Acute cystitis

Answer: C
A 2-year-old boy presents to the Outpatient Clinic with a 2-week history of facial swelling, poor dietary Intake and reduced activity with positive sacral oedema (see lab results).
Blood pressure 95/65 mmHg
Heart rate 95 /min
Respiratory rate 28 /min
Temperature 36.8° C
Test Result Normal Values
Hb 94 112-160 g/L
Alanine aminotransferase 52 10-35 U/L
Albumin 25 36-52 g/L
Total Cholesterol 7.0 <5.2 mmol/L
Which of the following is the most likely diagnosis?
A. Congestive heart failure
B. Chronic liver disease
C. Nephrotic syndrome
D. Protein malnutrition

Answer: C
A 5-year-old child presents with features of glomerulonephritis. A week later, she developed haemoptysis.
Which of the following is the most likely diagnosis?
A. Rapidly progressive glomeruionephrits
B. Henoch Schönlein purpura
C. Goodpastures syndrome
D. Lupus nephrits

Answer: C
An 8-year-old boy presents to the Paediatric Clinic with weakness, headache, nausea and anorexia that began 1 week before. Physical examination shows pitting edema, grade 2+ on the edema scale, over the eyelids and ankles. There is pallor of the skin, mucous membranes and nail beds. A 24-hour urine collection reveals a urine production of 20 ml per hour while the boy's bodyweight is 27 kg. the urine appears dark brown.
Blood pressure 140/100 mm
Heart rate 120 /min
Respiratory rate 36 /min
Temperature 38.6° C
What clinical history would be most important for the diagnosis?
A. Diagnosed sickle cell disease

B. Recurrent urinary tract infections
C. Abdominal trauma 1 week before
D. Impetigo contagiosa 1 month before

Answer: D
A 7-year-old child presents to the Medical Outpatients with a parent who reports edema over the face. examination of urine shows selective proteinuria (3.5 g/day).
Which of the following is the most likely diagnosis?
A. Kwashiorkor
B. Nephritic syndrome
C. Nephrotic syndrome
D. Protein losing diarrhea

Answer: C
A 14-year-old presents with photosensitivity, a malar rash, joint pains and purpuric rashes on both legs. Urine examination shows red cells.
Which of the following is the most likely diagnosis?
A. Idiopathic thrombocytopenic purpura
B. Rheumatoid arthritis
C. Aplastic anaemia
D. Lupus nephritis

Answer: D
A 10-year-old child presents with a high-grade fever, generalized body swelling, decreased urine output, and dark-coloured urine.
Which of the following is the most appropriate first investigation?
A. Abdominal ultrasound
B. Renal function tests
C. Urinary analysis
D. Kidney biopsy

Answer: C
A 7-year-old child presents with a history of fever and pain in the lower abdomen for the past 3 days. The parent reports that the child's urine has been foul smelling for the past 2 days (see report).
Urine culture:
Gram Negative bacteria, > 100,000 cells/mi.
What is the most likely organism?
A. Staphylococcus saprophytcus
B. Klebsiella pneumonia
C. Proteus mirabilis
D. Escherichia coli

Answer: D

Which of the following methods of collection of urine for culture is most sensitive and reliable for establishing the diagnosis of a urinary tract infection?
A. Midstream urine
B. Catheter collection
C. Suprapubic aspiration
D. Bag applied to genitals

Answer: C
What would be the best treatment of an uncomplicated cystitis in a 5-year-old girl?

A. Oral amoxicillin
B. Oral sodium bicarbonate
C. Intravenous ciprofloxacin
D. Intramuscular ceftriaxone

Answer: A
A 9-year-old boy presents with abdominal pain and swelling of his right ankle. He had had an upper respiratory tract infection 3 weeks earlier. On examination, he had multiple maculopapular lesions (see image and lab result).
Test Result Normal Value
Erythrocytes 15 0-2 per high power field
Which of the following is the most appropriate management?
A. antibiotics
B. Platelet transfusion
C. Supportive treatment
D. Factor VIll transfusion

Answer: C
A 6-year-old girl developed reddish urine. Her parents stat that she had been constipated for the last 2 weeks and recently became incontinent. On examination, there is abdominal tenderness, and no organomegaly (see lab results).
Blood pressure 115/76 mmg
Heart rate 76/min
Respiratory rate 27 /min
Temperature 38.9° C

Test Result Normal Values
WBC 9.5 4.5-10.5 x 109/L
Hb 79 112-165 g/L
Platelets count 325 150-400 × 109/L

Urine Analysis Result Normal Values
Leukocytes 25-30 0-3 per high power field
Erythrocytes 30-35 0-2 per high power field
Protein + Absent
Blood +++ Absent
Which of the following is most likely diagnosis?
A. Urinary tract infection
B. Henoch-schönleinpurpura
C. Autoimmune haemolytic anaemia
D. Post streptococcal glomerulonephritis

Answer: A

A 9-year-old girl is brought to the Emergency Department with abnormal movements and a recent history of tea coloured urine. Her parents described a preceding skin eruption. On examination, the child looks unwell, unresponsive and lower limb swelling is noted (see lab results).
Blood pressure 155/105 mmHg
Heart rate 110/min
Respiratory rate 25 /min
Temperature 37.4° C
Test Result Normal Values
Hb 78 112-165 g/L

Platelets count 165 150-400 x 109/L
Blood urea nitrogen 13.1 1.8 to 6.4 mol/L
Creatinine 101 27-88 mol/L
Complement C3 0.5 0.7-1.5 g/L
Which of the following is the most likely diagnosis?
A. Post streptococcal glomerulonephritis
B. Hemolytic ureamic syndrome
C. Henoch-schönleinpurpura
D. Acute renal failure

Answer: A

A 5-year-old boy is evaluated for generalized swelling of the face, abdomen and extremities over the past 3 weeks. The mother does not report any significant past medical problems and there is no history of fever or sore throat. There is no shortness of breath or cough. On examination, he appears comfortable and has significant edema of his face and lower extremities. The child will be started on a low sodium diet (see lab results).
Temperature 37.1°C
Heart rate 88/min
Blood pressure 100/60 mmHg Respiratory rate 16/min

Serum Test Results Normal Value
Serum albumin 25 36-52 g/L
Urinalysis Result Normal Values
Proteins 3+ 0
Erythrocytes 0 0-2 per high power field
24 Urine Analysis Results Normal Value
24 hr urine protein 140 0.8-14 mg/di
Which of the following should be recommended in addition to the diet?
A. Furosemide
B. Renal biopsy
C. Corticosteroid trial
D. Admit and start dialysis
Answer: C
A 14-year-old boy presents to the clinic complaining of "brown urine" for the past 3 days. 2 weeks ago, he had 2 days of fever and a sore throat, but he improved spontaneously and has been well ever since. On physical examination, only periorbital edema is noted (see lab results).
Blood pressure 136/92 mmHg
Heart rate 98 /min
Respiratory rate 28 /min Temperature 37 ° C

Urinalysis Result Normal Values
Specific gravity 1.035 1001-1030
Proteins ++ Absent
Blood ++ Absent
Which of the following is the most likely diagnosis?
A. IgA nephropathy
B. Acute pyelonephritis
C. Hemolytc-uremic syndrome
D. Post-streptococcal glomerulonephritis

Answer: D

A 4-year-old girl presents to the clinic for the 4-6-year vaccinations. She has steroid-dependent nephrotic syndrome. She has been on daily steroids 2 mg/kg/day for the last 6 weeks. examination reveals cushingoid facies and abdominal striae.
Which of the following is the best approach for immunizations?
A. Defer live vaccines until off steroid 1 month
B. Give immunoglobulins instead of vaccines
C. Decrease the dose of live vaccines
D. Give all vaccinations as scheduled

Answer: A
An 8-year-old girl is referred to the pediatric nephrologist for increasing lethargy and extremely puffy eyelids for a week. Urine analysis showed 4+ proteins but glucose, red blood cells and ketones were absent.
Which of the following is the most likely diagnosis?
A. Membrano-proliferatve glomerulonephritis
B. Focal segmental glomerulosclerosis
C. Minimal change disease
D. Berger's disease

Answer: C
A 5-year-old child has a 1-day history of cola-coloured urine with red blood cell casts. 2 weeks ago, the child had a culture-positive streptococcal tonsillitis. A diagnosis acute post- streptococcal glomerulonephritis is suspected.
Which of the following is the best clinical evidence to support the diagnosis?
A. Mildly elevated blood creatnine
B. Negative antnuclear antbody
C. positive streptozyme test
D. Low C3

Answer: D

A child is brought to the clinic with chronic proteinuria renal disease and hypertension. The parents are concerned about progression to end-stage renal disease.
Which of the following drugs would be most useful in preventing this progression?
A. Enalapril
B. Clonidine
C. Propranolol
D. Amlodipine

Answer: A

A 3-year-old child presents with history of increasing eye puffiness for 1 week especially in the morning. He has had flu-like illness 2 week ago. On examination, he looks well, not in distress and no lower limb edema. Eye, chest and abdomen exams are normal (see lab results).
Blood pressure 110/70 mmg
Heart rate 76 /min Respiratory rate 18/min
Temperature 36.6 °C
Oxygen saturation 95 %
Test Result Normal Values
Color Clear clear or light yellow
Protein +4 Absent
Glucose Absent Absent
Leukocytes 0 0-3 per high power field

Erythrocytes 0 0-2 per high power field
Which of the following has highest diagnostic value?
A. Serum albumin
B. Renal ultrasound
C. Serum electrolytes
D. Unine total protein

Answer: A
A 5-year-old child presents with history of increasing eye puffiness for 1 week especially in the morning. He has had flu-like illness 2 week ago. On examination, he looks well, not in distress and no lower limb edema. Eye, chest and abdomen exams are normal (see lab results).
Blood pressure 100/70 mmHg
Heart rate 88 /min Respiratory rate 18 /min Temperature 36.6 °C Oxygen saturation 95 % Test Result Normal Values Total Proteins 35 56-80 g/L
Albumin 16 36-52 g/L
Protein +4 Absent
Which of the following is the most likely diagnosis?
A. Nephritic syndrome
B. Nephrotic syndrome
C. Hereditary angioedema
D. Protein losing enteropathy

Answer: B
A 6-year-old child presents with history of increasing eye puffiness for 1 week especially in the morning. He has had flu-like illness 2 week ago. On examination, he looks well, not in distress and no lower limb edema. Eye, chest and abdomen exams are normal (see lab result).
Test Result Normal Values
Total Cholesterol 3 <5.2 mmol/L
Triglycerides 10 <1.7 mmol/L
Protein +4 Absent
Leukocytes 0 0-3 per high power field
Erythrocytes 0 0-2 per high power field
Which of the following is the most appropriate management?
A. Antihistamine
B. Renal biopsy
C. Systematic steroids
D. Albumin replacement

Answer: C

A 7-year-old girl presents with fever, abdominal pain and vomiting. On examination, she looks unwell with suprapubic tenderness, but no rebound tenderness nor guarding. Bed side dipstick done (see lab results).
Blood pressure 110/70 mmH
Heart rate 76 /min
Respiratory rate 18 /min Temperature 39 °C
Oxygen saturation 95 %
Test Result Normal Values
Color turbid clear or light yellow Appearance cludy clear
Nitrite pesent Absent
Leukocytes 22 0-3 per high power field
Erythrocytes 3 0-2 per high power field
Which of the following is the most appropriate management?

A. Admit for surgery
B. Admit for iv antibiotics
C. Discharge with oral antibiotics
D. Call gynecologist on duties

Answer: B

A healthy 8-month-old circumcised boy brought to the clinic by his parents with history of fever since morning. He has not had nasal congestion, cough, or diarrhea. He had 1 episode of emesis when he was given acetaminophen. He has continued to breastfeed and has had a normal number of wet diapers. He is mildly ill appearing. There is no clear focus of infection on examination.
Temperature 39.4 °C
which of the following factors is most strongly associated with an increased probability of having a urinary tract infection?
A. Being male
B. Episode of vomiting
C. Temperature of 39.4°C
D. Duration of the fever

Answer:C
A healthy 3-month-old uncircumcised boy brought to the clinic by his parents with history of fever since morning. He has not had nasal congestion, cough, or diarrhea. He had 1 episode of emesis when he was given acetaminophen. He has continued to breastfeed and has had a normal number of wet diapers. He is mildly ill appearing. There is no discernible focus of infection on examination.
Temperature 38.4 °C
Which of the following factors is most strongly associated with an increased probability of having a urinary tract infection?
A. Being male
B. Duration of fever
C. Being uncircumcised
D. Temperature of 38.4°C

Answer: C
A 4-month-old uncircumcised boy brought to the Emergency Department with a 2-day history of fever. He has not had vomiting, diarrhea, cough, or nasal congestion. There has been no known contact with ill persons. He has continued to breastfeed and has not had a decrease in the number wet diapers. He is mildly ill appearing but alert and interactive. Examination shows no focus of infection. A bag urine sample is collected and sent for urinalysis (see lab results). Temperature 39.1 °C
Test Result Normal Values
Color light yellow clear or light yellow
pH 8 4.3-8
Specific gravity 1030 1001-1030
Protein Absent Absent
Glucose Absent Absent
Ketones Absent Absent
Nitrite Positive Absent
Leukocytes 11 0-3 per high power field
Which of the following results of urinalysis would increase the probability of having a urinary tract infection?
A. Nitrite
B. Urine pH
C. Leukocytes

D. Specific gravity

Answer: A
A 3-month-old boy presented history of fever for the last 2 days. On examination, he is circumcised and has no focus of infection. Urine sample obtained by bladder catheterization is performed. He was started on oral trimethoprim-sulfamethoxazole. After being at home, his mother is called and states that he is back to his normal activity with no more fevers. The mother is advised to have him complete the 10-day course of trimethoprim-sulfamethoxazole (see report)
Urine culture: Grows 80,000 colony-forming units (FU) per milliliter of Escherichia coli that is susceptible to trimethoprim-sulfamethoxazole.
Which of the following is the most appropriate next step in management?
A. Renal ultrasonography
B. Voiding cystourethrogram
C. No further imaging or treatment is indicated
D. Antibiotic prophylaxis with trimethoprim-sulfamethaxole until 2 years of age

Answer: A
A previously healthy 4-month-old uncircumcised boy is brought to the Emergency Department with a 2-day history of fever. He has not had vomiting, diarrhea, cough, or nasal congestion. There has been no known contact with ill persons. He has continued to breastfeed and has not had a decrease in the number wet diapers. He is mildly ill appearing but alert and interactive. Examination shows no focus of infection. Urinary catheterization is performed (see lab results). Temperature 39.1 °C
Test Result Normal Values
pH 7.5 4.3-8
Specific gravity 1010 1001-1030 Protein Absent Absent
Glucose Absent Absent
Ketones Absent Absent
Nitrite Positive Absent
Leukocytes 30-40 0-3 per high power field
Erythrocytes 5-10 0-2 per high power field
which of the following is the most appropriate empirical therapy?
A. Oral cephalexin
B. Oral azithromycin
C. Intravenous cefepime
D. Intravenous ceftriaxone

Answer: A

Which of the following is not necessary to diagnose Nephrotic Syndrome?
A. Edema
B. Proteinuria
C. Hypertension
D. Low serum albumin

Answer: C
A 5-year-old boy presented with eye puffiness and lower limb edema. He has no fever, vomiting or abdominal pain. His physical examination revealed well perfused extremities and lower limb pitting edema. The rest of his exam is unremarkable (see lab results).
Color Clear clear or light yellow
pH 6 4.3-8
Specific gravity 1010 1001-1030
Protein ++ Absent
Glucose Absent Absent

Ketones Absent Absent
Nitrite Absent Absent
Hemoglobin Absent Absent
Leukocytes 0 0-3 per high power field
Erythrocytes 0 0-2 per high power field
Which of the following is the most common cause of nephrotic syndrome in such patient?
A. Finish type
B. Minimal change disease
C. Membranous nephropathy
D. Focal segmental glomeruloscierosis

Answer: B
A 5-year-old boy who is known to have nephrotic syndrome presents to the clinic with severe abdominal pain and distention. He has omitted once a clear stomach content of his last meal. No diarrhea or constipation. Which of the following this patient is at higher risk to have?
A. Hepatitis
B. Gastritis
C. Peritonitis
D. Appendicitis

Answer: C
A 5-year-old boy presented with eye puffiness and lower limb edema. He has no fever, vomiting or abdominal pain. His physical examination revealed well-perfused extremities and lower limb pitting edema. The rest of his exam is unremarkable. His laboratory testing confirmed the diagnosis of nephrotic syndrome. After starting him on treatment, which of the following defines complete remission?
A. Normal serum albumin
B. Complete resolution of his edema
C. Urine dipstick negative or trace reading for 3 consecutive days
D. Urine dipstick negative or trace reading for 5 consecutive days

Answer: C
A 5-year-old boy presented with eye puffiness and lower limb edema. He has no fever, vomiting or abdominal pain. His physical examination revealed well-perfused extremities and lower limb pitting edema. The rest of his exam is unremarkable. His laboratory testing confirmed the diagnosis of nephrotic syndrome. He is started on steroid treatment.
Which of the following used to define steroid resistant nephrotic syndrome?
A. Inability to induce a remission within 4 weeks of daily steroid therapy
B. Inability to induce a remission within 4 month of daily steroid therapy
C. Inability to induce a remission within 12 days of daily steroid therapy
D. Inability to induce a remission within 12 weeks of daily steroid therapy

Answer: A
A 5-year-old boy presented with eye puffiness and lower limb edema. He has no fever, vomiting or abdominal pain. His physical examination revealed well-perfused extremities and lower limb pitting edema. The rest of his exam is unremarkable. His laboratory testing confirmed the diagnosis of nephrotic syndrome.
Which of the following is the first line of treatment in this patient?
A. Rituximab
B. Prednisone
C. Cyclosporine
D. Cyclophosphamide

Answer: B

A 5-year-old boy is brought to the clinic by his mother. She is concerned of bed-wetting overnight despite toilet training, compared to his sister when she was in his age she controls herself. Until what age this is considered as a normal behaviour?
A. 5
B. 6
C. 7
D. 8

Answer: B
An 8-year-old boy is brought to the clinic by his mother. She is concerned of bed-wetting overight despite toilet training and compered to his sister when she was in his age with good control. Which of the following pelvic muscle is affected and possible target therapy?
A. Rectus
B. Detrusal
C. Perianal
D. Pelvic floor

Answer:B
An 11-year-old boy is brought to the clinic by his mother. She is concerned of bed-wetting overnight despite toilet training and several sessions of discussion. He feels a shamed and asked help from his family and physician. Which of the following is the best management?
A. Avoid punishment
B. Desmopressin intra-nasal
C. Alarm beds with positive reinforcement
D. Family centred approach and techniques

Answer: D
A 9-year-old boy brought to the clinic by his mother. She is concered of bed-wetting overnight for one month. He was dry since age of 6 years and the mother gave history of fever and foul smell of his urine recently. Urine dipstick done (see lab result).
Blood pressure 90/60 mmH
Heart rate 76 /min
Respiratory rate 18 /min Temperature 37.6 °C
Test Result Normal Values
Color dark yellow clear or light yellow
Appearance cloudy clear
Specific gravity 1030 1001-1030
Nitrite present Absent
Leukocytes 7 0-3 per high power field
Which of the following is the most likely diagnosis?
A. Nocturnal enuresis
B. Urinary tract infection
C. Improper personal hygiene
D. Normal pattem at this age

Answer: B

A 3-year-old child presents to the clinic with history of passing red-brown urine. There is a positive history of flu-like prodrome 2 weeks ago. Which of the following has the highest diagnostic value?
A. High complement C3

B. Low complement C4
C. High complement C4
D. Low complement C3

Answer: D

Neurology

A 12-year-old girl presents with tiredness, frequent ptosis and diplopia. She has no myalgia or fasciculation and her deep tendon reflexes are 1-2+. She reports that the symptoms become worse as the day progresses and that she feels better in the morning.
Which of the following is the most likely pathophysiology?
A. Ingestion of botulinum toxin
B. Antibodies against the AChRs
C. Mutation in the dystrophin gene
D. Antibodies against thyroid peroxidase

Answer: B
A 15-month-old child is brought to the Emergency Department after having a generalized tonic-clonic seizure that lasted approximately 5 minutes. The parents say that the child had been previously well but developed cough and rhinorrhea earlier that day with a temperature of 39.2° C.
Which of the following is the most appropriate management?
A. Diazepam
B. Phenytoin
C. Head CT scan
D. Control of the fever

Answer: D
Which of the following findings is expected in a 6-year-old boy with Duchenne muscular dystrophy?
A. Pes cavus
B. Claw hands
C. Muscle atrophy
D. positive gower's sign

Answer: D

A 9-month-old baby is brought to the Well-Baby Clinic. He is unable to sit, and on examination his hands are fisted and his legs are crossed. His mother said that he did not cry after the birth and resuscitation.
Which of the following is consistent with these findings?
A. Normal child
B. Cerebral palsy
C. Down syndrome
D. Congenital hypothyroidism

Answer: B
A full term baby born to a diabetic mother is brought to the clinic for assessment. The baby was born by vaginal delivery. The Moro reflex is absent on the right side. Baby weight is 4.8 Kg. Which of the following is the most likely cause?
A. Intracranial haemorrhage
B. Neonatal hypoglycaemia
C. Neonatal sepsis
D. Erb's palsy

Answer: D
A 3-year-old child falls out of a bunk bed. The child cries immediately, and there is no apparent loss of consciousness. Then he vomited twice and complains of a headache. The child has a normal mental status, normal neurological examination, and no evidence of a skull fracture. Which of the following is the most appropriate next step in management?

A. Brain magnified resonance imaging
B. Brain computed tomography scan
C. Neurosurgical consultation
D.Close observaton

Answer: D
A 6-month-old baby is brought by his mother for assessment of feeding difficulties and constipation. On examination, the baby's growth curve is below the 25th percentile, and there is a marked delay in his milestones. He had spastc tone all over, with crossed legs, but both arms were less spastic.
Blood pressure 90/50
Heart rate 120 /min
Respiratory rate 22 /min
Temperature 36.6° C
Which of the following cerebral palsy types is most likely to have?
A. Mixed
B. Diplegic
C. Hemiplegic
D. Quadriplegic

Answer: B
A 6-year-old child is brought to the clinic with sudden onset of a clumsy gait and an inability to stand or walk without support. The child is irritable and has been vomiting over the last 3 days and had chickenpox 3 weeks before. Muscle strength testing does not reveal any abnormality, but there is a resistance to neck flexion.
Which of the following disorders is most likely present?
A. Friedrich's ataxia
B. Meningoencephalitis
C. Acute cerebellar ataxia
D. Guillain-Barré syndrome

Answer: C

A 9-year-old child has been declining in school performance for several months due to lack of attention and "daydreaming" in class. The child, of average intelligence, has brief spells of staring into space that last 10 or 15 seconds. They occur several times an hour but the child seems quite unaware of these incidences. EEG reveals frontally predominant, generalized 3-Hz spike and wave complexes.
Which of the following drugs is most likely to be effective?
A. Phentoin
B. Imipramine
C. Ethosucimide
D. Catbamazepine

Answer: C

A 12-year-old boy presents with difficulty walking for the last 2 days. He states that this weakness was preceded by pain in the muscles and a tingling sensation that started down the lower limb but kept ascending upwards to the upper limbs. He gave a history of a preceding upper respiratory tract infection 2 weeks before. On examination, he had total areflexia of both upper and lower limbs, with moderate sensory involvement.
Which of the following is the most likely diagnosis?
A. Viral myelitis
B. Epidural abscess

C. Transverse myelitis
D. Guillain-Barré syndrome

Answer: D
A 16-year-old girl presents with throbbing headaches that last for 2 hours and occurs 3 to 4 times a week. The headaches are usually unilateral, begins with nausea, and exacerbated by noise and light. A recent brain CT scan was normal.
Which of the following should be used to treat acute symptoms?
A. Levofloxacin
B. Sumatriptan
C. Meclizine
D. Aspirin

Answer: B
A 12-year-old girl comes to the clinic after developing progressive weakness and inability to walk. She has intermittent tingling of her fingers and toes. History reveals she had 3 days of diarrhea about 2 weeks ago. Physical examination confirms marked peripheral muscle weakness without atrophy or fasciculation. The ankle and patella reflexes are absent, and a sensory examination is normal. Motor involvement is symmetric.
Which of the following is the most likely diagnosis?
A. Guillain-Barré syndrome
B. Mononeurits multiplex
C. Myasthenia gravis
D. Polio

Answer: A
A 12-year-old child presents to the clinic with a worsening history of severe headaches that are unilateral, pulsating, exacerbated by movement, and aggravated by light. Which of the following is the most likely type of headache?
A. Migraine
B. Cluster
C. Stress
D. Sinus

Answer: A
A 16-year-old boy comes to the clinic complaining of severe headaches twice a week. He describes the pain like a tight band around the head with pressure behind the eyes with a throbbing and burning sensation. The patent has been stressed with school work and sports. Which of the following is the most likely diagnosis?
A. Sinus headache
B. Tension headache
C. Vascular headache
D. Migraine headache

Answer: B

A 6-month-old boy is brought to the clinic for follow up. He has active seizures and is on anti-epileptic medications. On examination, he has dysmorphic features; is hypotonic and does not roll over or sit up with support. He does not seem to be fixing and following.
How shall the immunizaton schedule be modified?
A. Give IPV instead of OPV
B. Defer the DTP vaccine
C. Defer all live vaccines

D. Defer all vaccines

Answer: B
Cerebral palsy is a non-progressive disorder of motion and posture. Which one of the following risk factors is the most significant?
A. Prematurity
B. Perinatal asphyxia
C. Birth weight less than 1.5 Kg
D. Genetic/Metabolic/Metabolic abnormalities

Answer: C
A 6-year-old girl presents to the Pediatric Emergency Department with an unsteady gait and progressive difficulty climbing stairs over the last 10 days. The parents reported that the child has been repeatedly choking on her saliva. The symptoms began with a prickly sensation in the fingers and toes that developed with an ascending and symmetric numbness. There is a history of viral gastroenteritis three weeks before. A physical examination confirms absence of the knee jerk reflex (see reports).
CSF Analysis:
Elevated protein concentration with a normal cell count.
Nerve Conduction Test:
Reduced velocity.
Which of the following is the most appropriate treatment?
A. Plasma exchange
B. Immunoglobulin
C. Methotrexate
D. Steroids

Answer: B
A 6-year-old girl presents to the Pediatrics Emergency Department with an unsteady gait and progressive difficulty climbing stairs over the last 10 days. The parents reports that the child has been repeatedly choking on her saliva. The symptoms began with a prickly sensation in the fingers and toes that came along with an ascending and symmetric numbness. There is a history of viral gastroenteritis 3 weeks before. Physical examination confirms absence of the knee jerk reflex (see reports).
CSF Analysis:
Elevated protein concentration with a normal cell count.
Nerve Conduction Test:
Reduced velocity.
Which of the following is most likely long term prognosis?
A. Respiratory failure
B. Residual weakness
C. Complete recovery
D. Progressive paralysis

Answer: C
A 9-year-old boy referred to his local hospital with a history of fever, headache, prostration and nuchal rigidity. Gram stain done on the CSF smears shows gram positive cocci.
Which of the following CSF findings is consistent with this diagnosis?
A. Numerous lymphocytes, decreased glucose, normal protein
B. Numerous neutrophils, increased glucose, normal protein
C. Numerous lymphocytes, normal glucose, decreased protein
D. Numerous neutrophils, decreased glucose, increased protein

Answer: D

Which of the following drugs would be most suitable for a 7-year-old child with absence seizures?
A. Phenytoin
B. Ethosuximide
C. Phenobarbital
D. Carbamazepine

Answer: B

A 6-year-old boy presents to the Outpatient Department with frequent and brief episodes of rapid eyes blinking for 2 weeks. The episodes are not associated with pain, redness, tearing or discharge. During the episode, he communicated well with his parents. The physical examination shows normal visual acuity.
Which of the following is the most likely diagnosis?
A. Tic disorder
B. Pett mal seizure
C. Dry eye syndrome
D. Tourette syndrome

Answer: A

A 6-year-old girl presents with sudden involuntary abnormal movement in her limbs that started 10 minutes ago. It was associated with inability to speak. Few hours ago, she took metoclopramide, hyoscine butylbromide and oral rehydration solution after being diagnosed with viral gastroenteritis. The physical examination showed facial grimacing and protrusion of tongue.
Which of the following is most appropriate treatment?
A. Tizanidine
B. Epinephrine
C. Cyproheptadine
D. Diphenhydramine

Answer: D

A 5-year-old boy is brought to the Emergency Department after an episode of convulsion that involved all limbs 1 hour ago. It Continued for 3 minutes and associated with drooling and loss of consciousness. He had runny nose, mild cough and fever for last 3 days. On physical examination, he was fully active with mild congested throat.
Blood pressure 120/80 mmhg
Heart rate 98 /min
Respiratory rate 22 /min
Temperature 37.9° C
Which of the following is the most appropriate treatment?
A. Phenytoin
B. Diazepam
C. Amoxicillin
D. Paracetamol
Answer: D

A 14-year-old boy with a history of epilepsy complains of progressive lower limbs weakness and tingling sensation since last two days. His face, trunk and upper limbs are not involved. He has had

fever and runny nose 10 days ago. On physical examination, there is 3/5 muscle strength in both lower limbs with absent reflexes.
Blood pressure 120/80 mmHg
Heart rate 85 /min
Respiratory rate 18 /min
Temperature 37.4° C
Which of the following is the most likely diagnosis?
A. Polymyositis
B. Multiple sclerosis
C. Acute poliomyelitis
D. Guillain-barre syndrome

Answer: D

A 2-day-old baby is brought to the hospital because of reluctance to feed and seizures since last night. On examination, the baby weighed 3 kg, is inactive and has a generalized increased tone. A cerebral spinal fluid sample was obtained (see lab results).
Test Result Normal Values
Gross appearance Clear Clear and colourless
Glucose 3.5 2.8-5 mmol/L
Total protein 0.22 0.22-0.33 g/L
Cells 10 Up to 30/pL
Which of the following is the most likely diagnosis?
A. Neonatal sepsis
B. Neonatal tetanus
C. Pyogenic meningitis
D. Hypoxic- ischemic encephalopathy
Answer: D

A 9-month-old infant presents to Emergency Room with active generalized tonic-clonic seizure for 5 minutes ago. He has flu-like illness today and the review of system was unremarkable. The child was connected to cardiopulmonary monitor and intravenous access immediately attained. Blood pressure 80/50 mmHg
Heart rate 132 /min
Respiratory rate 31 /min
Temperature 39.6 °C
Oxygen saturation 95 %
Which of the following is the most appropriate next step in management?
A. Oral lorazepam
B. Nasal diazepam
C. Rectal diazepam
D. Intravenous lorazepam

Answer: D

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Heart rate 85 /min
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A. Tizanidine
B. Epinephrine
C. Cyproheptadine
D. Diphenhydramine

Answer: D

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A. Tic disorder
B. Petit mal seizure
C. Dry eye syndrome
D.Tourette syndrome

Answer: A

Which of the following cerebral palsy type is associated with Kernicterus?
A. Spastic
B. Ataxic
C. Hypotonic
D. Dyskinetic

Answer: D
A 4-year-old child presented to the clinic with delayed language and speech, and learning difficulties. Which of the following is the most appropriate test?
A. ECG
B. EEG
C. CT brain
D. Hearing test

Answer: D

Pediatric urology/ Optha/ ortho

A 10-year-old child presents with pain over the tibial tubercle. The child has recently had a growth spurt.
Which of the following is the most likely diagnosis?
A. Legg-Calve-Perthes disease
B. Osgood-Schlatter disease
C. Stress fracture
D. Shin splints

Answer: B

During physical examination of a child, the physician asks the patient to face the wall, bend forward and let the arms hang down unsupported.
What does this test screen for?
A. Scoliosis
B. Sexual abuse
C. Rectal prolapse
D. Leg length symmetry

Answer: A

A 12-year-old boy is brought to the Primary Health Care Clinic with knee pain for the past 3 weeks. The pain increases when he runs, jumps and climbs stairs. He plays basketball and football at school and has no history of injury or trauma. Physical examination is normal except for mild edema over the tibial tuberosity.
Which of the following is the most likely diagnosis?
A. Septic arthritis
B. Patellar dislocation
C. Osgood-schlatter disease
D. Iliotbial band friction syndrome

Answer: C

A 3-year-old boy fell on his outstretched hand while playing in the yard. On examination, there is tenderness and swelling of the forearm, which seemed to be abnormally curved (see image). Which of the following is the most appropriate management?
A. Perform a CT scan of forearm
B. Rest. ice, compression and elevation
C. Administer paracetamol and discharge
D. Reduce the fracture and cast the forearm

Answer: D

A 4-year-old child is brought to the clinic for a check-up. When performing a cover test, the child's left eye moves laterally when the right eye is covered.
Which of the following is the most serious complication?
A. Strabismus
B. Nystagmus
C. Amblyopia
D. Leukocoria

Answer: C

A 2-month-old infant seen in the screening clinic and the red reflex eye exam shown (see image). Which of the following is the most appropriate next step in management?
A. Send for brain imaging
B. Next available appointment
C. Reassurance as its normal finding
D. Immediate referral to ophthalmology

Answer: D

A 2-year-old boy with Down syndrome is brought to the clinic after his mother noticed that his eyes are turning in. Which of the following is the most appropriate method to screen for strabismus?
A. Red Reflex
B. Cover test
C. Visual Acuity
D. Corneal light reflex

Answer: D; aka Hirchberg test
Which of the following is the most appropriate age to do Red reflex test?
A. All ages
B. 3-4 years
C. Pre-school at age 6
D. 6 months until cooperative for vision testing

Answer: A

A 2-month-old girl brought to the Well Baby Clinic. On examination, she has absent red reflex in her left eye. She is otherwise healthy and growing well with no concerns from her parents. Which of the following is the most appropriate next step evaluation?
A. CT of the brain and orbits
B. MRI of the brain and orbits
C. Reassurance of the parent as this is a normal variation
D. Dilated eye examination under sedation by an ophthalmologist

Answer: D
A 2-month-old girl brought to the Well Baby Clinic. On examination, she has absent red reflex in her left eye. She is otherwise healthy and growing well with no concerns from her parents. Which of the following is the most likely tumour in such patient?
A. Fibrosarcoma
B. Retinoblastoma
C. Teratoma
D. Soft tissue sarcoma

Answer:B

What is the most common disorder of sexual differentiation in boys?
A. Micropenis
B. Buried penis
C. Hypospadias
D.Cryptorchidism

Answer: D

A newborn boy is brought to the Paediatrics clinic. examination shows abdominal distension with palpable kidneys and a distended bladder (see report).
Voiding cystourethrogram:
Shows bilateral hydronephrosis and distension of the bladder and proximal urethra.
Which of the following is the most likely diagnosis?
A. Prune-belly syndrome
B. Posterior urethral valves
C. Polycystic kidney disease
D. Ureteropelvic junction obstruction

Answer: B
A 6-year-old boy presents to the Emergency Room with severe scrotal pain. The parents report that he started complaining 5 hours before and that the pain has become increasingly worse. He is otherwise healthy and there is no history of trauma. A physical examination confirms a swelling of the scrotum and tenderness of the left testicle. The cremasteric reflex is absent. Which of the following actions should be taken next?
A. Surgical consult
B. Testicular ultrasound
C. Urinalysis and urine culture
D.Analgesics and apply ice to scrotum

Answer: A
Which of the following physical findings in boys is the earliest indicator that puberty has begun?
A. Increased penile size
B. Increased prostate size
C. Increased testicular size
D. Appearance of pubic hair

Answer: C
A 2-year-old boy presents to the clinic with abdominal pain, malaise, fever, nausea and vomiting. He has a history of 2 urinary tract infections. A contrast voiding cysto-urethrogram is requested (see image).
Which of the following is the most likely diagnosis?
A. Ureterocele
B. Urinary lithiasis
C. Vesicoureteral reflux
D. Agenesia of the left ureter

Answer: C

A 4-month-old boy brought by his mother asking for circumcision. History revealed that he was diagnosed with mid-shaft hypospadias. Which of the following is the most appropriate counseling should be given?
A. No link between hypospadias and circumcision
B. It is late for circumcision as correction is often undertaken before 4 month old
C. Infant with hypospadias must not be circumcised as the skin will used in the surgery
D. Proceed with circumcision as there is no need for surgical correction for this type of hypospadias

Answer: C

In which conditions circumcision should be avoided?

A. Phimosis
B. Hypospadias
C. Unilateral Cryptorchidism
D. Posterior urethral valve

Answer: B
in which conditions circumcision should be avoided?
A. Phimosis
B. Hypospadias
C. Unilateral Cryptorchidism
D. Posterior urethral valve

Answer: B
A 2-year-old boy presents to the paediatrician with severe testicular pain and swelling that had just started 2 hours before. The child is nauseous and has vomited twice. The parents report a similar episode that had happened 3 weeks ago but was of a very short duraton. The scrotum is swollen and tender and does not transilluminate. No contraction of the cremaster muscle can be provoked by stroking the inner part of the thigh.
Blood pressure 110/70 mmhg Heart rate 160 /min Respiratory rate 50 /min Temperature 36.6° C
Which of the following is the most appropriate treatment?
A. NSAID
B. Antibiotic therapy
C. Surgical exploration
D. Elevated scrotal position

Answer: C
A newborn boy is brought to the Paediatric Clinic for a check-up. examination shows a tense fluctuant, painless mass on the right side of the scrotum. Shining a bright light toward the enlarged right side of the scrotum by torch, a clear illumination is observed.
Which of the following is the most likely diagnosis?
A. Varicocele
B. Hydrocele
C.Haematocele
D. Indirect inguinal hernia

Answer: B

Practice / Ethics

A 7-year-old child is brought to the clinic. The parent reports yellowish-brown discoloraton and caries on the occlusive surfaces of her teeth.
Which of the following is the most appropriate treatment?
A. Diet modification
B. Antiseptic mouth wash
C. Topical intraoral antibiotics
D. Systemic fluoride supplementation

Answer: D

When should breastfeeding begin after the delivery?
A. Immediately
B. 8 hours
C. 24 hours
D. 48 hours

Answer: A
A 5-day-old cyanotic infant with a diagnosis of Tricuspid Atresia. Which of the following is the most important issue to consider during the discussion with parents?
A. Make sure that you have left no question unanswered
B. Make sure the parents understand the dismal prognosis
C. Make sure the parents understand that you are doing everything you can
D. Make sure you leave the parents with a follow-up plan and provide them cardiac consultation

Answer: D
A 3-month-old infant diagnosed with congestive heart failure with follow-up by cardiology. Mother asks regarding the nutritional needs for her infant. What is the most appropriate counseling should be given?
A. Less than a healthy infants
B. The same as 2-year-old infant
C. The same as a healthy infants
D. Greater than a healthy infants

Answer: D
A 7-year-old boy case of congestive heart failure on digoxin. Which of the following is the most important step to improve his compliance?
A. Explain to child's parents
B. Send a nurse to his home to check on compliance
C. Having the child participate in devising the plan
D. Inform the family about side effects of medication

Answer: C
Which of the following is the best advice to decrease anxiety of a child's parents who is scheduled for cardiac surgery?
A. Obtain an order for anti-anxiety medication for the parents
B. Explain the steps that will occur before and after surgery
C. Send education material to parents and the child about the surgery
D. Tell the parents not to worry, because the surgeon performs this procedure daily

Answer: B

A child with tetralogy of fallot present for regular check-up. Mother asks why her child squats after exertion. Which of the following is the best explanation?
A. Reduces muscle aches
B. Increases cardiac efficiency
C. Enhances the pull of gravity
D. Decreases blood volume in the extremities

Answer: B
A 6-week-old infant presents to Well Baby Clinic by the mother complains that her baby "spitting up" all for julia since birth. The baby weighs 5.5 kg and birth weight was 3.4 Kg What is the most appropriate counseling should be given?
A. Refer the child to a pediatric gastroenterologist
B. Return home and relax; the child will grow out of it
C. Start investigating the child for pyloric stenosis immediately
D. Increase the time spent burping the infant and keep the infant semi upright after feedings

Answer: D

A 4-year-old child known to have bronchial asthma presents with the mother for primary clinic checkup. There is family history of asthma and father smokes at home.
What is the most appropriate counselling should be given?
A. Reassure the mother for now
B. Refer the child for allergy test
C. Prescribe inhaled steroid for regular use
D. Counseling his parents regarding smoking cessations

Answer: D
A 12-year-old boy complains of bilateral leg pain at night time. He is diagnosed with growing pain. What would be the best management?
A. Refer to orthopedic clinic
B. Use aspirin to relive the pain
C. Request bilateral lower limp X-ray
D. Reassure the mother and the child

Answer: D
A 2-day-old baby seen in clinic with mother for counselling about breastfeeding. Which of the following indicates adequate breastfeeding in a baby?
A. Stool 4 times/day
B. Urine 2-3 times/ day
C. Does not cry frequently
D. Baby sleeps 2 hour after feeds

Answer: A
A 20-day-old premature baby brought to the clinic with the mother for counselling about breast feeding. Which of the following is low in mother's milk after premature delivery?
A. Fat
B. Protein
C. Lactose
D. Sodium

Answer: C
A 2-week-old baby girl brought by her mother to the clinic because she noticed mucus vaginal discharge. Baby is product of spontaneous vaginal delivery with normal urine and stool. Diagnosis is

mucoid vaginal discharge.
Which of the following is the best response to this mother's concern?
A. Take swab
B. Request urinalysis
C. Refer to pediatrician
D. Reassure the mother

Answer: D
A 7-year-old boy presented to Emergency Department with newly diagnosed acute diabetic ketoacidosis. which of the following is most essential element in management plan to ensure compliance?
A. Refer to health educator
B. Provide written management plan
C. Involve the child in type diabetic patient's group
D. Listen closely to the child's concerns and questions

Answer: D
A 7-year-old boy admitted with acute diabetic ketoacidosis. Mother states that her son was diagnosed with type diabetes mellitus started on insulin but she decided to stop it, as she was not sure of her son's diagnosis and thought to give chance for his pancreas to recover.
Which of the following term describes the mother action?
A. Denial
B. Anxiety
C. Neglect
D. Depression

Answer: A
A 9-year-old boy just diagnosed with growth hormone deficiency. In family meeting with the child and his patents after making introductions, the doctor makes the following comment, "I'd like to go over the diagnosis, then talk about what might happen next, that's to say how you might progress. then I'll try to answer all your questions"
Which of the following describes doctor phrase?
A. Probing
B. Agenda setting
C. Safety netting
D. Checking understanding

Answer: B
An 8-year-old girl complaints of recurrent periumbilical pain lasting for 10-15 min morning time, no vomiting or loose stools. Abdomen examination is normal. Urine and stool analysis are normal. Mother says this pain is only during school days. Counseling to the mother and the teacher to manage school phobia/refusal includes behavior-management strategies such as providing positive reinforcements.
Which of the following is an example of positive reinforcements?
A. Allow student to have more TV time at home
B. Comments focus on what the student did right
C. Give the student moderately challenging task to perform
D. Keep records of number of absent days during the semester

Answer: B
A 6-year old girl presents to the clinic with malnutrition. She seems to be neglected with poor hygiene. Her mother express feeling anxious and busy with other children at home with poor social

circumstances. During consultation, it is essential to explore mother perspective, feeling. ideas and beliefs. Which of the following is the best next step?
A. Refer to social worker
B. Use close ended questions
C. Refer mother to psychiatry
D. Encourage the mother to use her own words to clarify the issues

Answer: D
A 6-year-old boy presents with his mother to outpatient clinic with malnutrition, he seems to be neglected with poor hygiene. Which of the following is an important skill needed to conduct consultation?
A. Lay out the consulting room with toys
B. Ask the mother to leave the room
C. Call the nurse to take vital sings, height and weight
D. Put aside time at the beginning to build rapport with him

Answer: D
A 42 hours-old full-term infant has not yet passed meconium was diagnosed with cystic fibrosis after Sweat chloride test. Family meetings with parents to discuss the diagnosis, management plan. The physician pulls together the main ideas, plan of management, prognosis and feelings of the parents to show understanding.
Which of the following describe the technique in this communication skill?
A. Reflection
B. Summarizing
C. Paraphrasing
D. Verbal mirroring

Answer: B
A 14-month-old child is brought by his father with 2nd degree burns on both feet. He states that the child dipped in very hot bathtub water. There is a clear distinction line on the burn with no splash burns. There are other burses on the child limps of different age. Child abuse suspected. Which of the following is the correct order of the four important advocacy skills for mandated reporters of child abuse?
A. Observing, documenting, reporting, self-care
B. Documenting, confirming, reporting. conferencing with parents
C. Observing. gathering information, making referrals, following up
D. Observing, making referrals, gathering information, following up

Answer: D
A 15-year-old boy brought by concerned mother for an evaluation her son problem of 3 months after the death of his friend in car accident, reluctantly mentioned to his mother that he wished he were dead, but would never do anything about these wishes. Which of the following is the most appropriate action?
A. Asking him about his suicide may increase the risk
B. Indirect inquiries about his suicide risk will foster rapport
C. Since this is a normal grief reaction, he should be left alone
D. Detailed questions about his suicidality are essential for prevention

Answer: D
Surgical intern is assigned to stay with a 6-year-old boy who is waiting to have surgery to repair a cleft palate. The boy does not speak Arabic and appears anxious. The charge nurse has requested a translator but the translator has not yet arrived.
Which of the following is the most appropriate action for the intern to do?

A. Insert IV cannula and start preanesthetic
B. Sedate the child to decrease her anxiety level
C. Suggest that the nurse stay with the child so that he can review her chart
D. Listen to the child's heart with a stethoscope and then let the child try to use the stethoscope to listen to his heart

Answer: D
An effective interviewing skill behaviors of listening carefully by entering the world of the patient and communicating that you imagine the patients world as the patient sees and experiences it. Yet, not becoming the patient and remain separate from the patient.
Which of the following is the communication skill described above?
A. Empathy
B. Transferce
C. Respect and warmth
D. Authenticity or congruence

Answer: A
An angry father comes to the clinic because his 16-year-old daughter request to join sport class. He claims that she was encouraged by her mother to do so and ask the doctor if he can convince her after physical examination that she is not fit. During consultation, the doctor needed to gather more information using confrontational communication skills.
Which of the following is an example of a confrontational response?
A. "I totally agree with your opinion*
B. "I can see that you have been worried a lot"
C. "you have been through this very thing myself*
D. "You are looking very upset. What is happening?"

Answer: D
Which of the following is the appropriate action when caring for parents immediately after a newborn death?
A. Discussing future risk of malformed baby
B. Ask the father if he wants to file complaint
C. Arrange for early follow up with primary care
D. Allowing the mother to spend time with her dead baby

Answer: D
A 12-month-old child with acute gastroenteritis seen in Emergency Department and being discharged with dietary advice and important worrying symptoms that need immediate care. Which of the following describes the technique in this step?
A. Reflection
B. paraphrasing
C. Safety netting
D. Verbal mirroring

Answer: C
Which of the following is a common reaction to a parent's illness in 12-year old girl considering her developmental stage?
A. Difficulty speaking directly about strong or difficult feelings
B. Unable to understand or appreciate details of diagnosis, prognosis and treatment
C. Incorporate themes of illness in their play, such as pretending to have their parent's symptoms
D. Likely to shift to earlier, less mature behaviors (e.g., fussiness, difficulty with separations or bedtime

Answer: A
Which of the following is the most successful strategy to prevent poisoning among young children?
A. Education about the risk using social media
B. Educating daycare providers on poisoning risks
C. Provide parents with education material regarding safety
D. Enforce legislation to require childproof caps on hazardous household cleaners and medicines

Answer:D
An 8-week-old infant presents by his mother to Well-Baby Clinic worried about her baby crying with diagnosis of infantile colic. Which of the following is the most appropriate counseling should be given?
A. Infants cry an average of 5 hours per day, peaking at 6 weeks of age and gradually decreasing
B. Infants cry an average of 6 hours per day, peaking at 6 weeks of age and gradually increasing
C. Infants cry an average of 2 hours per day, peaking at 6 month of age and gradually decreasing
D. Infants cry an average of 2.2 hours per day, peaking at 6 weeks of age and gradually decreasing

Answer: D
A 2-year-old boy brought by his mother to clinic asking for help to control her child behavior. The child seems to be stubborn and he throws his toys away. When she walks him to daycare, he refuses to hold her hand, attempts to run away from her. Which of the following is the most appropriate technique dealing with his behavior?
A. Spanking
B. Time-out
C. Scolding
D. Verbal waming

Answer: B;
Which of the following best describes time-out technique to control child behavior?
A. Last 1 min/yr. of age
B. Last at least 10-15 min
C. If accompanied by penalty
D. If accompanied by verbal warnings

Answer: A
A 7-year-old boy with a diagnosis of separation anxiety. The mother asks for consultation. Which of the following is the most appropriate strategy that the mother should do?
A. Let your child focus on what unlikely bad things might happen
B. Let your child stay home when he does not want to go to school
C. Punish for behaviors that are a result of separation anxiety/fears
D. Help him think of ways a favorite superhero might handle the situation

Answer: D
Which of the following is most appropriate counseling regarding storage and handling of breast milk?
A. Can be frozen for up to 12-18 months
B. After freezing can be refrozen if not used
C. Can be stored up to 4-6 months in the refrigerator
D. Can be safely stored up to 4-10 hours at room temperature

Answer: D
In consulting a new mother about vaccination, the doctor mentioned, "Giving your child a vaccination against a disease dramatically lowers the risk of him catching the illness which could be extremely harmful to him both in the short and long-term. By stopping the disease developing in

your child, we also stop your child being able to spread the disease." The doctor asked the mother "Does everything I've said so far make sense?" Which of the following describes doctor last phrase?
A. Paraphrasing
B. Safety netting
C. Verbal mirroring
D. Check understanding

Answer: D

Which of the following carry a high risk to induce fabricated illness in the child?
A. Father
B. Mother
C. House maid
D. Step mother

Answer: B

While counseling a new mother about the importance of the breastfeeding, she mentioned that currently she had a yellow-white substance coming from the breast and she is worried about feeding her baby. Which of the following is correct regarding this yellow-white substance?
A. It has high content of proteins
B. It will continue for 4 month normally
C. It should be discarded because the color
D. It has high content of lipid and glucose

Answer: A

What is the best time to start educating new mothers about breastfeeding?
A. After delivery
B. Last trimester
C. Second trimester
D. Before planning for pregnancy

Answer: D

Which of the following is the most appropriate advice to a mother about breastfeeding schedule for her full term newborn in the first year?
A. On demand for first 4 month
B. On demand every 2-4 hr for first 2 month
C. On demand 6 times per day for first 2 month
D. On demand 6-8 times per day at age 8-12 month

Answer:B

A 19-month-old boy known case of spinal muscular atrophy admitted with deterioration of pulmonary functions that mandate intubation. The pediatrician discussed the decision of intubation with the family but they refused because of their previous experience with their child who died with the same condition. Which of the following is the most appropriate next action?
A. Call the hospital's child abuse team
B. Search the Internet for similar cases
C. Consult the hospital's clinical ethics committee

D. Ignore the family objection and intubate the patient

Answer: C
A 15-year-old boy is a suspected case of acute meningitis was rushed to the Emergency Room by his parents. The pediatrician decided to do a lumber puncture. Which of the following is the most appropriate next action?
A. Take parent consent but not patient assent
B. Take patient assent but not parent consent
C. Take both patient assent and parent consent
D. Do not take patient assent or parent consent because it is an emergency case

Answer: C
A 12-year-old girl is brought to the Emergency Room with multiple bruises attributed by her parents to an accident while she was playing. After thorough history and physical examination, the pediatrician suspected child abuse. Which of the following is the most appropriate next action?
A. Call the hospital's child protection team
B. Confront the parents with her suspicions
C. Consult the hospitals clinical ethics committee
D. Ask the family to take the patient to another hospital

Answer: A
A newly born baby resulted from out of wedlock pregnancy was abandoned by his mother. On examination, the baby found to have inguinal heria and the pediatric surgeon advised to do hernia repair now. Which of the following is the most appropriate action?
A. Refer the case to the police
B. Refer the case to the social worker
C. Consult the hospitals clinical ethics committee
D. Perform the operation without consent form

Answer: B
A 7-year boy diagnosed with measles. When his father was asked about the child immunization record, he replayed that he does not believe in immunization. Which of the following is the most appropriate action?
A. Inform the child protection team
B. Consult the hospital's clinical ethics committee
C. Explain to the father the importance of immunization
D. Ignore the father's response and focus on treating the child

Answer: C
A 6-year-old child a known case of bronchial asthma presents to the Emergency Room with asthmatic attack. His father is a heavy smoker and was advised many times to avoid exposing his son to smoking. The father seems to be careless and not interested to listen to the pediatrician. Which of the following is the most appropriate action?
A. Inform the child protection team
B. Consult the hospital's clinical ethics committee
C. Explain to the father the importance of immunization
D. Ignore the father's response and focus on treating the child

Answer: A
A premature baby at 23-weeks, with fatal congenital anomalies is born and in need for intubation and resuscitation. His mother is one of the physicians in the hospital and she expressed her wish to do not resuscitate the baby. Which of the following is the most appropriate action?
A. Honor the mother's wish

B. Refer the case to another colleague
C. Consult the hospital's clinical ethics committee
D. Ignore the mother's wish and do full resuscitation

Answer: C
Which of the following is the minimum recommended daily minutes of physical activity for children?
A. 30
B. 60
C. 120
D. 150

Answer: B
A 19-month-old boy was brought to the Pediatric Clinic for ear infection. The nurse noticed the boy has not received any vaccination as recommended. The parents mentioned that they read that vaccinations are "bad to the boy brain". Which of the following is the most appropriate action?
A. Refer the parent for psychiatric evaluation
B. Accept since parent have full right to choose
C. Call social services to take over the custody
D. Correct the myth and explain the great benefit of vaccinations

Answer: D